Mass Gap of Four-Dimensional Lattice Yang–Mills Theory for Every Compact Simple Gauge Group

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We fix the gauge of four-dimensional lattice Yang–Mills theory on lines that meet every plaquette once, so that each free link interacts with twelve others. For every compact simple gauge group the theory then has a unique infinite-volume state, exponential clustering and an explicit mass gap at fixed lattice spacing, past the curvature limit of the ungauged theory; for SU(2) and SU(3) the ranges reach $\beta_W \le 0.345$ and $\beta_W \le 0.76$. A mass gap in physical units for every continuum limit follows from one estimate of Dobrushin–Shlosman type in a cube of fixed physical size; that estimate and the existence of the limit remain open.

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I. INTRODUCTION

Wilson's lattice gauge theory [63] assigns an element Q_e of a compact group $G \subset \mathrm{U}(N)$ to every link e of $\mathbb{Z}^4$ and weighs a configuration by $\exp(N\beta \sum_p \mathrm{Re}\,\mathrm{Tr}\,Q_p)$, where Q_p is the ordered product of the four link variables around the plaquette p.[1] At small $|\beta|$ the theory is under control: it has a unique infinite-volume state, correlations decay exponentially, and the transfer matrix has a gap [52, 54].[2] The question is how far into the

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An earlier version, titled "Mass Gap of Four-Dimensional Lattice Yang–Mills Theory in a Forest Gauge", appeared as Hexagon preprint 2610.00341. This version adds the theorem for every compact simple gauge group, the reduction of the continuum gap to one physical scale, and complete proofs of the results earlier taken from our first paper. The LaTeX source, the checks and the Lean certificate are in github.com/creelie/4Y-M, release v3.5.0.

[1] This is the normalisation of Shen, Zhu and Zhu [55] and of our earlier paper [13]. The conventional inverse coupling is $\beta_W = N^2\beta$, so the weight is $\exp(\frac{\beta_W}{N}\sum_p \mathrm{Re}\,\mathrm{Tr}\,Q_p)$; for SU(2), identified with the unit quaternions, $\frac{1}{2}\mathrm{Tr}\,Q_p = \mathrm{Re}\,Q_p$ and $\beta_W = 4\beta$. Standard accounts of the lattice theory are the books of Montvay and Münster [49] and of Gattringer and Lang [31]; the Hamiltonian version is due to Kogut and Susskind [43].

[2] Strong-coupling results of this kind go back to Osterwalder and Seiler [52], who proved exponential clustering and a gap by a convergent expansion, and to Seiler's monograph [54]; the general framework for lattice gauge fields and their continuum limits is that of Brydges, Fröhlich and Seiler [19]. More recent work treats the strong-coupling regime by probabilistic methods: the survey of Chatterjee [22], his large-N solution of $\mathrm{SO}(N)$ lattice gauge theory at strong coupling [23] and its planar extension by Basu and Ganguly [9],

coupling range such statements can be proved with explicit constants, since the problem of the continuum limit [39] begins where strong coupling ends and Bałaban's weak-coupling analysis [4–6] takes over.

Two methods give explicit ranges. The Bakry–Émery method [2, 3] of Shen, Zhu and Zhu [55] bounds the Hessian of the action; in [13] we showed that the sharp bound is $4d = 16$ and that no curvature argument for the ungauged measure goes beyond $|\beta| < 1/32$. The Dobrushin method [26, 27] bounds the influence of each link on the conditional law of another; for $\mathrm{SU}(2)$ the single-link conditional law is a von Mises–Fisher law on the three-sphere, the sharp Kantorovich modulus of that family is one quarter [13] (Theorem 56 in Appendix G), and every link has eighteen partners, which gives $\beta_W < 2/9$ and nothing beyond, because the coefficients are attained.

Both limits come from the same count. A link of $\mathbb{Z}^4$ lies in six plaquettes and shares each with three other links. Gauge invariance makes part of this redundant: in a box, the variables on a forest of links can be set to the identity without changing any gauge-invariant expectation [24].[3] The forest we use is a field of straight lines with one line through every site, chosen so that every plaquette contains exactly one fixed link. Then every free link has exactly twelve free partners, two in each of its plaquettes (Lemma 9), and both methods improve.

Theorem 1 ($\mathrm{SU}(2)$)**.** *Let $G = \mathrm{SU}(2)$ and $0 \le \beta_W < \frac{1}{3}$. For every gauge-invariant local observable O the expectations $W_\Lambda(O)$ in boxes with free boundary converge, along every sequence of boxes that exhausts $\mathbb{Z}^4$, to a limit $\omega(O)$ that does not depend on the sequence. The state ω is invariant under translations and the hypercubic group and reflection positive; truncated correlations of Lipschitz observables decay at every rate below $-\log(3\beta_W)$; and the transfer matrix has spectral gap*

$$m \ge -\log(3\beta_W) > 0.$$

For $-\frac{1}{3} < \beta_W < 0$ all statements except reflection positivity and the gap hold.

The step behind Theorem 1 is short once the line field is in place.[4] The number $\frac{1}{3}$ is also the end of the road for this kind of argument.

Theorem 2 (Ceiling)**.** *Let F' be any set of links of $\mathbb{Z}^4$ without cycles, and give two distinct links outside F' that share a plaquette the coefficient $|\beta|$, the largest single-link Kantorovich coefficient of $\mathrm{SU}(2)$. Then the weighted Dobrushin condition with these coefficients fails for $\beta_W \ge \frac{1}{3}$. For the line field the coefficients are attained, so no single-link Kantorovich argument in that gauge reaches $\beta_W = \frac{1}{3}$.*

The ceiling concerns coefficients between single links. Take instead as a block the three free links of one plaquette. The free links of the line field split into such triples with period two (Lemma 12), the coupling inside the block plaquette is no longer paid for, and each link keeps ten outside partners. The price is the sensitivity of a law on three spheres instead of one, which we bound analytically by integrating out the middle link of the block.

Theorem 3 ($\mathrm{SU}(2)$ with plaquette blocks)**.** *Let $G = \mathrm{SU}(2)$ and $0 \le \beta_W \le 0.345$. The number $\alpha(\beta_W)$ of (4) is increasing, $\alpha(\frac{1}{3}) \le 0.9561$ and $\alpha(0.345) \le 0.9973$. The conclusions of Theorem 1 hold with decay at every rate below $-\frac{1}{2}\log\alpha(\beta_W)$ and a transfer-matrix gap*

$$m \ge -\tfrac{1}{2}\log\alpha(\beta_W) > 0;$$

at $\beta_W = \frac{1}{3}$ this gives $m \ge 0.0224$. For $-0.345 \le \beta_W < 0$ all statements except reflection positivity and the gap hold.

At small coupling the rate of Theorem 1 is the better one, so both bounds stand side by side; Theorem 3 matters past $\beta_W = \frac{1}{3}$, where the gap of Theorem 1 has closed.

The curvature method gains as well, and for every N. Let $\mathrm{Hess}\, S_0$ be the Hessian of $\sum_p \mathrm{Re}\,\mathrm{Tr}\, Q_p$ on G^E and F the set of fixed links.

Theorem 4 (Curvature in the gauge)**.** *For every N, every field Q and every tangent vector $v = (X_e Q_e)$ with $X_e = 0$ for $e \in F$,*

$$|\mathrm{Hess}\, S_0(v,v)| \le \tfrac{29}{2}|v|^2,$$

against the sharp constant 16 without gauge fixing. Consequently, for $G = \mathrm{SU}(N)$ and $|\beta| < \frac{1}{29}$, or $G = \mathrm{SO}(N)$, $N \ge 3$, and $|\beta| < \frac{N-2}{58N}$, the conclusions of Theorem 1 hold with the rate and gap of Theorem 5. For $\mathrm{SU}(3)$ this is $\beta_W < \frac{9}{29} = 0.310$.

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the Langevin dynamics of Shen, Zhu and Zhu [55], and the area law in an enlarged region by Cao, Nissim and Sheffield [21]. For finite gauge groups Adhikari and Cao proved correlation decay at weak coupling [1]. In two dimensions the continuum Yang–Mills measure has been constructed by Gross, King and Sengupta [34], Driver [29] and Lévy [44]; in four dimensions Bałaban's renormalisation group, as expounded by Dimock [25], gives ultraviolet stability in finite volume, and Magnen, Rivasseau and Sénéor constructed the theory with an infrared cutoff [45]. None of these results gives a mass gap in physical units.

[3] Creutz used a maximal tree to remove the gauge freedom completely; the temporal gauge, with all links in one direction fixed, is the special case of a family of straight lines. In a maximal tree almost every link is fixed in some region and none in others, so the number of partners of a free link is not uniform, and the temporal gauge leaves fourteen partners per link. The point here is a pattern that fixes exactly one link in every plaquette.

[4] The combination of a forest gauge fixing with the sharp quarter of [13] was first assembled during a run of Anaxia, the reasoning engine of the BLLM system [11], release v3.0.5. In that run the engine combines proof steps along typed links (a step that produces a single-link Kantorovich bound may only feed a step that consumes one), and in every run with typed crossing it reached a composite accepting a lattice mass gap for four-dimensional $\mathrm{SU}(2)$ at $\beta_W < \frac{1}{3}$, which no untyped or random run reached. The composite was first proposed by the language model inside the engine, and the engine's network reproduces it. The engine did not supply the line field, the free-boundary limit, the ceiling, the curvature bound or anything in the $\mathrm{SU}(3)$ part; the authors wrote and checked every proof in this paper. Release v3.1.0 of the engine [12] records its Yang–Mills runs in the failure log: the narrowing of the target from the continuum gap to explicit couplings in this gauge, the change from analytic one-link bounds to bounds certified box by box, and the change to blocks of several links. These records follow the course of the work and supply no step of the proofs.

TABLE I. Least Ricci curvature $\kappa_G = h^\vee/(2\ell_\rho)$ of the compact simple groups in a faithful representation ρ of least dimension N, and the range of Theorem 5 for the conventional inverse coupling $\beta_W = N^2\beta$.

Group	ρ	$h^\vee$	ℓ_ρ	κ_G	$\beta_W <$
$\mathrm{SU}(N)$, $N \geq 2$	$\mathbb{C}^N$	N	1	$\frac{N}{2}$	$\frac{N^2}{29}$
$\mathrm{SO}(N)$, $N \geq 5$	$\mathbb{R}^N$	$N-2$	2	$\frac{N-2}{4}$	$\frac{N(N-2)}{58}$
$\mathrm{SO}(3)$	$\mathbb{R}^3$	2	4	$\frac{1}{4}$	$\frac{3}{58}$
$\mathrm{Sp}(n)$, $n \geq 2$	$\mathbb{C}^{2n}$	$n+1$	1	$\frac{n+1}{2}$	$\frac{2n(n+1)}{29}$
G_2	7	4	2	1	$\frac{14}{29}$
F_4	26	9	6	$\frac{3}{4}$	$\frac{39}{29}$
E_6	27	12	6	1	$\frac{54}{29}$
E_7	56	18	12	$\frac{3}{4}$	$\frac{84}{29}$
E_8	248	30	60	$\frac{1}{4}$	$\frac{124}{29}$

The constant $\frac{29}{2}$ is a statement about $\mathrm{U}(N)$, so it holds on every closed subgroup, and the curvature of the group does the rest. This gives the theorem for every compact simple gauge group, the exceptional groups included, with an explicit gap. Let $\rho : G \to \mathrm{U}(N)$ be a faithful unitary representation of a compact connected Lie group G, weigh a configuration by $\exp(N\beta \sum_p \mathrm{Re}\,\mathrm{Tr}\,\rho(Q_p))$, and let κ_G be the least Ricci curvature of G for the bi-invariant metric $\langle X, Y\rangle = -\mathrm{Tr}(\rho(X)\rho(Y))$; we identify G with $\rho(G) \subset \mathrm{U}(N)$.

Theorem 5 (Every compact simple group)**.** *Suppose that the Lie algebra of G is semisimple, so that $\kappa_G > 0$, and that*

$$N|\beta| < \tfrac{2}{29}\,\kappa_G.$$

Then the conclusions of Theorem 1 hold, for $\beta < 0$ all of them except reflection positivity and the gap, with decay at every rate below

$$m_G(\beta) = \operatorname{arccosh}\Big(1 + \frac{1}{12}\Big(\frac{\kappa_G}{N|\beta|} - \frac{29}{2}\Big)\Big) > 0$$

and a transfer-matrix gap $m \geq m_G(\beta)$. For a simple G, $\kappa_G = h^\vee/(2\ell_\rho)$, where $h^\vee$ is the dual Coxeter number and ℓ_ρ the Dynkin index of ρ; Table I lists the values.

For $\mathrm{SU}(N)$, $\kappa_G = \frac{N}{2}$, and for $\mathrm{SO}(N)$, $\kappa_G = \frac{N-2}{4}$, so Theorem 5 contains the ranges of Theorem 4 and supplies their rate.[5] For small $|\beta|$ the existence of a gap is classical for every compact group, with no explicit range [52, Thm. 3.6]; what is new is an explicit range for every group, past the range $N|\beta| < \kappa_G/16$ that the Hessian bound 16 gives without gauge fixing, and an explicit gap. As $\beta \to 0$ the gap grows like $\log \frac{\kappa_G}{6N|\beta|}$.

No Bakry–Émery argument for the ungauged measure can pass $|\beta| = \frac{1}{32}$ [13, Thm. C]; the gauge fixing changes the measure, not the metric, which is why it can. For $\mathrm{SU}(2)$ the curvature range $\beta_W < \frac{4}{29}$ lies inside that of Theorem 1. For $\mathrm{SU}(3)$ the Dobrushin route goes much further.

Theorem 6 ($\mathrm{SU}(3)$)**.** *Let $G = \mathrm{SU}(3)$ and $0 \leq \beta_W \leq \frac{3}{4}$. Then the conclusions of Theorem 1 hold, with decay at every rate below $-\log(1.33319\,\beta_W)$ and a transfer-matrix gap*

$$m \geq -\log(1.33319\,\beta_W) > 0.$$

The proof of Theorem 6 bounds the Kantorovich modulus of the one-link measure $\nu_b(dU) \propto e^{\mathrm{Re}\,\mathrm{Tr}(Ub)}dU$ on $\mathrm{SU}(3)$ uniformly over staple sums b. Its analytic part is a lower bound for the spectral gap of ν_b through Bochner's formula with a cross term (Lemma 29), a corrector chosen box by box among polynomials of degree two in the matrix entries (Lemma 30), the reduction of b to diagonal matrices (Lemma 31), the character expansion of the moments of ν_b (Lemma 32) and an interpolation inequality on boxes of staple sums (Lemma 33). The remaining step evaluates the resulting bound on 2341 boxes with rigorous error bounds; the theorem rests on that computation, which is described in full in Sec. VII G and Appendix E.

The plaquette blocks of Theorem 3 work for $\mathrm{SU}(3)$ as well. They use the same modulus of the one-link measure, but only through the three links of a block, so they need $\Lambda(2\beta_W)$ small and not $2r\Lambda(r) < 1$: near the end of Theorem 6 they give a much larger gap, and they go past it.

Theorem 7 ($\mathrm{SU}(3)$ with plaquette blocks)**.** *Let $G = \mathrm{SU}(3)$ and $0 \leq \beta_W \leq \frac{19}{25} = 0.76$. Then $\alpha_3(\beta_W) < 1$ for the number of (15), with $\alpha_3(0.72) \leq 0.9182$, $\alpha_3(\frac{3}{4}) \leq 0.9653$ and $\alpha_3(0.76) \leq 0.99950$. The conclusions of Theorem 1 hold with decay at every rate below $-\frac{1}{2}\log\alpha_3(\beta_W)$ and a transfer-matrix gap*

$$m \geq -\tfrac{1}{2}\log\alpha_3(\beta_W) > 0;$$

at $\beta_W = \frac{3}{4}$ this gives $m \geq 0.0177$, against $1.1 \cdot 10^{-4}$ from Theorem 6, and at $\beta_W = 0.76$, where Theorem 6 no longer applies, $m \geq 2.5 \cdot 10^{-4}$.

All seven theorems are statements at fixed lattice spacing. They do not touch the continuum limit, the crossover to Bałaban's regime, or a gap in physical units, which together are the Clay problem; Sec. IX says where each method stops and what remains of the four open problems of [13]. The obstruction proved in [14], that a lattice gap uniform in lattice units forces trivial continuum two-point functions, applies to all of them. Section X goes past it in one respect: a Dobrushin–Shlosman condition on one cube of fixed physical size, uniform along the continuum tuning, gives a mass gap in physical units for every continuum limit (Theorems 43 and 45). That condition and the existence of the limit are the two statements left open.

[5] The dual Coxeter number is $h^\vee = N$ for $\mathrm{SU}(N)$, $N-2$ for $\mathrm{SO}(N)$, $n+1$ for $\mathrm{Sp}(n)$, $4, 9, 12, 18, 30$ for G_2, F_4, E_6, E_7, E_8, and the Dynkin index of a representation is normalised to one on the defining representation of $\mathrm{SU}(N)$ [41, Ch. 6]. The identity $\kappa_G = h^\vee/(2\ell_\rho)$ is the statement that the Ricci tensor of a compact simple group with a bi-invariant metric is one quarter of the negative Killing form, $\mathrm{Ric}(X,X) = -\frac{1}{4}B(X,X)$ [48], together with $B(X,X) = 2h^\vee\,\mathrm{Tr}(\rho(X)^2)/\ell_\rho$. Lower Ricci bounds of the same kind, for special holonomy and for Riemannian submersions, are studied in [17, 32].

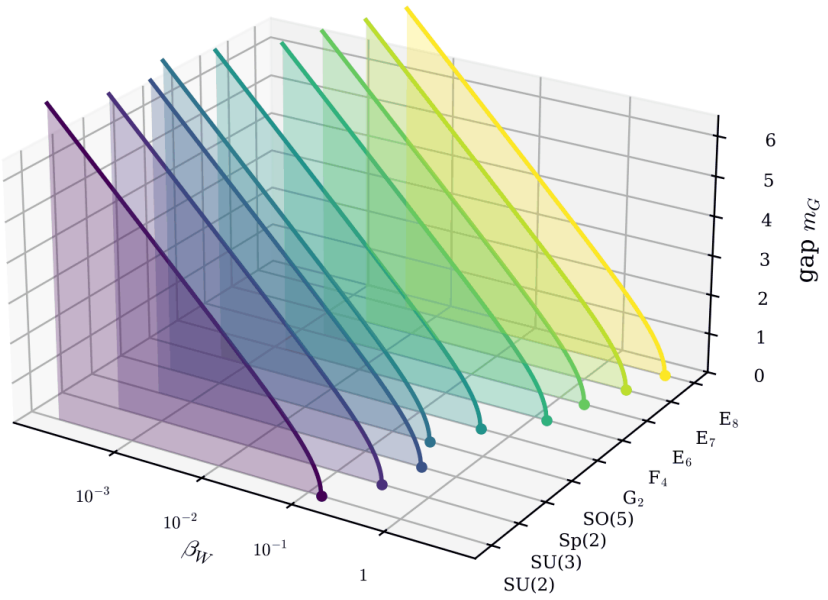


FIG. 1. The gap $m_G(\beta)$ of Theorem 5 as a function of $\beta_W = N^2\beta$ on a logarithmic scale, for nine groups in the representations of Table I. Each curve ends at the edge $\beta_W = 2N\kappa_G/29$ of the range, marked by a dot, where the gap tends to zero; as $\beta_W \to 0$ it grows like $\log(\kappa_G/(6N|\beta|))$. The curves differ only by the factor $N\kappa_G$, since m_G depends on $N|\beta|/\kappa_G$ alone.

TABLE II. Explicit ranges of uniqueness, decay and a transfer-matrix gap for four-dimensional lattice Yang–Mills theory. The first three rows hold for the ungauged measure; the last three are the results of this paper.

Method	SU(2), β_W	SU(3), β_W
Curvature, Shen–Zhu–Zhu [55]	$< \frac{1}{12}$	$< \frac{3}{16}$
Curvature, sharp [13]	$< \frac{1}{8}$	$< \frac{9}{32}$
Dobrushin, sharp quarter [13]	$< \frac{2}{9}$	
Curvature, forest gauge	$< \frac{4}{29}$	$< \frac{9}{29}$
Dobrushin, forest gauge	$< \frac{1}{3}$	$\leq \frac{3}{4}$
Plaquette blocks, forest gauge	≤ 0.345	≤ 0.76

Section II constructs the line field. Section III shows that a Dobrushin bound for the gauge-fixed specification gives the full conclusion of Theorem 1. Sections IV, V, VI, VII and VIII prove Theorems 1–7, and Sec. IX measures what is left; Sec. X reduces the continuum gap to one cube and the existence of the limit. The appendices contain the residue table of the line field, the comparison lemma and its version for blocks, the blocks of the local Hessian inequality, the details of the certified computation, the reconstruction of the transfer matrix with the equivalence of a gap and exponential clustering (Appendix F), and a complete proof of the sharp quarter for the single-link law of SU(2) (Appendices G–I), so that the paper does not depend on our earlier one for any step.

II. THE LINE FIELD

Write $e_0, \dots, e_3$ for the unit vectors of $\mathbb{Z}^4$ and (x,μ) for the link from x to $x+e_\mu$. A *direction field* is a map $\sigma : \mathbb{Z}^4 \to \{0,1,2,3\}$; its links are $F = \{(x,\sigma(x)) : x \in \mathbb{Z}^4\}$. At each site put $\pi_x(\mu) = \sigma(x+e_\mu)$. The plaquette at x in the plane $\{\mu,\nu\}$ has the links (x,μ), $(x+e_\mu,\nu)$, $(x+e_\nu,\mu)$, (x,ν), so it contains

$$[\sigma(x) \in \{\mu,\nu\}] + [\pi_x(\mu) = \nu] + [\pi_x(\nu) = \mu] \qquad (1)$$

links of F.

Lemma 8. *Every plaquette of $\mathbb{Z}^4$ contains exactly one link of F if and only if, at every site x, π_x fixes $\sigma(x)$ and permutes the other three directions cyclically.*

Proof. Fix x, let $s = \sigma(x)$ and let R be the set of the other three directions. For a plane $\{s,\nu\}$ the count (1) is $1 + [\pi_x(s) = \nu] + [\pi_x(\nu) = s]$; it equals one for all $\nu \in R$ exactly when $\pi_x(s) = s$ and $\pi_x(\nu) \neq s$ for $\nu \in R$, that is, when π_x fixes s and maps R into R. For a plane $\{\mu,\nu\} \subset R$ the count is $[\pi_x(\mu) = \nu] + [\pi_x(\nu) = \mu]$. Draw an arrow $\mu \to \pi_x(\mu)$ for every $\mu \in R$ with $\pi_x(\mu) \in R \setminus \{\mu\}$. Each vertex of R has at most one outgoing arrow, and the condition says that each of the three pairs of R carries exactly one arrow. Three arrows on three vertices with out-degree at most one give out-degree exactly one everywhere, and a tournament on three vertices with all out-degrees one is a directed three-cycle. □

From now on σ is the field of period two given in Table III of Appendix A, read from x modulo two. The table lists π_x at

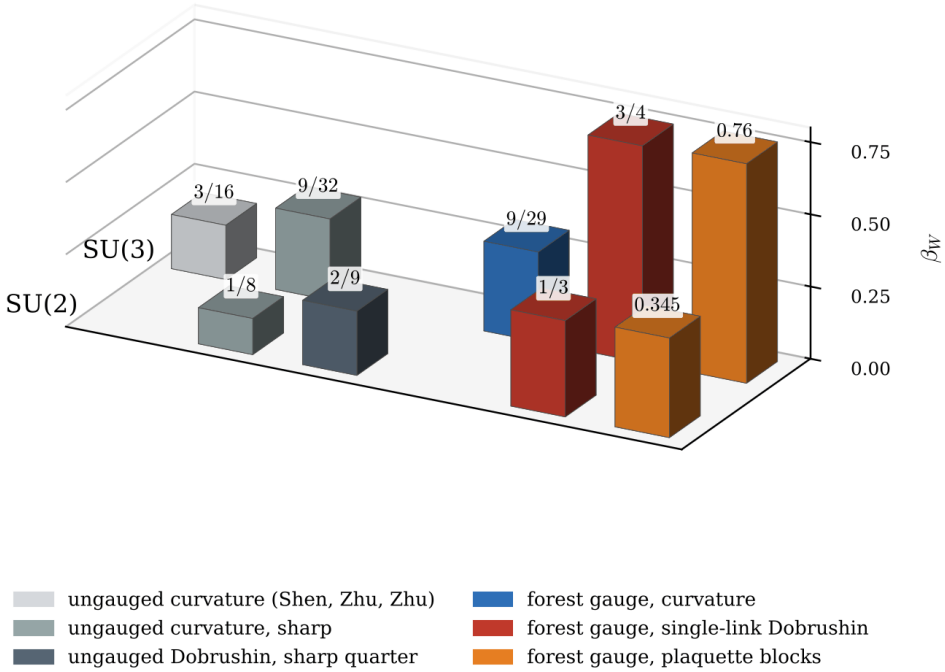


FIG. 2. Ranges of Table II. The height of each bar is the largest inverse coupling β_W covered by the method. Grey bars are methods for the ungauged measure; blue, red and orange bars use the forest gauge. For SU(2) the red bar ends at the ceiling of Theorem 2 and the orange bar, plaquette blocks, passes it (Theorem 3); for SU(3) the red bar is Theorem 6 and the orange bar, plaquette blocks, gives a larger gap there and passes it (Theorem 7).

each of the sixteen residue classes; in each row π_x fixes $\sigma(x)$ and is a three-cycle on the rest, so every plaquette contains exactly one link of F by Lemma 8.[6]

Lemma 9. (i) *F is a disjoint union of bi-infinite straight lines, one through every site: every site has exactly one outgoing and one incoming link of F, and both point in the direction $\sigma(x)$.*

(ii) *For every box Λ (a product of four finite intervals, with the links and plaquettes it contains), $F \cap E(\Lambda)$ is a disjoint union of segments, hence a forest.*

(iii) *A link $e = (x,\mu) \notin F$ lies in six plaquettes. Four of them contain a link of F that shares an end point with e, and these lie in the planes $\{\mu,\sigma(x)\}$ and $\{\mu,\pi_x(\mu)\}$. In the other two, which lie in the remaining plane $\{\mu,\rho\}$, the link of F is opposite to e.*

(iv) *Every link outside F shares a plaquette with exactly twelve other links outside F, two in each of its plaquettes.*

[6] There are exactly eight period-two direction fields with this property, all equivalent to σ under the symmetries of the lattice. A field with one fixed link at every site and one in every plaquette exists only in four dimensions: a link lies in $2(d-1)$ plaquettes and a site carries $d(d-1)/2$ of them, and $2(d-1) = d(d-1)/2$ forces $d = 4$. The statements of this section are also checked by `decide` in the Lean file `GaugeFix.lean` of the archive.

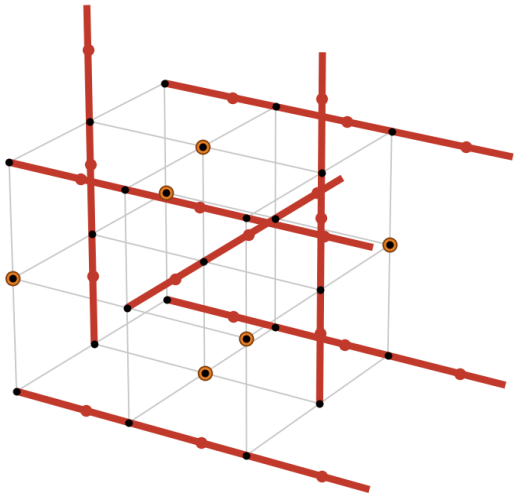
Proof. (i) Since π_x fixes $\sigma(x)$, $\sigma(x+e_{\sigma(x)}) = \sigma(x)$. Because σ has period two, $x - e_\mu \equiv x + e_\mu$, so also $\sigma(x - e_{\sigma(x)}) = \sigma(x)$, and the link $(x - e_{\sigma(x)}, \sigma(x))$ is in F. Conversely, if $(x - e_\nu, \nu) \in F$, then $\sigma(x - e_\nu) = \nu$, and straightness at $x - e_\nu$ gives $\sigma(x) = \nu$. So the incoming link is unique and continues the outgoing one.

(ii) A line meets a box in a segment, and distinct lines are disjoint.

(iii) The plaquettes of e are, for each $\nu \neq \mu$, the one at x and the one at $x - e_\nu$ in the plane $\{\mu,\nu\}$. A link of F sharing an end point with e is the outgoing or the incoming link of F at x or at $x + e_\mu$. At x both point in the direction $s = \sigma(x) \neq \mu$, and they lie in the plaquettes at x and at $x - e_s$ in the plane $\{\mu,s\}$. At $x + e_\mu$ both point in the direction $t = \pi_x(\mu)$, which lies in $R \setminus \{\mu\}$ because π_x is a three-cycle on R, and they lie in the plaquettes at x and at $x - e_t$ in the plane $\{\mu,t\}$. These four plaquettes are distinct because $s \neq t$. Each plaquette contains exactly one link of F, so in the two plaquettes of the third plane $\{\mu,\rho\}$ the link of F is a link of direction μ or ρ not touching e; the links of direction ρ in such a plaquette touch e, so it is the opposite link.

(iv) Each of the six plaquettes contains e, one link of F and two other links, and two distinct links share at most one plaquette. □

(a) the line field in the slice $x_3 = 0$



(b) one plaquette in the gauge

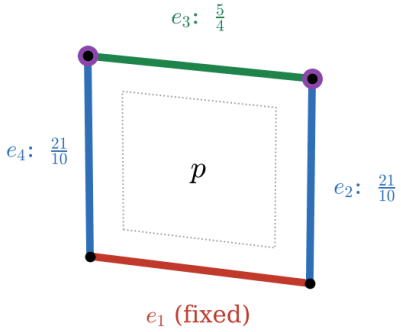


FIG. 3. (a) The line field in the slice $x_3 = 0$ of a box of side three. Thick red segments are the links of F in the directions $0, 1, 2$, with a dot marking the direction of travel; an orange ring marks a site whose line leaves the slice in direction 3. Every line is straight (Lemma 9(i)), and the segments that leave the box are its boundary stubs. (b) One plaquette with its fixed link e_1. The weights are those of the local inequality of Lemma 20 for the three free links: $\frac{21}{10}$ on the two links adjacent to e_1 and $\frac{5}{4}$ on the opposite one. The two purple corners are the ones that enter the corner form (7) once $Y_1 = 0$.

III. GAUGE FIXING AND THE FREE-BOUNDARY STATE

Let $G \subset \mathrm{U}(N)$ be compact with Haar measure dU, and let Λ be a box. The Wilson measure with free boundary is

$$W_\Lambda(dQ) = \frac{1}{Z_\Lambda} \exp\Big(N\beta \sum_{p \in P(\Lambda)} \mathrm{Re}\,\mathrm{Tr}\, Q_p\Big) \prod_{e \in E(\Lambda)} dQ_e .$$

The measure is invariant under the gauge transformations $Q_{(x,y)} \mapsto g_x Q_{(x,y)} g_y^{-1}$ with $g \in G^\Lambda$. Call the links outside F *active* and write A for their set. For a function O of the links, O^F is O with every link of F set to the identity.

Lemma 10 (Gauge fixing)**.** *Let μ_Λ^F be the probability measure on $G^{A \cap E(\Lambda)}$ whose density with respect to Haar measure is proportional to $\exp(N\beta \sum_{p \in P(\Lambda)} \mathrm{Re}\,\mathrm{Tr}\, Q_p^F)$. For every gauge-invariant O on $G^{E(\Lambda)}$,*

$$W_\Lambda(O) = \mu_\Lambda^F(O^F).$$

Proof. Each segment of $F \cap E(\Lambda)$ is a path $x_0, x_1, \dots, x_k$ along one direction. Given the variables on F, put $g_{x_0} = 1$ and $g_{x_{i+1}} = g_{x_i} Q_{(x_i, x_{i+1})}$ on every segment. The transformed field has $Q_e^g = 1$ on $F \cap E(\Lambda)$, and for fixed values on F the map $Q_A \mapsto Q_A^g$ is a product of left and right translations, which preserves Haar measure. Gauge invariance of the action and of O and the change of variables give the claim. □

No leftover holonomy appears because $F \cap E(\Lambda)$ is a forest. The measures μ_Λ^F are finite-volume versions of the specification γ^F on G^A: for $e \in A$, γ_e^F resamples Q_e from the law with density proportional to $\exp(N\beta \sum_{p \ni e} \mathrm{Re}\,\mathrm{Tr}\, Q_p^F)$ given all other active links. No infinite-volume gauge transformation is performed; γ^F is a specification in its own right. Fix a metric δ on G of diameter D that is bi-invariant (for $\mathrm{SU}(2)$ the chordal metric on S^3, $D = 2$; for $\mathrm{SU}(N)$ the Hilbert–Schmidt distance). The Kantorovich coefficient of e' on e is

$$C_{ee'} = \sup \frac{W_1\big(\gamma_e^F(\cdot \mid Q), \gamma_e^F(\cdot \mid Q')\big)}{\delta(Q_{e'}, Q'_{e'})},$$

the supremum over pairs of configurations that differ only at e'. It vanishes unless e and e' share a plaquette.

Proposition 11. *Suppose that every row and every column of $(C_{ee'})$ sums to at most $\alpha < 1$. Then:*

(i) *γ^F has exactly one Gibbs measure μ_F;*

(ii) *for every gauge-invariant local O the limit $\omega(O) = \lim W_\Lambda(O)$ exists along every sequence of boxes exhausting $\mathbb{Z}^4$ and equals $\mu_F(O^F)$;*

(iii) *ω is invariant under translations and the hypercubic group;*

(iv) *if $\beta \geq 0$, ω is reflection positive for reflections in hyperplanes through sites and in hyperplanes bisecting links;*

(v) *for gauge-invariant local O, O' with O Lipschitz in each link, $|\mathrm{Cov}_\omega(O, O')| \leq C_{O,O'} e^{-\lambda\, \mathrm{dist}(O,O')}$ for every $\lambda < -\log\alpha$;*

(vi) *if $\beta \geq 0$, the transfer matrix of ω has spectral gap at least $-\log\alpha$.*

Proof. (i) Gibbs measures exist by compactness, and the comparison lemma of Appendix B with $S=\emptyset$ shows that any two coincide.

(ii) Extend μ_Λ^F by Haar measure on the active links outside Λ, and let S be the set of active links with a plaquette outside $P(\Lambda)$; this includes every active link outside $E(\Lambda)$. For an active link $e\notin S$ all six plaquettes of e lie in $P(\Lambda)$, so the conditional law of Q_e under the extended measure is γ_e^F. Let T be the support of O^F and $\lambda<-\log\alpha$. The comparison lemma with weights $e^{-\lambda\,\mathrm{dist}(\cdot,T)}$ gives

$$|\mu_\Lambda^F(O^F)-\mu_F(O^F)|\le L\,\frac{\theta D}{1-\theta}\sum_{z\in S}e^{-\lambda\,\mathrm{dist}(z,T)},$$

with $\theta=\alpha e^{\lambda}<1$ and L a Lipschitz constant of O^F; the sum is at most $C_T R^3 e^{-\lambda R}$ when R is the distance from T to the complement of Λ. For O continuous but not Lipschitz, approximate uniformly. Lemma 10 gives $W_\Lambda(O)=\mu_\Lambda^F(O^F)$.

(iii) $W_\Lambda(\tau_v O)=W_{\Lambda-v}(O)$, and the boxes $\Lambda-v$ also exhaust $\mathbb{Z}^4$; by (ii) the limit does not depend on the sequence. The line field is a device for computing W_Λ and is never shifted. The same argument applies to the hypercubic group.

(iv) For $\beta\ge 0$ the measure W_Λ is reflection positive for every box symmetric under the reflection [52], and reflection positivity passes to the limit along such boxes, which is ω by (ii).

(v) $\mathrm{Cov}_\omega(O,O')=\mathrm{Cov}_{\mu_F}(O^F,O'^F)$, and setting the links of F to the identity keeps the Lipschitz property. The bound is Corollary 49.

(vi) Let ϑ be a time reflection and τ_t a time translation. For gauge-invariant local O that are Lipschitz in each link, (v) gives $|\omega(\vartheta(\bar O)\tau_t O)-|\omega(O)|^2|\le C_O e^{-\lambda t}$. On the support of O these functions are dense in the gauge-invariant part of $L^2(\omega)$: gauge-invariant functions are functions on the compact quotient of G^E by the gauge group, Lipschitz functions for the quotient metric are dense in L^2 of every finite measure on it, and they lift to gauge-invariant Lipschitz functions. By Proposition 52 the spectrum of the transfer matrix on the complement of the vacuum lies in $[0,e^{-\lambda}]$, for every $\lambda<-\log\alpha$. □

The proposition replaces the torus limits and the uniqueness of the ungauged Gibbs measure used in [13, Thm. F] by the free-boundary limit, which has all the properties the reconstruction of the transfer matrix needs.[7]

IV. SU(2) UP TO ONE THIRD

Proof of Theorem 1. Identify $\mathrm{SU}(2)$ with the unit quaternions and use the chordal metric. For an active link e and a plaquette $p\ni e$, the staple s_p is the product of the other three link variables in the order that makes $\mathrm{Re}(Q_e s_p)=\frac12\mathrm{Tr}\,Q_p$, with links of F equal to one. Since $N\beta\,\mathrm{Re}\,\mathrm{Tr}\,Q_p=4\beta\langle Q_e,\bar s_p\rangle$ in $\mathbb{R}^4$, the conditional law of Q_e is the von Mises–Fisher law $\nu_b(dx)\propto e^{\langle x,b\rangle}dx$ on S^3 with $b=4\beta\sum_{p\ni e}\bar s_p$. Changing a partner e' by δ in the chordal metric changes one staple by an isometric image of that displacement, so b moves by $4|\beta|\delta$, and the sharp quarter (Theorem 56), $W_1(\nu_b,\nu_{b'})\le\frac14|b-b'|$, gives $C_{ee'}\le|\beta|$. By Lemma 9(iv) every row and column has twelve nonzero entries, so $\alpha\le 12|\beta|=3|\beta_W|$. Proposition 11 applies for $|\beta_W|<\frac13$. □

Proof of Theorem 2. For the line field, every staple of an active link contains two active links, so the six staples can be chosen with sum zero while the staple through a given partner e' moves freely. At $b=0$ the derivative of ν_b in the direction h has Kantorovich norm $\frac14|h|$ by Lemma 55(i), so $C_{ee'}=|\beta|$ exactly, and every row sum is $12|\beta|=3|\beta_W|$.

Now let F' be any forest and C' the matrix of the statement. Suppose a positive weight w satisfies $C'w\le\alpha w$. For a box Λ_n of side n let C_n be the principal submatrix on the active links of $E(\Lambda_n)$; then $C_n w\le\alpha w$ on these links, and the Collatz–Wielandt inequality for nonnegative matrices gives $\rho(C_n)\le\alpha$. Since C_n is symmetric, $\rho(C_n)$ is at least the mean row sum. The box has $4n^4-O(n^3)$ links and n^4 sites, and a forest has fewer links than sites, so the number a of active links is at least $3n^4-O(n^3)$. Let a_p be the number of active links of the plaquette p; the box has $P=6n^4-O(n^3)$ plaquettes and $\sum_p a_p=6a-O(n^3)$. Two links share at most one plaquette, so the row sums add up to $|\beta|\sum_p a_p(a_p-1)$, and by convexity

$$\frac{|\beta|}{a}\sum_p a_p(a_p-1)\ge\frac{|\beta|P}{a}\,\bar a(\bar a-1)=6|\beta|(\bar a-1)-o(1),$$

with $\bar a=\sum_p a_p/P=6a/P-o(1)\ge 3-o(1)$. Hence $\alpha\ge 12|\beta|-o(1)$ for every n, and $\alpha\ge 12|\beta|=3|\beta_W|$. □

Without gauge fixing the same count gives eighteen partners and the ceiling $\beta_W=\frac29$ of [13, Prop. 13]; the temporal gauge, a family of lines in one direction, leaves fourteen. The line field is optimal among forests: twelve is the mean number of partners forced by the number of free links, and it is attained at every link.

V. SU(2) PAST ONE THIRD: PLAQUETTE BLOCKS

Throughout this section $G=\mathrm{SU}(2)$ is the group of unit quaternions with the chordal metric $|x-y|$, $\bar x$ is the conjugate, $\langle x,y\rangle=\mathrm{Re}(x\bar y)$ is the inner product of $\mathbb{R}^4$, and dx is the normalised Haar measure on S^3. Left and right multiplication by unit quaternions and conjugation are isometries, and $\mathrm{Re}(xyz)=\mathrm{Re}(yzx)=\langle y,\bar z\bar x\rangle$. Put

$$Z_0(c)=\int_{S^3}e^{\langle c,x\rangle}dx=\sum_{j\ge0}\frac{(|c|^2/4)^j}{j!\,(j+1)!},\qquad c\in\mathbb{R}^4,$$

let $\nu_c(dx)\propto e^{\langle c,x\rangle}dx$ be the von Mises–Fisher law, and write $g=\log Z_0$, so that $\nabla g(c)$ is the mean and $\mathrm{Hess}\,g(c)$ the covariance of ν_c.

[7] On a torus the line field closes up into cycles, and the holonomies along them cannot be removed by a gauge transformation; the argument never uses a torus. The state ω is defined only on gauge-invariant observables, which is all that the Osterwalder–Schrader reconstruction [50–52] uses.

A. The blocks

A *block plaquette* is a plaquette of the set $\mathscr{B}$ described in Table IV of Appendix C, and its *block* is the set of its three active links.

Lemma 12. *The set $\mathscr{B}$ is invariant under translations by $2\mathbb{Z}^4$ and:*

(i) *the blocks partition the active links;*

(ii) *every plaquette $p \notin \mathscr{B}$ meets three distinct blocks, one in each of its active links;*

(iii) *every active link lies in exactly five plaquettes outside $\mathscr{B}$, so it shares a plaquette outside $\mathscr{B}$ with exactly ten active links, none of them in its own block;*

(iv) *if an active link j and a link of a block B lie in a common plaquette outside $\mathscr{B}$, then every link of B has its midpoint within distance 2 of the midpoint of j.*

Proof. All four statements are invariant under $2\mathbb{Z}^4$ and involve plaquettes at distance at most two from one site, so they hold on $\mathbb{Z}^4$ as soon as they hold on the torus $(\mathbb{Z}/P\mathbb{Z})^4$ with $P \geq 6$, where no such configuration wraps around. They are checked on the tori of side four and six by the program described in Appendix C; the side four already suffices for (i)–(iii). For (iii), an active link lies in six plaquettes, one of which is its block plaquette, and each of the other five contains two further active links, which lie in other blocks by (ii). □

Fix a block plaquette p with active links e_1, e_2, e_3. Reading the plaquette product cyclically and deleting its fixed link, $\frac{1}{2}\operatorname{Tr} Q_p = \operatorname{Re}(y_1 y_2 y_3)$, where y_i is Q_{e_i} or $Q_{e_i}^{-1}$. Every other plaquette containing e_i meets the block only in e_i by Lemma 12(ii) and contributes $4\beta\langle y_i, \bar{s}\rangle$ to the exponent, where s is the staple of y_i in it. Hence the conditional law of the block given all other active links is

$$\mu_b(dy) \propto \exp\Big(\sum_{i=1}^{3} \langle y_i, b_i\rangle + 4\beta \operatorname{Re}(y_1 y_2 y_3)\Big)\, dy_1\, dy_2\, dy_3, \quad (2)$$

with b_i the sum of $4\beta\bar{s}$ over the five staples, so that $|b_i| \leq 20|\beta|$. Inversion is an isometry of S^3 and preserves Haar measure, so we may work with y instead of Q. On $(S^3)^3$ we use the metric $d(y,y') = \sum_i |y_i - y_i'|$.

B. The sensitivity of one block

Let λ_* and τ be a lower and an upper bound for the eigenvalues of $\operatorname{Hess} g(c)$ on the ball $|c| \leq 24|\beta|$, and M an upper bound there for the norm of the third derivative $D^3 g(c)$ as a trilinear form. Put

$$\begin{aligned} w &= 8\beta^2(\tau - \lambda_*), \qquad s = \tfrac{1}{4} + (e^w - 1)e^w\sqrt{\tau}, \\ C &= 20|\beta|\big(4|\beta|\tau s + 16\beta^2 M\big), \qquad K = \frac{(1+|\beta|)\, s}{1-C}, \end{aligned} \quad (3)$$

assuming $C < 1$.

Lemma 13. *Let ν be a probability measure and $\lambda = e^R\nu/\nu(e^R)$ with $\operatorname{osc} R \leq \omega$. For bounded f, h,*

$$\operatorname{Cov}_\lambda(f,h) \leq \operatorname{Cov}_\nu(f,h) + (e^\omega - 1)e^\omega \operatorname{sd}_\nu(f)\operatorname{sd}_\nu(h).$$

Proof. We may assume $0 \leq R \leq \omega$. The density $\rho = e^R/\nu(e^R)$ lies in $[e^{-\omega}, e^\omega]$, so $\delta = \|\rho - 1\|_\infty \leq e^\omega - 1$. With $\tilde{f} = f - \nu f$ and $\tilde{h} = h - \nu h$,

$$\begin{aligned} \operatorname{Cov}_\lambda(f,h) - \operatorname{Cov}_\nu(f,h) = \nu\big((\rho-1)\tilde{f}\tilde{h}\big) \\ - \nu\big((\rho-1)\tilde{f}\big)\,\nu\big((\rho-1)\tilde{h}\big), \end{aligned}$$

and the right side is at most $(\delta + \delta^2)\operatorname{sd}_\nu(f)\operatorname{sd}_\nu(h)$ by the Cauchy–Schwarz inequality. □

Lemma 14. *Let $\beta \geq 0$, $a, c_0 \in \mathbb{R}^4$ with $|a|, |c_0| \leq 20\beta$, and*

$$\kappa(dx) \propto \exp\big(\langle x, a\rangle + g(c_0 + 4\beta\bar{x})\big)\, dx \quad \text{on } S^3.$$

For every 1-Lipschitz f and every $h \in \mathbb{R}^4$:

(i) $\operatorname{Cov}_\kappa(f, \langle x, h\rangle) \leq s|h|$;

(ii) *if c_0 moves in the direction k, then $|\frac{d}{dt}\kappa(f)| \leq (4\beta\tau s + 16\beta^2 M)|k|$.*

Proof. (i) Taylor's formula at c_0 gives $g(c_0 + 4\beta u) = g(c_0) + 4\beta\langle \nabla g(c_0), u\rangle + R(u)$ with $R(u) = 8\beta^2 \operatorname{Hess} g(\xi)(u,u)$ for some ξ on the segment, $|\xi| \leq 24\beta$. For $u = \bar{x}$ of norm one, $R(\bar{x}) \in [8\beta^2\lambda_*, 8\beta^2\tau]$, so $\operatorname{osc} R \leq w$. Since $\langle \nabla g(c_0), \bar{x}\rangle = \langle x, \overline{\nabla g(c_0)}\rangle$, κ is the tilt $e^R\nu/\nu(e^R)$ of $\nu = \nu_c$ with $c = a + 4\beta\overline{\nabla g(c_0)}$. The sharp quarter, Theorem 56, gives $\operatorname{Cov}_\nu(f, \langle x, h\rangle) = \frac{d}{dt}\nu_{c+th}(f) \leq \frac{1}{4}|h|$. A 1-Lipschitz function on a set of diameter two has $\operatorname{sd} \leq 1$, and $\operatorname{sd}_\nu\langle x, h\rangle \leq \sqrt{\tau}|h|$ because $|c| \leq |a| + 4\beta \leq 24\beta$. Lemma 13 gives (i).

(ii) The derivative is $\operatorname{Cov}_\kappa(f, \langle \nabla g(c_0 + 4\beta\bar{x}), k\rangle)$. Write $\nabla g(c_0 + 4\beta u) = \nabla g(c_0) + 4\beta \operatorname{Hess} g(c_0)u + N(u)$. The constant term does not contribute. The linear term is $\langle x, h\rangle$ with $h = 4\beta\overline{\operatorname{Hess} g(c_0)k}$, $|h| \leq 4\beta\tau|k|$, and contributes at most $4\beta\tau s|k|$ by (i). For $|u|, |u'| \leq 1$, $\|\operatorname{Hess} g(c_0 + 4\beta v) - \operatorname{Hess} g(c_0)\| \leq 4\beta M|v| \leq 4\beta M$ on the segment between them, so $|N(u) - N(u')| \leq 16\beta^2 M|u - u'|$; the function $\langle N(\bar{x}), k\rangle$ is therefore $16\beta^2 M|k|$-Lipschitz, its standard deviation is at most that constant, and its covariance with f is at most $16\beta^2 M|k|$. □

Lemma 15. *Let $C < 1$. If b and b' satisfy $|b_i|, |b_i'| \leq 20|\beta|$ and differ only in one coordinate b_i, then*

$$W_1(\mu_b, \mu_{b'}) \leq K\,|b_i - b_i'|$$

in the metric d.

Proof. The map $y_1 \mapsto -y_1$ turns μ_b for β into the law for $-\beta$ with b_1 replaced by $-b_1$ and is an isometry, so we may take $\beta \geq 0$; the cyclic symmetry of $\operatorname{Re}(y_1 y_2 y_3)$ lets us take $i = 1$. Let f be 1-Lipschitz for d and $\varepsilon = |b_1 - b_1'|$.

The middle link. Given y_1, y_3, the law of y_2 is ν_c with $c = b_2 + 4\beta\overline{y_3 y_1}$, which does not involve b_1. Changing y_1 by η moves c by $4\beta\eta$, and the sharp quarter moves ν_c by at most

$\beta\eta$ in W_1. Hence $F(y_1,y_3)=\mathbb{E}[f\mid y_1,y_3]$ is $(1+\beta)$-Lipschitz in each variable, and $\mu_b(f)=\mu_b^{13}(F)$, where μ_b^{13} is the law of (y_1,y_3):

$$\mu_b^{13}\propto e^{\langle y_1,b_1\rangle+\langle y_3,b_3\rangle}Z_0\big(b_2+4\beta\,\overline{y_3y_1}\big).$$

Since $|b_2+4\beta\overline{y_3y_1}|=|b_2y_3+4\beta\bar{y}_1|=|y_1b_2+4\beta\bar{y}_3|$, its conditional laws are

$$\kappa_1(dy_1\mid y_3)\propto e^{\langle y_1,b_1\rangle+g(b_2y_3+4\beta\bar{y}_1)}dy_1,$$
$$\kappa_3(dy_3\mid y_1)\propto e^{\langle y_3,b_3\rangle+g(y_1b_2+4\beta\bar{y}_3)}dy_3,$$

both of the form of Lemma 14, with $|c_0|=|b_2|$.

Two sites. Integrating Lemma 14(i) along the segment from b_1 to b_1' gives $W_1(\kappa_1,\kappa_1')\le s\varepsilon$ for the same y_3. Moving y_3 to y_3' moves $c_0=b_2y_3$ along a segment of length $|b_2||y_3-y_3'|\le 20\beta|y_3-y_3'|$ inside the ball $|c_0|\le 20\beta$, so Lemma 14(ii) gives $W_1(\kappa_1(\cdot\mid y_3),\kappa_1(\cdot\mid y_3'))\le C|y_3-y_3'|$, and the same for κ_3, which does not depend on b_1. Let π be a coupling of μ_b^{13} and $\mu_{b'}^{13}$ minimising d_1+d_3, $d_k=\mathbb{E}_\pi|y_k-y_k'|$. Replacing (y_1,y_1') by an optimal coupling of $\kappa_1(\cdot\mid y_3)$ and $\kappa_1'(\cdot\mid y_3')$ keeps a coupling, so minimality gives $d_1\le s\varepsilon+Cd_3$; in the same way $d_3\le Cd_1$. Hence $d_1\le s\varepsilon/(1-C^2)$ and

$$|\mu_b(f)-\mu_{b'}(f)|\le(1+\beta)(d_1+d_3)\le\frac{(1+\beta)s}{1-C}\,\varepsilon=K\varepsilon.\ \square$$

C. The constants

Lemma 16. *Write $g(c)=\varphi(|c|^2)$ and $r=|c|$. For $c\neq 0$ the eigenvalues of* Hess $g(c)$ *are $2\varphi'(r^2)$, three times, and $2\varphi'(r^2)+4r^2\varphi''(r^2)$, and the norm of $D^3g(c)$ is*

$$\max\Big(|A+\Gamma|,\tfrac{2}{3}|A|\Big),\qquad A=12r\varphi''(r^2),\quad \Gamma=8r^3\varphi'''(r^2).$$

On the ball $r\le 2.07$ every eigenvalue of Hess g *lies between* 0.158491 *and* 0.25000001*, and the norm of D^3g is at most* 0.0615586.

Proof. Differentiating $\varphi(|c|^2)$ gives $D^2g(v,v)=2\varphi'|v|^2+4\varphi''\langle c,v\rangle^2$ and $D^3g(v,v,v)=12\varphi''\langle c,v\rangle|v|^2+8\varphi'''\langle c,v\rangle^3$. For $|v|=1$ and $t=\langle c/r,v\rangle\in[-1,1]$ the cubic form is $At+\Gamma t^3$, whose largest absolute value is attained at $t=\pm1$ or at an interior critical point, where it equals $\frac{2}{3}|A||t|\le\frac{2}{3}|A|$; by Banach's theorem [7] the supremum of a symmetric trilinear form on the diagonal of the unit sphere is its norm. The numerical bounds are obtained in ball arithmetic [40] from the series of Z_0, truncated after forty terms with a proved bound on the tail, on $2\cdot10^4$ subintervals of $[0,2.07]$; the program is described in Appendix C. $\square$

D. The proof of Theorem 3

Let $\gamma^{\mathscr{B}}$ be the block specification: for a block B, $\gamma_B^{\mathscr{B}}$ resamples the three links of B from (2). It comes from the same interaction as γ^F, so the two specifications have the same Gibbs measures. Define the block coefficients

$$C_{Bj}=\sup\frac{W_1\big(\gamma_B^{\mathscr{B}}(\cdot\mid Q),\gamma_B^{\mathscr{B}}(\cdot\mid Q')\big)}{|Q_j-Q_j'|}$$

for an active link $j\notin B$, the supremum over pairs of configurations that differ only at j, with W_1 for the metric d.

Proposition 17. *Suppose that $\sum_B C_{Bj}\le\alpha<1$ for every active link j. Then the conclusions (i)–(vi) of Proposition 11 hold with $-\frac{1}{2}\log\alpha$ in place of $-\log\alpha$.*

Proof. The proof of Proposition 11 goes through with Lemma 50 of Appendix B in place of Lemma 48, and with the set S of part (ii), and the support of H in Corollary 49, enlarged to the union of the blocks that meet it. The enlargement multiplies the size of these sets by at most three and moves them by at most one, which changes only the constants. $\square$

Proof of Theorem 3. Let $|\beta|\le 0.345/4$ and take the constants of Lemma 16, whose ball $r\le 2.07$ contains $|c|\le 24|\beta|$. With these fixed constants every quantity in (3) is increasing in $|\beta|$, and so is

$$\alpha(\beta_W)=40|\beta|K.\tag{4}$$

Changing an active link j by η changes one staple of each of its ten partners outside its block by an isometric image of the displacement, so the field b_i of each such partner i moves by at most $4|\beta|\eta$, and Lemma 15, applied once for each of these ten fields, gives $\sum_B C_{Bj}\le 40|\beta|K$. Evaluating (3) in ball arithmetic at $\beta_W=0.345$ gives $w\le 0.0054459$, $s\le 0.252746$, $C\le 0.050243$, $K\le 0.289069$ and $\alpha\le 0.99729$; at $\beta_W=\frac{1}{3}$ it gives $\alpha\le 0.95602$. Proposition 17 applies. $\square$

Remark 18. Most of the range is kept. A second bound for the same sensitivity, which keeps the law of the middle link instead of linearising it, evaluated in floating point on a quadrature grid and maximised over the fields b by a random search followed by the Nelder–Mead method, gives 0.2783 at $\beta_W=0.34$, against 0.2881 from Lemma 15, and $40|\beta|$ times it crosses one near $\beta_W=0.357$. That computation is not certified and is not used in the proof. Past this point the gain from integrating out one link is spent, and larger blocks are needed, such as the free links of a cube of the line field, whose sensitivity no longer reduces to two sites.

VI. CURVATURE IN THE GAUGE

In this section $G\subset\mathrm{U}(N)$ is a closed connected subgroup with Lie algebra $\mathfrak{g}$ and the inner product $\langle X,Y\rangle=-\mathrm{Tr}(XY)$; the bound of Theorem 4 is proved first for $\mathrm{U}(N)$. A tangent vector at a field Q is $v=(X_eQ_e)$, $|v|^2=\sum_e|X_e|^2$. For the plaquette at x in the plane $\mu<\nu$ number its links $e_1=(x,\mu)$, $e_2=(x+e_\mu,\nu)$, $e_3=(x+e_\nu,\mu)$, $e_4=(x,\nu)$, transport their vectors to x,

$$Y_1=X_{e_1},\quad Y_2=Q_{e_1}X_{e_2}Q_{e_1}^*,\quad Y_3=Q_{e_4}X_{e_3}Q_{e_4}^*,\quad Y_4=X_{e_4},\tag{5}$$

and write $W = Q_p$. The second derivative of $\operatorname{Re}\operatorname{Tr} Q_p$ along the geodesic $Q_e(t) = e^{tX_e}Q_e$ is

$$f_p = \sum_{i=1}^{4} \operatorname{Re}\operatorname{Tr}(Y_i^2 W) + 2\operatorname{Re}\operatorname{Tr}(Y_1 Y_2 W) + 2\operatorname{Re}\operatorname{Tr}(Y_3 Y_4 W) - 2 \sum_{i\in\{1,2\},\, j\in\{3,4\}} \operatorname{Re}\operatorname{Tr}(Y_i W Y_j), \tag{6}$$

so that $\operatorname{Hess} S_0(v,v) = \sum_p f_p$. To see this, write $Q_p = Q_{e_1} Q_{e_2} Q_{e_3}^* Q_{e_4}^*$. Since $Q_{e_1} e^{tX_{e_2}} = e^{tY_2} Q_{e_1}$ and $e^{-tX_{e_3}} Q_{e_4}^* = Q_{e_4}^* e^{-tY_3}$, the plaquette variable along the geodesic is $Q_p(t) = e^{tY_1} e^{tY_2} W e^{-tY_3} e^{-tY_4}$. Differentiating twice at $t = 0$ gives the four squares, the two products of neighbouring factors and the four products across W, and the squares and the product WY_3Y_4 are moved cyclically under the trace. The corner form of p is

$$c_p = 2\langle Y_1, Y_4\rangle - 2\langle Y_1, Y_2\rangle + 2\langle Y_2, WY_3W^*\rangle - 2\langle Y_3, Y_4\rangle. \tag{7}$$

It collects the inner products of the transported vectors of two links that meet at a right angle at a corner of p.

Lemma 19. *On $\mathbb{Z}^4$, or on a box with free boundary, for every field Q and every square-summable X,*

$$\sum_y \Big[|D_y X|^2 + \sum_\mu \big| X_{y,\mu} + Q^*_{y-e_\mu,\mu} X_{y-e_\mu,\mu} Q_{y-e_\mu,\mu} \big|^2 \Big] = 4|X|^2 + \sum_p c_p,$$

where $D_y X = \sum_\mu [X_{y,\mu} - Q^*_{y-e_\mu,\mu} X_{y-e_\mu,\mu} Q_{y-e_\mu,\mu}]$ is the covariant divergence and terms with a link outside the box are omitted.

Proof. At a site y write o_μ for the vector of the outgoing link in direction μ and i_μ for the transported vector of the incoming one, so that $D_y X = \sum_\mu (o_\mu - i_\mu)$. Every link is outgoing at one end and incoming at the other, so the squares $|o_\mu|^2 + |i_\mu|^2$, which appear once in $|D_y X|^2$ and once in $|o_\mu + i_\mu|^2$, add up to $4|X|^2$. The collinear cross term $-2\langle o_\mu, i_\mu\rangle$ of $|D_y X|^2$ cancels the term $+2\langle o_\mu, i_\mu\rangle$ of $|o_\mu + i_\mu|^2$. The remaining cross terms of $|D_y X|^2$ pair two perpendicular ends at y, and each such pair is a corner of exactly one plaquette, with the sign and the transport of (7) (this is the proof of [13, Lemma 1]). In a box two perpendicular ends at y span a plaquette of the box, because a box is a product of intervals. □

Lemma 20. *Let $k \in \{1,2,3,4\}$, let $k \pm 1$ (mod 4) be the positions adjacent to k and $k+2$ the opposite one. For every N, every $W \in \mathrm{U}(N)$ and all $Y_1, \dots, Y_4 \in \mathfrak{u}(N)$ with $Y_k = 0$,*

$$\pm f_p + \tfrac{9}{10} c_p \le \tfrac{21}{10}\big(|Y_{k-1}|^2 + |Y_{k+1}|^2\big) + \tfrac{5}{4}|Y_{k+2}|^2.$$

Proof. Replacing W by $-W$ changes the sign of f_p and fixes c_p, so the sign $+$ suffices. Conjugation by a unitary matrix preserves both sides, so we may take $W = \operatorname{diag}(w_a)$. The form $\mathscr{F} = \mathrm{RHS} - f_p - \frac{9}{10} c_p$ then splits into a block for each diagonal position a, in the real variables $y_i = -i(Y_i)_{aa}$, and a block for each pair $a \neq b$, in the complex variables $z_i = (Y_i)_{ab}$, since $(Y_i)_{ba} = -\bar{z}_i$. Appendix D shows that every block is positive definite. □

Proof of the bound in Theorem 4. Every plaquette has exactly one link of F, and X vanishes on it. By Lemma 9(iii) each active link is adjacent to the fixed link in four of its plaquettes and opposite to it in two. Summing Lemma 20 over the plaquettes,

$$\sum_p \big(\pm f_p + \tfrac{9}{10} c_p\big) \le \big(4 \cdot \tfrac{21}{10} + 2 \cdot \tfrac{5}{4}\big)|X|^2 = \tfrac{109}{10}|X|^2.$$

By Lemma 19, $\frac{9}{10}\sum_p c_p = \frac{9}{10}\Sigma - \frac{36}{10}|X|^2$, where $\Sigma \ge 0$ is the left side of the lemma. Hence

$$\pm \operatorname{Hess} S_0(v,v) + \tfrac{9}{10}\Sigma \le \big(\tfrac{109}{10} + \tfrac{36}{10}\big)|X|^2 = \tfrac{29}{2}|X|^2.$$

In a box, links near the boundary lie in fewer plaquettes and receive less. □

The factor $\frac{9}{10}$ is chosen so that the corners are paid by the divergence identity at the price $\frac{36}{10}$ while the local blocks stay positive; Lemma 19 is the divergence identity of [13, Lemma 1] with the collinear term written as a square, so nothing is lost by Cauchy–Schwarz.

The two lemmas concern vectors in $\mathfrak{u}(N)$ and fields in $\mathrm{U}(N)$, so the bound holds without change on every closed subgroup $G \subset \mathrm{U}(N)$, the vectors X_e ranging over its Lie algebra $\mathfrak{g}$. It is also the Hessian of S_0 on G^E: the metric is bi-invariant, the geodesics of G through Q_e are the curves $e^{tX}Q_e$ with $X \in \mathfrak{g}$, which are geodesics of $\mathrm{U}(N)$, and the Hessian is read off from second derivatives along geodesics.

Lemma 21. *Let $G \subset \mathrm{U}(N)$ be closed and connected with simple Lie algebra $\mathfrak{g}$, let $\mathfrak{t} \subset \mathfrak{g}$ be a maximal abelian subalgebra, and write the eigenvalues of $\operatorname{ad} H$ on $\mathfrak{g} \otimes \mathbb{C}$ and of H on $\mathbb{C}^N$, for $H \in \mathfrak{t}$, as $i\alpha(H)$ and $i\mu(H)$, with α running over the roots and μ over the weights of $\mathbb{C}^N$ counted with multiplicity. Then $\mathrm{Ric}_G = \kappa_G \langle\cdot,\cdot\rangle$ with*

$$\kappa_G = \frac{\sum_\alpha \alpha(H)^2}{4\sum_\mu \mu(H)^2} \qquad (0 \neq H \in \mathfrak{t}).$$

Proof. For a bi-invariant metric the curvature is $R(X,Y)Z = -\frac{1}{4}[[X,Y],Z]$ on left-invariant fields [48, Sect. 7], so for an orthonormal basis (E_i) of $\mathfrak{g}$,

$$\mathrm{Ric}(X,X) = \sum_i \langle R(E_i,X)X, E_i\rangle = \tfrac{1}{4}\sum_i |[X,E_i]|^2 = -\tfrac{1}{4}\operatorname{Tr}(\operatorname{ad} X)^2,$$

since $\operatorname{ad} X$ is skew. The Killing form $B(X,Y) = \operatorname{Tr}(\operatorname{ad} X \operatorname{ad} Y)$ and $\langle\cdot,\cdot\rangle$ are invariant under the adjoint action, and $\mathfrak{g}$ is simple, so $B = -4\kappa\langle\cdot,\cdot\rangle$ for a constant κ and $\mathrm{Ric} = \kappa\langle\cdot,\cdot\rangle$. On $H \in \mathfrak{t}$ the two forms are $B(H,H) = -\sum_\alpha \alpha(H)^2$ and $\langle H,H\rangle = -\operatorname{Tr} H^2 = \sum_\mu \mu(H)^2$. □

Table I evaluates the ratio of Lemma 21 for every compact simple Lie algebra and a faithful representation of least dimension, on the root systems of [38, Sect. 12.1]; the weights of the representations of dimensions 27 and 56 of E_6 and E_7 are the roots of E_8 with prescribed inner products with a subsystem A_2,

respectively A_1, projected to its orthogonal complement, by the decompositions $248=(78,1)+(27,3)+(\overline{27},\bar{3})+(1,8)$ and $248=(133,1)+(56,2)+(1,3)$ [56]. Taking for H the coroot $\theta^\vee$ of the highest root gives $\sum_\alpha \alpha(\theta^\vee)^2 = 4h^\vee$ and $\sum_\mu \mu(\theta^\vee)^2 = 2\ell_\rho$, so $\kappa_G = h^\vee/(2\ell_\rho)$, with the Dynkin index normalised to one on the defining representation of $\mathrm{SU}(N)$. The ratio depends on $\mathfrak{g}$ and ρ only, so it is the same for every compact group with Lie algebra $\mathfrak{g}$, simply connected or not.

Lemma 22. *Let $e \neq e'$ be two links of a plaquette p, and write*

$$\operatorname{Hess}\operatorname{Re}\operatorname{Tr} Q_p(v,v) = \sum_{e,e'\in p} H^p_{ee'}(X_e, X_{e'})$$

with bilinear forms satisfying $H^p_{ee'}(a,b) = H^p_{e'e}(b,a)$. Then $|H^p_{ee'}(a,b)| \le |a|\,|b|$ for all $a,b \in \mathfrak{u}(N)$ and every field.

Proof. By (6) the terms of f_p that mix two positions $i \neq j$ are $2\operatorname{Re}\operatorname{Tr}(Y_1Y_2W)$, $2\operatorname{Re}\operatorname{Tr}(Y_3Y_4W)$ and $-2\operatorname{Re}\operatorname{Tr}(Y_iWY_j)$, so $H^p_{ee'}$ is $\pm\operatorname{Re}\operatorname{Tr}(abW)$ or $\pm\operatorname{Re}\operatorname{Tr}(aWb)$ after an isometric transport of a and b, and $|\operatorname{Tr}(abW)| \le \|a\|_{\mathrm{HS}}\|bW\|_{\mathrm{HS}} = |a|\,|b|$. □

Corollary 23. *Let $G \subset \mathrm{U}(N)$ be closed and connected with semisimple Lie algebra and $N|\beta| < \frac{2}{29}\kappa_G$. For every box the measure μ^F_Λ satisfies the Bakry–Émery condition with $K = \kappa_G - \frac{29}{2}N|\beta| > 0$, uniformly in the box, and hence a logarithmic Sobolev inequality with constant $2/K$ and a Poincaré inequality with constant $1/K$.*

Proof. The set $\{Q_F = 1\}$ is a product factor, so its Ricci curvature is at least κ_G on every link, and the Hessian of the restricted action is the restriction of $N\beta \operatorname{Hess} S_0$ to $X_F = 0$. Apply the bound and [3, Cor. 5.7.2]. □

A. Decay with an explicit rate

Let A' be a finite set of active links and P' a finite set of plaquettes whose active links lie in A'. An action $S = \sum_{p\in P'} c_p \operatorname{Re}\operatorname{Tr} Q^F_p$ with $0 \le c_p \le 1$ is *admissible* if $|\operatorname{Hess} S(v,v)| \le \frac{29}{2}|v|^2$ for all v tangent to $G^{A'}$. Two links lie in at most one common plaquette, so by Lemma 22 and Lemma 9(iv) the mixed blocks of its Hessian satisfy $|\operatorname{Hess} S_{ee'}(a,b)| \le |a|\,|b|$, and each link has at most twelve partners. The action $S_\Lambda = \sum_{p\in P(\Lambda)} \operatorname{Re}\operatorname{Tr} Q^F_p$ of a box is admissible by the bound proved above, and so is every convex combination of two admissible actions. Let μ be the probability measure on $G^{A'}$ with density proportional to $e^{N\beta S}$ with respect to Haar measure, let $L = \Delta + N\beta\langle \nabla S, \nabla\cdot\rangle$ be its Langevin generator, which is self-adjoint in $L^2(\mu)$, and $P_t = e^{tL}$. Write ∇_e for the gradient in the factor of the link e, and for weights $w : A' \to (0,\infty)$ put $\Gamma^w(f) = \sum_e w_e |\nabla_e f|^2$; Γ is the case $w \equiv 1$.

Lemma 24. *Let $\lambda \ge 0$ and $w_e = e^{\lambda d_e}$, where $|d_e - d_{e'}| \le 1$ whenever e and e' share a plaquette, and put*

$$K_\lambda = \kappa_G - N|\beta|\Big(\tfrac{29}{2} + 12\big(\cosh\tfrac{\lambda}{2} - 1\big)\Big).$$

Then $\Gamma^w(P_t f) \le e^{-2K_\lambda t} P_t \Gamma^w(f)$ for every smooth f and $t \ge 0$.

Proof. Put $V = -N\beta S$. On the product manifold $M = G^{A'}$ the projection to the tangent space of one factor is parallel, the rough Laplacian commutes with it, and the Ricci tensor is block diagonal. Hence the Bochner–Weitzenböck formula $\nabla\Delta g = \bar\Delta \nabla g + \mathrm{Ric}(\nabla g)$, applied factor by factor, together with the derivative of the drift, gives for smooth g

$$\begin{aligned}\tfrac12 L\Gamma^w(g) - \Gamma^w(g, Lg) &= \sum_e w_e\big(|\nabla\nabla_e g|^2 + \mathrm{Ric}_G(\nabla_e g, \nabla_e g)\big)\\ &\quad + \sum_{e,e'} w_e \operatorname{Hess} V_{ee'}(\nabla_e g, \nabla_{e'} g),\end{aligned}$$

where $\Gamma^w(g,h)$ is the polarised form.[8] Put $u_e = \nabla_e g$ and $v_e = \sqrt{w_e}\,u_e$. By the symmetry of $\operatorname{Hess} V$,

$$\begin{aligned}\sum_{e,e'} w_e \operatorname{Hess} V_{ee'}(u_e, u_{e'}) &= \operatorname{Hess} V(v,v)\\ &\quad + \sum_{e\neq e'} \Big(\cosh\tfrac{\lambda(d_e - d_{e'})}{2} - 1\Big)\operatorname{Hess} V_{ee'}(v_e, v_{e'}),\end{aligned}$$

because $\frac{w_e + w_{e'}}{2\sqrt{w_e w_{e'}}} = \cosh\frac{\lambda(d_e - d_{e'})}{2}$. The first term is at least $-\frac{29}{2}N|\beta|\,|v|^2$. In the second only partners contribute, each with a coefficient at most $\cosh\frac{\lambda}{2} - 1$ and a block of norm at most $N|\beta|$, and $2|v_e|\,|v_{e'}| \le |v_e|^2 + |v_{e'}|^2$ with twelve partners per link bounds it by $12(\cosh\frac{\lambda}{2} - 1)N|\beta|\,|v|^2$. Since $\mathrm{Ric}_G \ge \kappa_G$ and $|v|^2 = \Gamma^w(g)$,

$$\tfrac12 L\Gamma^w(g) - \Gamma^w(g, Lg) \ge K_\lambda \Gamma^w(g).$$

For fixed t let $\varphi(s) = P_s\Gamma^w(P_{t-s}f)$, $0 \le s \le t$. Then $\varphi'(s) = P_s\big(L\Gamma^w(g_s) - 2\Gamma^w(g_s, Lg_s)\big) \ge 2K_\lambda \varphi(s)$ with $g_s = P_{t-s}f$, and $\varphi(0) \le e^{-2K_\lambda t}\varphi(t)$ is the claim. □

Lemma 25. *Suppose $K_\lambda > 0$. Let f and g be smooth functions on $G^{A'}$ that depend only on the links of A_f and of B_g, and let D be the least number of steps from A_f to B_g in the graph on A' whose edges join links that share a plaquette. Then*

$$|\operatorname{Cov}_\mu(f,g)| \le \frac{e^{-\lambda D/2}}{K_\lambda}\,\|\nabla f\|_\infty \sum_{e\in B_g} \|\nabla_e g\|_\infty,$$

where $\|\nabla f\|^2_\infty = \sup\sum_e |\nabla_e f|^2$.

Proof. Since $K_0 \ge K_\lambda > 0$, Lemma 24 with $\lambda = 0$ gives $\Gamma(P_t f) \le e^{-2K_0 t}\|\nabla f\|^2_\infty$, so $P_t f$ converges uniformly to a constant, which is $\mu(f)$ by invariance, M being connected. Since $\frac{d}{dt}\mu(gP_t f) = \mu(gLP_t f) = -\mu(\Gamma(g, P_t f))$,

$$\operatorname{Cov}_\mu(f,g) = \int_0^\infty \mu\big(\Gamma(P_t f, g)\big)\,dt.$$

[8] For $w \equiv 1$ this is Bochner's identity $\Gamma_2 = |\nabla^2 g|^2 + (\mathrm{Ric} + \operatorname{Hess} V)(\nabla g, \nabla g)$; the weights enter only through the factor index, which the connection, the Laplacian and the Ricci tensor preserve, so only the off-diagonal blocks of $\operatorname{Hess} V$ see them.

Let d_e be the smaller of D and the graph distance from e to A_f, so that $|d_e - d_{e'}| \le 1$ for partners, $w_e = 1$ on A_f, $w_e = e^{\lambda D}$ on B_g, and $\Gamma^w(f) \le \|\nabla f\|_\infty^2$. For $e \in B_g$, Lemma 24 gives $|\nabla_e P_t f|^2 \le w_e^{-1}\Gamma^w(P_t f) \le e^{-\lambda D}e^{-2K_\lambda t}\|\nabla f\|_\infty^2$. Hence $|\Gamma(P_t f, g)| \le e^{-\lambda D/2}e^{-K_\lambda t}\|\nabla f\|_\infty \sum_{e\in B_g}\|\nabla_e g\|_\infty$, and integration in t gives the bound. □

Proof of Theorem 5. Let $\lambda^* > 0$ solve $K_{\lambda^*} = 0$, so that $\frac{\lambda^*}{2} = m_G(\beta)$, and let $0 < \lambda < \lambda^*$; then $K_\lambda > 0$.

Decay in boxes. Lemma 25 with $S = S_\Lambda$ gives

$$|\operatorname{Cov}_{\mu_\Lambda^F}(f,g)| \le C_{f,g}e^{-\lambda D/2},$$

uniformly in the box.

Limit. For boxes $\Lambda \subset \Lambda'$ let $S_s = (1-s)S_\Lambda + sS_{\Lambda'}$ on the active links of Λ', $s \in [0,1]$, and μ_s the corresponding measure; S_s is admissible, μ_0 is μ_Λ^F times Haar measure on the active links outside Λ, and $\mu_1 = \mu_{\Lambda'}^F$. For smooth f of the active links of Λ,

$$\frac{d}{ds}\mu_s(f) = N\beta \sum_{p\in P(\Lambda')\setminus P(\Lambda)} \operatorname{Cov}_{\mu_s}\big(f, \operatorname{Re}\operatorname{Tr} Q_p^F\big),$$

and $|\nabla_e \operatorname{Re}\operatorname{Tr} Q_p| \le \|Q_p\|_{\mathrm{HS}} = \sqrt{N}$. A step in the graph of Lemma 25 moves a link by at most one lattice unit, so a plaquette at graph distance k from the support T of f lies within distance $k+1$ of T, there are at most $C_T(k+1)^4$ of them, and every plaquette outside $P(\Lambda)$ is at graph distance at least $R-1$, R the distance from T to the complement of Λ. Lemma 25 gives

$$|\mu_{\Lambda'}^F(f) - \mu_\Lambda^F(f)| \le C_f R^4 e^{-\lambda R/2}.$$

The limit therefore exists along every increasing sequence of boxes, and two sequences are compared through their intersections, which are boxes. Continuous gauge-invariant O are uniform limits of smooth gauge-invariant ones, for instance of $e^{\tau\Delta}O$, since the heat semigroup of G^E commutes with left and right translations; with Lemma 10 the limit $\omega(O)$ of $W_\Lambda(O)$ exists for all of them.

Properties. Invariance and, for $\beta \ge 0$, reflection positivity follow as in Proposition 11(iii)–(iv). For Lipschitz O the functions $e^{\tau\Delta}O$ are smooth with the same Lipschitz constant, since $\mathrm{Ric}_G \ge 0$ [3, Sect. 3.3], so the bound of the first step passes to ω and to Lipschitz observables. Two links of one plaquette have time coordinates that differ by at most one, so observables at time separation t are at graph distance at least $t-c$, and truncated correlations decay at every rate below $m_G(\beta)$. The gap follows from Proposition 52 as in Proposition 11(vi). □

Proof of the rest of Theorem 4. By Lemma 21 and Table I, $\kappa_{\mathrm{SU}(N)} = \frac{N}{2}$ and $\kappa_{\mathrm{SO}(N)} = \frac{N-2}{4}$; for SO(4), whose Lie algebra is $\mathfrak{so}(3)\oplus\mathfrak{so}(3)$ acting on $\mathbb{R}^4\otimes\mathbb{C} = \mathbb{C}^2\otimes\mathbb{C}^2$, each ideal gives $\frac12$ by the same computation. The conditions of Theorem 4 are those of Theorem 5. □

Remark 26. Let h_F be the supremum of $\pm\operatorname{Hess} S_0$ on the gauge-fixed directions, so that uniform curvature reaches $|\beta| < 1/(2h_F)$ and no further. At the identity field $-\operatorname{Hess} S_0$ restricted to $X_F = 0$ has the eigenvalue $9+\sqrt{17} = 13.123$ on the torus of side two, so $9+\sqrt{17} \le h_F \le \frac{29}{2}$. A gradient search over SU(2) fields of periods two, four and six gives $h_F = 13.97401$, attained by an abelian field; certificates made of plaquette forms and vertex forms on the period-two cell cannot go below 14.384. With this gauge the curvature method stops between $|\beta| = \frac{1}{29}$ and $\frac{1}{26.25}$.

VII. SU(3) UP TO 3/4

In this section $G = \mathrm{SU}(3)$, $\beta \ge 0$ and $\beta_W = 9\beta$. The distance on SU(3) is $d(U,V) = \|U-V\|_{\mathrm{HS}}$, which is at most the geodesic distance for $\langle X,Y\rangle = -\operatorname{Tr}(XY)$; $K = \operatorname{conv}\mathrm{SU}(3)$; $A(M) = \frac12(M-M^*)$ and P removes the trace. With this inner product $\mathrm{Ric} = \frac32$, the Casimir of the fundamental representation is $\lambda_F = \frac83$ and that of the adjoint representation is 6.

A. Reduction to one link

The conditional law of an active link is

$$\nu_b(dU) \propto e^{\operatorname{Re}\operatorname{Tr}(Ub)}dU, \qquad b = N\beta\sum_{p\ni e} S_p,$$

where S_p is the staple of e in p, a product of SU(3) matrices and identities, or its adjoint. Six staples give $b \in rK$ with $r = 6N\beta = 2\beta_W$. For a matrix h put $\varphi_h(U) = \operatorname{Re}\operatorname{Tr}(Uh)$ and

$$L_h(b) = \sup\{\operatorname{Cov}_{\nu_b}(g,\varphi_h) : g \text{ 1-Lipschitz for } d\},$$

$$\Lambda(r) = \sup_{b\in rK}\ \sup_{\|h\|_{\mathrm{HS}}=1} L_h(b).$$

Changing a partner e' changes one staple by a matrix of norm $d(Q_{e'}, Q'_{e'})$. Along the segment b_t from b to b', which stays in rK, $\frac{d}{dt}\nu_{b_t}(g) = \operatorname{Cov}_{\nu_{b_t}}(g, \operatorname{Re}\operatorname{Tr}(U(b'-b)))$, so $W_1(\nu_b,\nu_{b'}) \le N\beta\Lambda(r)\,d(Q_{e'},Q'_{e'})$. With twelve partners per link,

$$\alpha \le 12N\beta\Lambda(r) = 2r\Lambda(r) = 4\beta_W\Lambda(2\beta_W). \tag{8}$$

Since K is convex and contains 0, the mean of $I, \omega I, \omega^2 I$ with $\omega^3 = 1$, $r'K \subset rK$ for $r' \le r$ and Λ is nondecreasing.

B. Gradients and one-forms

Lemma 27. *In the left trivialisation, $\nabla\varphi_c(U) = -PA(cU)$; if $V = \varphi_b$ then $\operatorname{Hess} V(X,X) = \operatorname{Re}\operatorname{Tr}(UX^2b)$; and $\Delta\varphi_c = -\lambda_F\varphi_c$.*

Proof. $\frac{d}{dt}\operatorname{Re}\operatorname{Tr}(Ue^{tX}c)|_{t=0} = \operatorname{Re}\operatorname{Tr}(cUX) = -\langle PA(cU), X\rangle$ for $X \in \mathfrak{su}(3)$. On a group with a bi-invariant metric $\nabla_X X = 0$ for left-invariant X, so the Hessian is the second derivative along e^{tX}. The function φ_c is a matrix coefficient of the fundamental representation, on which the Casimir acts by λ_F. □

Lemma 28. *The bottom of the spectrum of the rough Laplacian $\nabla^*\nabla$ on L^2 one-forms of* SU(3) *is $\mu_0 = \frac76$.*

Proof. Trivialise one-forms on the left, so that a one-form is a function $f:\mathrm{SU}(3)\to\mathfrak{su}(3)$. For an orthonormal basis X_a acting as left-invariant vector fields, $\nabla_{X_a}Y=\frac12[X_a,Y]$ and $\nabla_{X_a}X_a=0$, so $\nabla^*\nabla=-\sum_a(\rho(X_a)+\frac12\operatorname{ad}X_a)^2$. On the isotypic part $V_\pi\otimes V_\pi^*\otimes\mathfrak{su}(3)$ of $L^2\otimes\mathfrak{su}(3)$ put $A_a=\rho_\pi(X_a)$, $B_a=\operatorname{ad}X_a$ and $T_a=A_a+B_a$, the action on the tensor product. Since A_a and B_a commute, $A_a+\frac12B_a=\frac12(A_a+T_a)$ and $A_aT_a+T_aA_a=A_a^2+T_a^2-B_a^2$, so on a component $\sigma\subset\pi\otimes\operatorname{ad}$

$$\nabla^*\nabla=\tfrac12(C_\pi+C_\sigma)-\tfrac14C_{\mathrm{ad}}=\tfrac12(C_\pi+C_\sigma)-\tfrac32,$$

with $C(p,q)=\frac23(p^2+q^2+pq+3p+3q)$ for the representation of highest weight (p,q) [30]. The components of $(p,q)\otimes(1,1)$ are $(p+1,q+1)$, $(p+2,q-1)$, $(p-1,q+2)$, $(p+1,q-2)$, $(p-2,q+1)$, $(p-1,q-1)$ and (p,q), the last with multiplicity two, one or zero as both, one or none of p,q are positive, and only those with nonnegative labels occur. For $\pi=(0,0)$ the only component is $(1,1)$, with value $\frac32$. For $\pi=(1,0)$ the components are $(2,1)$, $(0,2)$, $(1,0)$, with values $\frac{31}{6}$, $\frac{19}{6}$, $\frac76$, and $(0,1)$ is conjugate. For $p+q\ge2$ the smallest values occur at $(p-1,q-1)$, (p,q), $(p-2,q+1)$ or $(p+1,q-2)$, and each is at least $\frac13(2(p^2+q^2+pq)+3(p+q)-3)-\frac32\ge\frac32$. □

C. The gap of the one-link measure

Lemma 29. *For $b\in rK$ the spectral gap of $L=\Delta+\langle\nabla V,\nabla\cdot\rangle$, $V=\varphi_b$, on $L^2(\nu_b)$ is at least*

$$\gamma(r)=\tfrac{39}{64}\mu_0+\tfrac32-\tfrac r2=\tfrac{283}{128}-\tfrac r2.$$

Proof. Let $-Lf=\lambda f$ with $\lambda>0$, $\omega=df$ and $\eta=e^{V/2}\omega$; norms are in $L^2(dU)$. Bochner's formula integrated against ν_b gives $\lambda\nu_b|\omega|^2=\nu_b|\nabla\omega|^2+\nu_b((\mathrm{Ric}-\mathrm{Hess}\,V)(\omega,\omega))$, and since $e^{V/2}\nabla\omega=\nabla\eta-\frac12dV\otimes\eta$,

$$\lambda\|\eta\|^2=\|\nabla\eta-\tfrac12dV\otimes\eta\|^2+\int(\mathrm{Ric}-\mathrm{Hess}\,V)(\eta,\eta).\quad(9)$$

Put $\mathscr{A}=\nabla\eta$ and $\mathscr{B}=\frac12dV\otimes\eta$. Then

$$\langle\mathscr{A},\mathscr{B}\rangle=\tfrac14\int\langle\nabla V,\nabla|\eta|^2\rangle=\tfrac{\lambda_F}{4}\int V|\eta|^2,$$

because $-\Delta V=\lambda_FV$. Writing $\|\mathscr{A}-\mathscr{B}\|^2=\|\frac58\mathscr{A}-\mathscr{B}\|^2+\frac{39}{64}\|\mathscr{A}\|^2-\frac34\langle\mathscr{A},\mathscr{B}\rangle$ and dropping the first term,

$$\|\mathscr{A}-\mathscr{B}\|^2\ge\tfrac{39}{64}\mu_0\|\eta\|^2-\tfrac12\int V|\eta|^2$$

by Lemma 28. With $\mathrm{Ric}=\frac32$, (9) gives $\lambda\|\eta\|^2\ge(\frac{39}{64}\mu_0+\frac32)\|\eta\|^2-\int(\mathrm{Hess}\,V(\eta,\eta)+\frac12V|\eta|^2)$, and the claim follows from the pointwise bound

$$\mathrm{Re}\,\mathrm{Tr}(MX^2)+\tfrac12\,\mathrm{Re}\,\mathrm{Tr}\,M\le\tfrac r2|X|^2,\qquad M=bU\in rK,\quad(10)$$

since $\mathrm{Hess}\,V(X,X)=\mathrm{Re}\,\mathrm{Tr}(MX^2)$ and $V=\mathrm{Re}\,\mathrm{Tr}\,M$. The maximum over $|X|=1$ of the left side is convex in M, so it suffices to take $M=rW$ with $W\in\mathrm{SU}(3)$ and, after a conjugation, $W=\mathrm{diag}(e^{i\theta_k})$ with $\sum_k\theta_k=0$; put $c_k=\cos\theta_k$. Write $X=iH$ with H Hermitian, traceless, $\mathrm{Tr}\,H^2=1$, with diagonal d_k and entries H_{jk}. Then

$$\mathrm{Re}\,\mathrm{Tr}(WX^2)=-\sum_kc_kd_k^2-\sum_{j<k}(c_j+c_k)|H_{jk}|^2,$$

with $\sum_kd_k^2+2\sum_{j<k}|H_{jk}|^2=1$, so the maximum is the larger of $q_{\mathrm{off}}=\max_{j<k}-\frac12(c_j+c_k)$ and $q_{\mathrm{diag}}=\max\{-\sum c_kd_k^2:\sum d_k=0,\ \sum d_k^2=1\}$. If l is the third index, $q_{\mathrm{off}}+\frac12\sum_kc_k=\frac12c_l\le\frac12$. The value q_{diag} is minus the smaller eigenvalue of $\mathrm{diag}(c)$ compressed to the plane $\sum d_k=0$; the eigenvalues of the compression solve $\sum_k(c_k-\mu)^{-1}=0$, that is $3\mu^2-2e_1\mu+e_2=0$ with e_1,e_2 the elementary symmetric functions of the c_k, so $q_{\mathrm{diag}}=\frac13(\sqrt{e_1^2-3e_2}-e_1)$. The inequality $q_{\mathrm{diag}}+\frac12e_1\le\frac12$ is $\sqrt{e_1^2-3e_2}\le\frac12(3-e_1)$, that is $e_1^2-4e_2+2e_1-3\le0$. When $\sum\theta_k=0$, $\sum c_k^2=1+2c_1c_2c_3$, and with this identity the left side equals $-2(1-c_1)(1-c_2)(1-c_3)\le0$. □

The weight $\frac58$ is the one that gives the coefficient $\frac12$ of V, and (10) is an equality at $W=I$. At $r=\frac32$, $\gamma=\frac{187}{128}$.[9]

D. A corrector of degree two

Lemma 30. *For $b\in rK$, $\|h\|=1$ and every smooth real function u on* $\mathrm{SU}(3)$,

$$L_h(b)\le\sqrt{\nu_b|\nabla u|^2}+\sqrt{\mathrm{Var}_{\nu_b}(\varphi_h+Lu)/\gamma(r)}.$$

Proof. Let $-L\psi=\varphi_h-\nu_b\varphi_h$. For a 1-Lipschitz g, $|\nabla g|\le1$ almost everywhere, because d is at most the geodesic distance, so $\mathrm{Cov}(g,\varphi_h)=\nu_b\langle\nabla g,\nabla\psi\rangle\le\nu_b|\nabla\psi|$. Write $\psi=u+\chi$. Since $\nu_b(Lu)=0$, χ solves $-L\chi=X-\nu_bX$ with $X=\varphi_h+Lu$, and

$$\nu_b|\nabla\chi|^2=\mathrm{Cov}(\chi,X)\le\sqrt{\mathrm{Var}\,\chi\,\mathrm{Var}\,X}\le\sqrt{\nu_b|\nabla\chi|^2\,\mathrm{Var}\,X/\gamma}$$

by Lemma 29. Hence $\nu_b|\nabla\chi|^2\le\mathrm{Var}\,X/\gamma$, and $\nu_b|\nabla\psi|\le\sqrt{\nu_b|\nabla u|^2}+\sqrt{\nu_b|\nabla\chi|^2}$. □

The best u is ψ itself, and then the second term vanishes. With $u=s\varphi_h$, Lemma 27 gives $\varphi_h+Lu=(1-s\lambda_F)\varphi_h+s\Gamma$ with $\Gamma=\langle\nabla V,\nabla\varphi_h\rangle$; the choice $s=1/\lambda_F$ removes φ_h and leaves the drift. This blended corrector reaches $\beta_W=\frac{45}{64}$ and stops near 0.715 (Sec. IX).[10] We take u in a space of polynomials of degree two.

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[9] The direct Bochner bound, with Minkowski's inequality in place of the cross term, gives $(\sqrt{\mu_0}-\frac g2r)^2+\frac32-r$ with $g^2=\max_W\|PA(W)\|^2=2.0655$; it is exact at $b=0$, where the gap is $\frac83$, but falls below γ for $r>0.24$ and vanishes before $r=1$.

[10] An earlier version of this paper proved Theorem 6 for $\beta_W\le\frac{45}{64}$ with $u=s\varphi_h$, s chosen box by box, on 865 boxes at $r=\frac{45}{32}$; the largest bound was 0.999657.

Let $\mathscr{P}_2$ be the real vector space of real polynomials of degree one or two in the entries u_{ij} of U and $\bar{u}_{ij}$ of $\bar{U}$. It has dimension 189: a basis (B_k) consists of $\frac{1}{2}(m+\bar{m})$ and $\frac{1}{2i}(m-\bar{m})$ for each pair $\{m,\bar{m}\}$ of conjugate monomials with $m \neq \bar{m}$, and m itself when $m = \bar{m}$. Identify $h \in \mathbb{C}^{3\times 3}$ with $\mathbb{R}^{18}$ through the real and imaginary parts of its entries. For a real 189×18 matrix C put

$$q_h = \sum_k \sum_{m=1}^{18} C_{km} h_m B_k \in \mathscr{P}_2,$$

linear in h. For diagonal $b = \operatorname{diag}(a_1, a_2, x_3 + iy_3)$ we have $V = \sum_k b_k \varphi_{F_k}$ with $(b_k) = (a_1, a_2, x_3, y_3)$ and $F = (e_{11}, e_{22}, e_{33}, ie_{33})$, so

$$X_h = \varphi_h + L q_h = \varphi_h + \Delta q_h + \sum_k b_k \langle \nabla \varphi_{F_k}, \nabla q_h \rangle \tag{11}$$

is a polynomial of degree at most three, affine in b, while ∇q_h does not depend on b. The squares of these functions have degree at most six, so the quadratic forms $h \mapsto \nu_b |\nabla q_h|^2$ and $h \mapsto \nu_b[(X_h - \kappa)^2]$ are combinations of moments of ν_b of degree at most six.

To compute Δ and $\langle \nabla f, \nabla g \rangle$ exactly we use the basis of $\mathfrak{su}(3)$

$$\begin{aligned} &e_{jk} - e_{kj}, \quad i(e_{jk} + e_{kj}) \quad (j<k), \\ &i(e_{11} - e_{22}), \quad i(e_{11} + e_{22} - 2e_{33}), \end{aligned}$$

acting as left-invariant fields $X_a f(U) = \frac{d}{dt} f(U e^{tX_a})|_{t=0}$, with weights $w_a = \frac{1}{2}$ for the first seven and $\frac{1}{6}$ for the last. Then $\sum_a w_a X_a \otimes X_a$ is the Casimir tensor of $\langle X, Y \rangle = -\operatorname{Tr}(XY)$, so

$$\langle \nabla f, \nabla g \rangle = \sum_a w_a X_a f \, X_a g, \qquad \Delta = \sum_a w_a X_a^2.$$

The fields map monomials to combinations of monomials of the same degree with coefficients in $\mathbb{Z}[i]$; for instance $\Delta \varphi_h = -\frac{8}{3}\varphi_h$, as in Lemma 27.

E. Orbits of staple sums

Lemma 31. *Let $V, W \in \mathrm{SU}(3)$.*

(i) *$L_h(VbW) = L_{V^{-1}hW^{-1}}(b)$ and $L_h(\bar{b}) = L_{\bar{h}}(b)$.*

(ii) *Every $b \in \mathbb{C}^{3\times 3}$ is in the orbit of some $\operatorname{diag}(a_1, a_2, z_3)$ with $a_1 \geq a_2 \geq |z_3|$ and $\operatorname{Im} z_3 \geq 0$.*

(iii) *A diagonal matrix lies in K if and only if it lies in the convex hull of the diagonal torus T of $\mathrm{SU}(3)$.*

Proof. (i) If U has law ν_{VbW} then WUV has law ν_b; the map preserves d and Haar measure, and $\varphi_h(U) = \varphi_{V^{-1}hW^{-1}}(WUV)$. For $\bar{b}$ use $U \mapsto \bar{U}$. (ii) Take a singular value decomposition $b = V_1 \Sigma W_1$ and move the determinants of V_1, W_1 into the last entry; a complex conjugation makes $\operatorname{Im} z_3 \geq 0$. (iii) Let c be diagonal with nonzero entries of distinct moduli and let U maximise $\operatorname{Re}\operatorname{Tr}(cU)$ on $\mathrm{SU}(3)$. The derivative along $\mathfrak{su}(3)$ vanishes, so the anti-Hermitian part of cU is a multiple $i\lambda I$ of the identity and $cU = H + i\lambda I$ with H Hermitian. Then $H^2 + \lambda^2 I = cc^*$ is diagonal with distinct entries, H commutes with it and is diagonal, and $U \in T$. Hence $\max_{\mathrm{SU}(3)} \operatorname{Re}\operatorname{Tr}(cU) = \max_T \operatorname{Re}\operatorname{Tr}(ct)$ for such c, and by continuity for every diagonal c. A diagonal matrix outside $\operatorname{conv} T$ is separated from it by such a functional, which then separates it from K. □

Since every $b \in rK$ has operator norm at most r,

$$\Lambda(r) = \sup\Big\{ \sup_{\|h\|=1} L_h(b) : b \in \mathscr{R}_r \Big\},$$

where $\mathscr{R}_r$ is the set of $\operatorname{diag}(a_1, a_2, x_3 + iy_3) \in r \operatorname{conv} T$ with $a_1 \geq a_2 \geq |x_3 + iy_3|$ and $y_3 \geq 0$, a subset of the box $[0,r]^2 \times [-r,r] \times [0,r]$ in the coordinates (a_1, a_2, x_3, y_3).

F. Exact moments and interpolation

Lemma 32. *For $D, D' \in \mathbb{C}^{3\times 3}$ let $\mathbb{E}_H$ be the Haar mean on $\mathrm{SU}(3)$ and*

$$Z(D, D') = \mathbb{E}_H \exp\big(\tfrac{1}{2}\operatorname{Tr}(UD) + \tfrac{1}{2}\operatorname{Tr}(\bar{U}D')\big).$$

Then

$$Z(D, D') = \sum_{p \geq q \geq 0} \frac{s_{(p,q)}(DD'^{\top})\, G_{p,q}(\det D)\, G_{p,q}(\det D')}{d_{p,q}},$$

with $G_{p,q}(z) = \sum_{k \geq 0} c_{(p+k,q+k,k)} z^k$, $c_\lambda = 2^{-|\lambda|} / \prod \operatorname{hooks}(\lambda)$, $d_{p,q} = \frac{1}{2}(p-q+1)(q+1)(p+2)$ and $s_{(p,q)}$ the Schur polynomial. The moments of degree j in U and k in $\bar{U}$ are $\mathbb{E}_H[u^\alpha \bar{u}^\beta e^{\operatorname{Re}\operatorname{Tr}(Ub)}] = 2^{|\alpha|+|\beta|} \partial_D^\alpha \partial_{D'}^\beta Z(b, \bar{b})$, the entry u_{ij} pairing with D_{ji}.

Proof. We use $(\operatorname{Tr} X)^n = \sum_{\lambda \vdash n} f^\lambda s_\lambda(X)$, where f^λ is the number of standard tableaux of shape λ, given by the hook length formula [57, Cor. 7.21.6]. Expanding the exponential gives

$$e^{\frac{1}{2}\operatorname{Tr} X} = \sum_{n \geq 0} \frac{(\operatorname{Tr} X)^n}{2^n n!} = \sum_\lambda c_\lambda s_\lambda(X),$$

the sum over partitions with at most three parts. For $U \in \mathrm{SU}(3)$ one has $s_{\lambda+(k,k,k)}(UD) = (\det D)^k s_\lambda(UD)$, so

$$e^{\frac{1}{2}\operatorname{Tr}(UD)} = \sum_{p \geq q} G_{p,q}(\det D)\, \chi_{p,q}(UD),$$

where $\chi_{p,q}$ is the irreducible character of highest weight $(p,q,0)$ and $d_{p,q}$ its dimension. Since $\bar{U} = (U^{-1})^\top$, we have $\chi_{p,q}(\bar{U}D') = \chi_{p,q}(U^{-1}D'^{\top})$, and Schur orthogonality [20, Ch. 2],

$$\mathbb{E}_H\big[\chi_\rho(UA)\chi_{\rho'}(U^{-1}B)\big] = \delta_{\rho\rho'} \frac{\chi_\rho(AB)}{d_\rho},$$

gives the expansion. The bound below gives absolute convergence. Finally $\operatorname{Re}\operatorname{Tr}(Ub) = \frac{1}{2}\operatorname{Tr}(Ub) + \frac{1}{2}\operatorname{Tr}(\bar{U}\bar{b})$. □

The computation keeps $p \le 34$ and $k \le 11$. Every omitted partition has more than 34 boxes, and since $|s_\lambda(X)| \le \dim(\lambda)\|X\|^{|\lambda|}$ and $\sum_{\lambda\vdash n} f^\lambda \dim\lambda = 3^n$ by Schur–Weyl duality, the omitted part of each factor is at most $\sum_{n>34}(\frac{3}{2}\|D\|)^n/n!$; Cauchy's estimate on a polydisc around b turns this into a bound for every derivative (Appendix E).

Lemma 33 (Interpolation on a box)**.** *Let $Q=\prod_k[l_k,l_k+w_k]$ in the coordinates (a_1,a_2,x_3,y_3), with corners b_i and multilinear weights $\omega_i(b)\ge 0$, $\sum_i\omega_i(b)=1$, and let $\varsigma = w_1^2+w_2^2+\max(w_3^2,w_4^2)$. For $b\in Q$:*

(i) *$Z(b)\ge e^{-\varsigma/8}\sum_i\omega_i(b)Z(b_i)$;*

(ii) *$\nu_b(f)\le e^{\varsigma/8}\max_i\nu_{b_i}(f)$ for every $f\ge 0$ that does not depend on b;*

(iii) *if $Y_b=(\mathscr{A}+\sum_k(b_k-\beta_k)\mathscr{B}_k)^2$ with functions $\mathscr{A},\mathscr{B}_k$ of U and a fixed point β, then*

$$\nu_b(Y_b)\le e^{\varsigma/8}\Big(\max_i\nu_{b_i}(Y_{b_i})+\sum_k\frac{w_k^2}{4}\,\frac{\max_i Z(b_i)\nu_{b_i}(\mathscr{B}_k^2)}{\min_i Z(b_i)}\Big).$$

Proof. Write $E(b)=e^{\operatorname{Re}\operatorname{Tr}(Ub)}=\prod_k e^{c_kb_k}$, where

$$c=(\operatorname{Re}u_{11},\operatorname{Re}u_{22},\operatorname{Re}u_{33},-\operatorname{Im}u_{33}),$$

so that $|c_k|\le 1$ and $c_3^2+c_4^2\le 1$. In each coordinate the chord of e^{c_kx} over $[l_k,l_k+w_k]$ lies above the function and, by Hoeffding's lemma [37] for a two-point variable, below $e^{c_kx}e^{c_k^2w_k^2/8}$. Multiplying,

$$E(b)\le\sum_i\omega_i(b)E(b_i)\le E(b)\,e^{\sum_k c_k^2w_k^2/8}\le E(b)\,e^{\varsigma/8}. \tag{12}$$

Integrating the right inequality gives (i), and (ii) follows from the left one and (i). For (iii) let $\mathscr{F}(b)=Y_bE(b)$ and I_k the linear interpolation in the k-th coordinate, so that $\mathscr{F}-I_1I_2I_3I_4\mathscr{F}=\sum_k I_1\cdots I_{k-1}(\mathscr{F}-I_k\mathscr{F})$. Along the k-th coordinate $\mathscr{F}$ is $\ell^2e^{c_kx}$ times a positive factor, with ℓ affine of slope $\mathscr{B}_k$, and $\partial_k^2\mathscr{F}=((c_k\ell+2\mathscr{B}_k)^2-2\mathscr{B}_k^2)E\ge -2\mathscr{B}_k^2E$. With the Green kernel of the interval, whose integral is at most $w_k^2/8$,

$$\mathscr{F}-I_k\mathscr{F}\le 2\mathscr{B}_k^2\int GE\le\tfrac{w_k^2}{4}\mathscr{B}_k^2\sum_i\omega_i^{(k)}E(b_i)$$

by the left inequality of (12), with weights $\omega_i^{(k)}\ge 0$ of sum one that do not depend on U. The operators I_j have nonnegative weights of sum one; integrate over U and divide by $Z(b)$, bounded below by (i). □

G. The cover and the theorem

Fix a box Q and a matrix C. For any constant κ, $\operatorname{Var}(X_h)\le\nu_b[(X_h-\kappa)^2]$; take $\kappa=\langle\kappa(b),h\rangle$ with $\kappa(b)$ affine in b, fitted to the means at the corners. By (11), $X_h-\kappa=\mathscr{A}+\sum_k(b_k-\beta_k)\mathscr{B}_k$ with β the centre of Q and $\mathscr{B}_k=\langle\nabla\varphi_{F_k},\nabla q_h\rangle-\langle\partial_k\kappa,h\rangle$, and Lemmas 30 and 33 give, for all $b\in Q\cap\mathscr{R}_r$,

$$\sup_h L_h(b)\le\sqrt{e^{\varsigma/8}\max_i\lambda_A(b_i)}+\sqrt{\frac{e^{\varsigma/8}}{\gamma(r)}\Big(\max_i\lambda_V(b_i)+\frac{e_N}{\min_i Z(b_i)}\Big)}, \tag{13}$$

where $e_N=\sum_k\frac{w_k^2}{4}\max_i Z(b_i)\lambda_k(b_i)$ and λ_A, λ_V, λ_k are the largest eigenvalues of the quadratic forms $h\mapsto\nu_{b_i}|\nabla q_h|^2$, $h\mapsto\nu_{b_i}[(X_h-\kappa)^2]$ at $b=b_i$, and $h\mapsto\nu_{b_i}[\mathscr{B}_k^2]$ on $\mathbb{R}^{18}$. Their entries are moments of ν_{b_i} of degree at most six, computed with rigorous error bounds (Appendix E); each largest eigenvalue is bounded above by an $LDL^\top$ factorisation of tI minus the computed form with certified positive pivots, plus the norm of its error matrix by Weyl's inequality [35, 40].

The matrix C is chosen in each box in floating point. At the mean of the corner moments and at $b=\beta$, Lemma 30 for $u=q_h$ is a quadratic problem in C: for $\theta>0$ we minimise the trace of $\nu|\nabla q|^2+\theta\operatorname{Var}(X)/\gamma$ over the 18 coordinate directions h, a least-squares problem of size 189, keep the θ on a geometric grid of 48 values from 0.05 to 80 that gives the smallest bound, and round C to multiples of 2^{-30}. Any choice gives a valid bound.

A box is discarded when it contains no point with $a_1\ge a_2\ge|z_3|$, or when a linear functional $\ell(b)=\operatorname{Re}\operatorname{Tr}(c^*b)$ satisfies $\min_Q\ell>r\max_T\ell$, which separates it from $r\operatorname{conv}T$ and hence, by Lemma 31(iii), from $\mathscr{R}_r$.

Proposition 34. *$2r\Lambda(r)\le 0.999887$ at $r=\frac{3}{2}$.*

Proof. The box $[0,r]^2\times[-r,r]\times[0,r]$ is cut into 8192 boxes of side $r/8$. Of these, 841 survive the two tests, and 581 satisfy $2rB_Q<1$ for the right side B_Q of (13), with $\gamma(r)=\frac{187}{128}$. Each of the other 260 is cut into sixteen boxes of side $r/16$; 1760 of these survive the tests, and all satisfy $2rB_Q<1$. The 2341 certified boxes cover $\mathscr{R}_r$. The largest bound is 0.999887, on the box $[1.3125,1.5]\times[0.9375,1.125]\times[0.5625,0.75]\times[0,0.1875]$, and the median is 0.983. Figure 4 shows the cover, and Appendix E the error analysis. □

The same cover at a larger r bounds $\Lambda(r)$ where $2r\Lambda(r)$ is no longer below one; the plaquette blocks of Section VIII use that bound.

Proposition 35. *$2r\Lambda(r)\le 1.029967$ at $r=\frac{38}{25}$.*

Proof. As in the proof of Proposition 34, with $\gamma(r)=\frac{4643}{3200}$ and the threshold $2rB_Q<1.03$: of the 8192 boxes of side $r/8$, 784 survive the two tests and 690 satisfy the threshold; each of the other 94 is cut into sixteen boxes of side $r/16$, of which 373 survive the tests, and all satisfy it. The 1063 certified boxes cover $\mathscr{R}_r$, and the largest bound is 1.029967. □

Proof of Theorem 6. Proposition 34 gives

$$\Lambda\big(\tfrac{3}{2}\big)\le\frac{0.999887}{3}<0.333296.$$

For $\beta_W \le \frac{3}{4}$ we have $2\beta_W \le \frac{3}{2}$, and (8) with the monotonicity of Λ gives

$$\alpha \le 4\beta_W \Lambda\big(\tfrac{3}{2}\big) < 1.333183\,\beta_W \le \tfrac{3}{4}\cdot 1.333183 < 1.$$

Proposition 11 applies with the Hilbert–Schmidt metric. □

VIII. SU(3) WITH PLAQUETTE BLOCKS

In this section $G = \mathrm{SU}(3)$, $\beta \ge 0$, $\beta_W = 9\beta$, $N\beta = \beta_W/3$, and the notation is that of Sec. VII: d is the Hilbert–Schmidt distance, $K = \operatorname{conv}\mathrm{SU}(3)$, $\nu_c(dU) \propto e^{\mathrm{Re}\,\mathrm{Tr}(Uc)}dU$, $\varphi_h(U) = \mathrm{Re}\,\mathrm{Tr}(Uh)$ and $\Lambda(r)$ is the modulus of (8). Put $Z(c) = \int e^{\mathrm{Re}\,\mathrm{Tr}(Uc)}dU$ and $g = \log Z$ on $\mathbb{C}^{3\times 3}$, viewed as $\mathbb{R}^{18}$ with the inner product $\langle x,y\rangle = \mathrm{Re}\,\mathrm{Tr}(xy^*)$. The gradient of g at c is the conjugate mean $m(c)^*$, $m(c) = \nu_c(U) \in K$, and $\operatorname{Hess} g(c)(h,h) = \mathrm{Var}_{\nu_c}(\varphi_h)$. Since $Z(VcW) = Z(c)$ for $V,W \in \mathrm{SU}(3)$, the same holds for g.

The blocks are those of Lemma 12, which concerns the line field only and holds for every group. For a block plaquette with active links e_1,e_2,e_3, $\mathrm{Re}\,\mathrm{Tr}\,Q_p = \mathrm{Re}\,\mathrm{Tr}(y_1y_2y_3)$ with $y_i = Q_{e_i}$ or $Q_{e_i}^{-1}$, and the conditional law of the block is

$$\mu_b(dy) \propto \exp\Big(\sum_i \mathrm{Re}\,\mathrm{Tr}(y_i b_i) + N\beta\,\mathrm{Re}\,\mathrm{Tr}(y_1y_2y_3)\Big)dy, \quad (14)$$

where b_i is $N\beta$ times the sum of the five staples of y_i outside the block plaquette, so $b_i \in 5N\beta K$ and $\|b_i\|_{\mathrm{op}} \le 5N\beta$. On $\mathrm{SU}(3)^3$ we use $d(y,y') = \sum_i \|y_i - y_i'\|_{\mathrm{HS}}$.

A. The sensitivity of one block

Let $r = 2\beta_W = 6N\beta$. Let λ_* and τ be a lower and an upper bound for the eigenvalues of $\operatorname{Hess} g$ on rK, M an upper bound on rK for $|D^3 g(c)(u,v,v)|$ over unit vectors u,v, and $\gamma = \gamma(r)$ the gap of Lemma 29. Put

$$\begin{aligned} w &= \tfrac{3}{2}(N\beta)^2(\tau-\lambda_*), \qquad s = \Lambda(r) + (e^w-1)e^w\sqrt{\tau/\gamma},\\ C &= 5(N\beta)^2\Big(\tau s + \frac{\sqrt{3}N\beta\,M\,e^w}{\gamma}\Big),\\ K_3 &= \frac{(1+N\beta\Lambda(r))\,s}{1-C}, \qquad \alpha_3 = 10N\beta K_3, \end{aligned} \quad (15)$$

assuming $C < 1$.

Lemma 36. *Let $a, c_0 \in 5N\beta K$ and*

$$\kappa(dx) \propto \exp\big(\mathrm{Re}\,\mathrm{Tr}(xa) + g(c_0 + N\beta x)\big)dx \quad \text{on } \mathrm{SU}(3).$$

For every 1-Lipschitz f and every $h \in \mathbb{C}^{3\times 3}$:

(i) $\mathrm{Cov}_\kappa(f,\varphi_h) \le s\|h\|_{\mathrm{HS}}$;

(ii) *if c_0 moves in the direction k inside $5N\beta K$, then*

$$\big|\tfrac{d}{dt}\kappa(f)\big| \le N\beta\big(\tau s + \sqrt{3}N\beta M e^w/\gamma\big)\|k\|_{\mathrm{HS}}.$$

Proof. (i) The segment from c_0 to $c_0 + N\beta x$ lies in $6N\beta K = rK$, and Taylor's formula gives

$$g(c_0 + N\beta x) = g(c_0) + N\beta\,\mathrm{Re}\,\mathrm{Tr}(x\,m(c_0)) + R(x)$$

with $R(x) = \frac{1}{2}(N\beta)^2 \operatorname{Hess} g(\xi)(x,x)$. Since $\|x\|_{\mathrm{HS}}^2 = 3$, $R \in [\frac{3}{2}(N\beta)^2\lambda_*, \frac{3}{2}(N\beta)^2\tau]$ and $\operatorname{osc} R \le w$. Hence κ is the tilt $e^R\nu/\nu(e^R)$ of $\nu = \nu_{c'}$ with $c' = a + N\beta m(c_0) \in rK$. By definition of Λ, $\mathrm{Cov}_\nu(f,\varphi_h) \le \Lambda(r)\|h\|$. The function f is 1-Lipschitz for the geodesic distance, which dominates d, so $\mathrm{sd}_\nu f \le \gamma^{-1/2}$ by Lemma 29, and $\mathrm{sd}_\nu \varphi_h \le \sqrt{\tau}\|h\|$. Lemma 13 gives (i).

(ii) The derivative is $\mathrm{Cov}_\kappa(f, \langle \nabla g(c_0 + N\beta x), k\rangle)$. Write $\nabla g(c_0 + N\beta u) = \nabla g(c_0) + N\beta \operatorname{Hess} g(c_0)u + N(u)$. The linear term is φ_h with $\|h\| \le N\beta\tau\|k\|$ and contributes at most $N\beta\tau s\|k\|$ by (i). On the segment from c_0 the Hessian moves by at most $N\beta M\|u\| \le \sqrt{3}N\beta M$, so $x \mapsto \langle N(x),k\rangle$ is $\sqrt{3}(N\beta)^2 M\|k\|$-Lipschitz for d. The law κ is a tilt of ν by a function of oscillation at most w, so its spectral gap is at least $e^{-w}\gamma$ by the Holley–Stroock lemma [36], and the covariance of two Lipschitz functions under κ is at most e^w/γ times the product of their constants. □

Lemma 37. *Let $C < 1$. If b, b' have all fields in $5N\beta K$ and differ only in b_i, then $W_1(\mu_b,\mu_{b'}) \le K_3\|b_i - b_i'\|_{\mathrm{HS}}$ for the metric d.*

Proof. By the cyclic symmetry of $\mathrm{Re}\,\mathrm{Tr}(y_1y_2y_3)$ we may take $i = 1$. Given y_1, y_3, the law of y_2 is ν_c with $c = b_2 + N\beta y_3 y_1 \in rK$. Changing y_1 by η in d moves c by $N\beta\eta$ and ν_c by at most $N\beta\Lambda(r)\eta$ in W_1, as in (8); so $F(y_1,y_3) = \mathbb{E}[f \mid y_1, y_3]$ is $(1 + N\beta\Lambda(r))$-Lipschitz in each variable. The law of (y_1,y_3) is proportional to $e^{\mathrm{Re}\,\mathrm{Tr}(y_1b_1) + \mathrm{Re}\,\mathrm{Tr}(y_3b_3)} Z(b_2 + N\beta y_3 y_1)$, and $Z(b_2 + N\beta y_3 y_1) = Z(y_3^{-1}b_2 + N\beta y_1) = Z(b_2 y_1^{-1} + N\beta y_3)$, so its two conditional laws have the form of Lemma 36, with $c_0 = y_3^{-1}b_2$ and $c_0 = b_2 y_1^{-1}$. Moving y_3 to y_3' moves $y_3^{-1}b_2$ along a segment in $5N\beta K$ of length at most $\|b_2\|_{\mathrm{op}}\|y_3 - y_3'\| \le 5N\beta\|y_3 - y_3'\|$. The coupling argument of Lemma 15 now gives $d_1 \le s\varepsilon + Cd_3$, $d_3 \le Cd_1$ and $|\mu_b(f) - \mu_{b'}(f)| \le (1 + N\beta\Lambda(r))(d_1 + d_3) \le K_3\varepsilon$. □

B. The constants and the proof of Theorem 7

Lemma 38. *At $r = \frac{3}{2}$ every eigenvalue of $\operatorname{Hess} g$ on rK lies in $[0.056015, 0.247316]$, and $|D^3 g(c)(u,v,v)| \le 0.182126$ for $c \in rK$ and unit vectors u,v. At $r = \frac{38}{25}$ the same holds with $[0.054978, 0.248280]$ and 0.183546.*

Proof. For $V,W \in \mathrm{SU}(3)$, $g(VcW) = g(c)$ and $x \mapsto VxW$ is an isometry, so the eigenvalues of $\operatorname{Hess} g$ and the norm of $D^3 g$ are constant on the orbits, and by Lemma 31 it suffices to bound them on $\mathscr{R}_r$. The derivatives of g are the cumulants of φ under ν_c. Cut the box $[0,r]^2 \times [-r,r] \times [0,r]$ into boxes of side $r/8$ and keep the 841 (784 at $r = \frac{38}{25}$) that survive the two tests of Sec. VII G. On each box Q we compute, with rigorous error bounds, the moments of φ of degree at most three at the sixteen corners b_i and at the centre b_0, and put $k = m(b_0)$, the mean of φ at the centre.

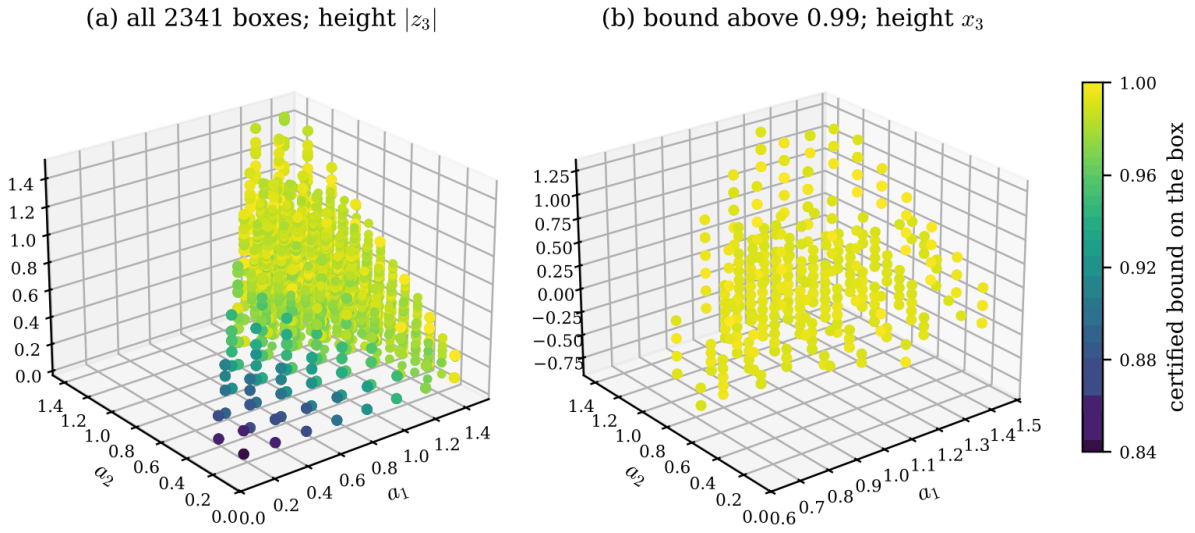


FIG. 4. The 2341 certified boxes at $r=\frac{3}{2}$, plotted at their centres in the coordinates (a_1,a_2) of the staple sum $\mathrm{diag}(a_1,a_2,x_3+iy_3)$, coloured by the certified bound on $2r\sup_h L_h$ over the box; the larger points are the boxes of side $r/8$. (a) All boxes, with height $|x_3+iy_3|$. (b) The 419 boxes with bound above 0.99, with height x_3; the largest bounds sit in the interior of $\mathscr{R}_r$, not at its extreme points.

Upper bound. For $b\in Q$ and every h, $\mathrm{Var}_{\nu_b}(\varphi_h)\le \nu_b[(\varphi_h-\langle k,h\rangle)^2]\le e^{\varsigma/8}\max_i \nu_{b_i}[(\varphi_h-\langle k,h\rangle)^2]$ by Lemma 33(ii).

Lower bound. Integrating the right inequality of (12) against $f\ge 0$ and dividing by $Z(b)\le\sum_i\omega_i Z(b_i)$, the left one, gives $\nu_b(f)\ge e^{-\varsigma/8}\min_i\nu_{b_i}(f)$. With $f=(\varphi_h-\nu_b\varphi_h)^2$ this gives $\mathrm{Var}_{\nu_b}(\varphi_h)\ge e^{-\varsigma/8}\min_i\mathrm{Var}_{\nu_{b_i}}(\varphi_h)$.

Third cumulant. Put $\delta=m(b)-k$. The mean value theorem and the upper bound give $\|\delta\|\le\tau\rho+e$, with ρ the largest distance from b_0 in Q and e the error of the computed mean. Expanding the central moment about k,

$$D^3g(b)(u,v,v)=T_b(u,v,v)-\delta_u\,\mathrm{Var}(\varphi_v)\\ -2\delta_v\,\mathrm{Cov}(\varphi_u,\varphi_v)-\delta_u\delta_v^2,$$

with $T_b=\nu_b[(\varphi-k)^{\otimes 3}]$, $\delta_u=\langle\delta,u\rangle$ and the moments under ν_b. For a function G of either sign, writing $G=G^+-G^-$ and applying the two bounds above to $G^\pm$ gives $|\nu_b(G)|\le(\max_i|\nu_{b_i}(G)|+\varepsilon\max_i\nu_{b_i}|G|)/(1-\varepsilon)$ with $\varepsilon=1-e^{-\varsigma/8}$. For $G=(\varphi_u-k_u)(\varphi_v-k_v)^2$ we use $|\nu_{b_i}(G)|\le\|T_{b_i}\|$, the largest singular value of the 18×324 unfolding of T_{b_i}, and $\nu_{b_i}|G|\le 2\sqrt{3}\,\nu_{b_i}[(\varphi_v-k_v)^2]$, since $\|U-k\|_{\mathrm{HS}}\le 2\sqrt{3}$. The largest eigenvalues and singular values are bounded as in Sec. VII G, and the maxima over these boxes are the numbers stated (Appendix C). □

Proof of Theorem 7. Let first $\beta_W\le\frac{3}{4}$, so that $r=2\beta_W\le\frac{3}{2}$. Every field met in Lemmas 36 and 37 lies in $\frac{3}{2}K$, so the constants of Lemma 38 at $\frac{3}{2}$, $\Lambda(\frac{3}{2})\le\frac{0.999887}{3}$ from Proposition 34 and $\gamma=\gamma(\frac{3}{2})=\frac{187}{128}$ apply; with these fixed constants every quantity in (15) is increasing in β_W. For $\frac{3}{4}<\beta_W\le\frac{19}{25}$ the fields lie in $\frac{38}{25}K$, and the constants of Lemma 38 at $\frac{38}{25}$, $\Lambda(\frac{38}{25})\le\frac{1.029967}{3.04}$ from Proposition 35 and $\gamma(\frac{38}{25})=\frac{4643}{3200}$ apply in the same way. Changing an active link j by η changes one staple of each of its ten partners outside its block by a matrix of norm η, so the field b_i of each such partner moves by $N\beta\eta$, and Lemma 37 gives $\sum_B C_{Bj}\le 10N\beta K_3=\alpha_3(\beta_W)$. In ball arithmetic, at $\beta_W=\frac{3}{4}$, $w\le 0.017935$, $s\le 0.34088$, $C\le 0.043520$, $K_3\le 0.38609$ and $\alpha_3\le 0.96521$; at $\beta_W=0.72$, $\alpha_3\le 0.91815$; at $\beta_W=0.76$, with the second set of constants, $w\le 0.018609$, $s\le 0.34673$, $C\le 0.045770$, $K_3\le 0.39454$ and $\alpha_3\le 0.99950$. Proposition 17, whose proof does not use the group, applies with the Hilbert–Schmidt metric. □

Remark 39. Theorem 6 ends where $2r\Lambda(r)$ reaches one. The blocks remove the factor $\frac{12}{10}$ of the single-link sum and keep most of it after paying for the three links: at $\beta_W=\frac{3}{4}$ the gap bound rises from $1.1\cdot 10^{-4}$ to 0.0177, and the block bound is the better one for $\beta_W\ge 0.715$. Before the corrections for the three links their sum is $\frac{10}{12}$ of the single-link sum $2r\Lambda(r)$, so it can stay below one while $2r\Lambda(r)$ is slightly above one; Proposition 35 certifies $2r\Lambda(r)\le 1.03$ at $r=1.52$, which is enough at $\beta_W=0.76$, with little room. Evaluated in floating point with the constants of a scan of $\mathscr{R}_r$, the block sum stays below one up to $\beta_W\approx 0.78$; that computation is not certified and is not used.

IX. WHERE THE METHODS STOP

SU(2). Theorem 2 closes the single-link route in every forest gauge at $\beta_W=\frac{1}{3}$. Blocks of several links, where the

influence of a boundary link on a whole block is bounded at once, in the spirit of the Dobrushin–Shlosman criterion [28], pass it: plaquette blocks reach $\beta_W = 0.345$ (Theorem 3), and the sharper floating-point bound of Remark 18 stops near 0.357. Larger blocks are the next step.

Curvature. Remark 26 locates the end of the curvature method in this gauge between $|\beta| = \frac{1}{29}$ and $\frac{1}{26.25}$.

SU(3). The route of Sec. VII bounds the $H^{-1}(\nu_b)$ norm of $\varphi_h - \nu_b\varphi_h$, which is at least $L_h(b)$, and needs it uniformly on rK. Its Galerkin values (polynomials of degree two) at matrices of rK give $2r$ times the norm equal to 0.79 at $r = 1.25$, 0.89 at $r = 1.4$, 0.97 at $r = 1.5$ and 1.009 at $r = 1.55$, so the route stops at or below $\beta_W \approx 0.77$ even with exact constants. With $u = s\varphi_h$, Lemma 30 stops earlier: in floating point its largest values on $\mathscr{R}_r$ lie on the face $\mathrm{diag}(r,\rho,\rho)$ with ρ between $0.75r$ and $0.8r$, and $2r$ times the bound is 0.9458 at $r = 1.36$, 0.9767 at $r = 1.40$ and 1.0080 at $r = 1.44$, so this corrector ends near $\beta_W \approx 0.715$. With $u = q_h$ of degree two the largest values on the same face are 0.9305 at $r = 1.44$, 0.9729 at $r = 1.50$ and 1.0018 at $r = 1.54$ (Fig. 5), close to the Galerkin values. Lemma 29 therefore costs little at these couplings, and most of the distance between the floating-point value at $r = 1.5$ and the certified 0.999887 of Proposition 34 is the interpolation over boxes. Between $\beta_W = \frac{3}{4}$ and 0.77 there is room only for smaller boxes. The plaquette blocks of Theorem 7 use $\Lambda(2\beta_W)$ as well, with a little more room: they reach 0.76 with the certified cover of Proposition 35 and near 0.78 in floating point (Remark 39); past that one needs larger blocks, whose sensitivity does not go through the one-link modulus.

Other groups. For groups other than SU(2) and SU(3) only the curvature route is carried out here (Theorem 5), and it stops where the bound $\frac{29}{2}$ uses up the curvature of G, at $N|\beta| = \frac{2}{29}\kappa_G$. The Dobrushin route needs a bound for the Kantorovich modulus of the one-link measure of each group, which is not attempted here.

The four open problems. Our earlier paper [13] listed four problems between the strong-coupling results and the Clay problem [39]: (1) uniqueness and a gap at intermediate fixed coupling; (2) the crossover from strong coupling to Bałaban's weak-coupling regime; (3) the continuum limit with the Osterwalder–Schrader axioms; and (4) a gap in physical units. The present paper moves the boundary of (1): for SU(2) from $\beta_W = \frac{2}{9}$ to 0.345, for SU(3) from $\frac{9}{32}$ to 0.76, with a gap at $\frac{3}{4}$ of 0.0177 in lattice units, and for every compact simple group, the exceptional groups included, past the curvature ceiling of the ungauged measure, with an explicit gap. It does not close (1); the region between these values and the scaling region, near $\beta_W \approx 2.2$ for SU(2) and $\beta_W \approx 6$ for SU(3), is open, and with it problem (2). Problems (3) and (4) are not solved here. Every bound proved here is uniform in lattice units, so by [14] none of them can by itself produce a continuum mass gap; that requires control of the gap in physical units along the renormalisation group flow [4, 16].[11]

The continuum problem. Two general facts limit what can be imported into problems (3) and (4) above from constructions of continuum measures. Both concern a random distribution F on $\mathbb{R}^d$, a Borel probability measure on the tempered distributions with finite moments, and the growth condition

$$\mathbb{E}\, e^{F(\chi)} \le e^{C(\|\chi\|_1 + \|\chi\|_2^2)} \qquad \text{for all real } \chi \in \mathscr{S}(\mathbb{R}^d), \quad (16)$$

which is the growth condition of the regularity axiom of Glimm and Jaffe at $p = 2$ [33, Sect. 6.1, (6.1.5)]. It is the hypothesis under which reflection positivity across coordinate and diagonal hyperplanes is shown in [59] to give rotation invariance of the Schwinger functions, a step proposed there for the continuum limit of lattice gauge theories.

Proposition 40. *Let F satisfy (16), and suppose that its connected two-point function is, off the diagonal, a locally integrable function $S_2(x,y)$ with $S_2(x,y) \ge c|x-y|^{-2\Delta}\big(\log\frac{1}{|x-y|}\big)^{-k}$ for $0 < |x-y| \le r_0$, where $c > 0$, $k \ge 0$ and $2\Delta > d$. Then F does not exist: no random distribution has both properties.*

Proof. Applying (16) to χ and $-\chi$ and using $\cosh x \ge 1 + \frac{x^2}{2}$ gives $\mathbb{E}F(\chi)^2 \le 2(e^{C(\|\chi\|_1+\|\chi\|_2^2)} - 1)$. Fix nonnegative $\varphi, \psi \in C_c^\infty(\mathbb{R}^d)$, not zero, with supports in a ball of radius $\frac{R}{2}$ and at distance $\delta > 0$ from each other, and put $\varphi_\varepsilon(x) = \varepsilon^{-d/2}\varphi(x/\varepsilon)$, and likewise ψ_ε. Then $\|\varphi_\varepsilon\|_2 = \|\varphi\|_2$ and $\|\varphi_\varepsilon\|_1 = \varepsilon^{d/2}\|\varphi\|_1 \le \|\varphi\|_1$ for $\varepsilon \le 1$, so the covariance of $F(\varphi_\varepsilon)$ and $F(\psi_\varepsilon)$ is bounded uniformly in ε by the Cauchy–Schwarz inequality. On the other hand the supports of φ_ε and ψ_ε are disjoint, and for u, v in the supports $\varepsilon\delta \le |\varepsilon u - \varepsilon v| \le \varepsilon R$. For $\varepsilon R \le r_0$ and $\varepsilon R < 1$ the lower bound for S_2 therefore gives

$$\begin{aligned}\mathrm{Cov}\big(F(\varphi_\varepsilon), F(\psi_\varepsilon)\big) &= \varepsilon^d \iint \varphi(u)\psi(v) S_2(\varepsilon u, \varepsilon v)\, du\, dv \\ &\ge c R^{-2\Delta} \varepsilon^{d-2\Delta} \Big(\log\frac{1}{\varepsilon\delta}\Big)^{-k} \|\varphi\|_1 \|\psi\|_1,\end{aligned}$$

which tends to infinity as $\varepsilon \to 0$. □

The action density $\mathrm{Tr}\, F_{\mu\nu}F_{\mu\nu}$ has dimension $\Delta = 4$ in four dimensions. A two-sided short-distance law of the form $|x-y|^{-8}$ times the fourth power of the running coupling, which is what asymptotic freedom predicts and what the conditional construction of [60] states for the smeared action density, has $2\Delta = 8 > 4$ and $k = 2$; so that field cannot satisfy (16), and rotation invariance for it has to come from another source, as in [60], where it is reduced to a separate infrared hypothesis. In three dimensions the free part of the action density already has $2\Delta = 6 > 3$. Condition (16) suits fields of dimension at most $\frac{d}{2}$, such as the field strength of the free Maxwell field, whose covariance is bounded in momentum space.

Positivity, symmetry and regularity do not produce a gap by themselves.

[11] Every one of these approaches needs an input at physical length scales. Renormalisation group approaches to the ultraviolet behaviour of gauge and gravitational theories, of which [4, 25] are the rigorous ones for Yang–Mills theory, have also been proposed for the effective dimension of spacetime [53] and for completions of quantum gravity [10, 15]; they do not bear on the infrared estimates needed here.

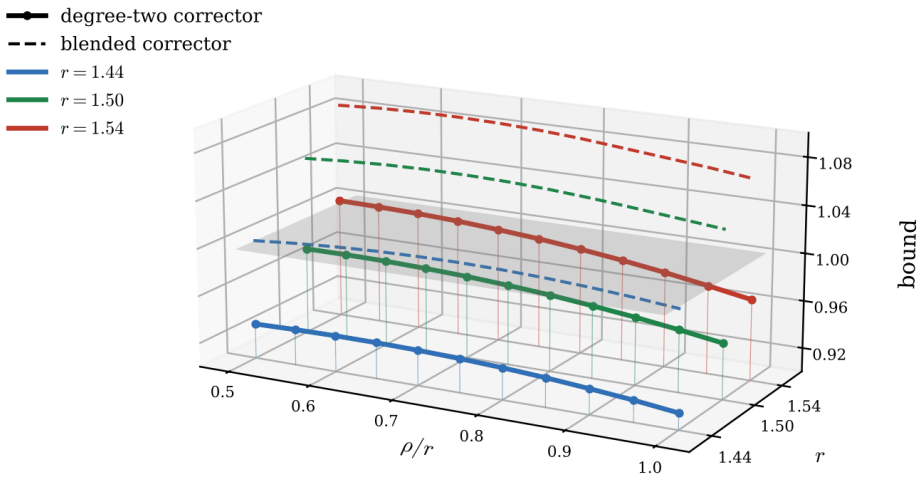


FIG. 5. The floating-point value of $2r\sup_h$ of the bound of Lemma 30 on the face $b = \mathrm{diag}(r,\rho,\rho)$, as a function of ρ/r, for $r = 1.44$, 1.50 and 1.54: dashed with the corrector $u = s\varphi_h$ for the best s, solid with $u = q_h$ of degree two for the best C. The grey plane is the Dobrushin threshold one. The first corrector is above it already at $r = 1.44$; the second stays below it at $r = 1.50$, where Proposition 34 certifies the bound on all of $\mathscr{R}_r$, and crosses it by $r = 1.54$.

Proposition 41. *On $\mathbb{R}^4$ let F be the centred Gaussian random distribution with covariance*

$$\widehat{C}(p) = \int_0^1 \frac{m^2}{|p|^2 + m^2}\, dm.$$

Then F is invariant under all Euclidean motions, reflection positive, satisfies (16) with $C = \frac{1}{2}$, and its Osterwalder–Schrader Hamiltonian has no gap above the vacuum.

Proof. Since $0 \le \widehat{C} \le 1$, $\mathbb{E}e^{F(\chi)} = e^{\frac{1}{2}\langle\chi, C\chi\rangle} \le e^{\frac{1}{2}\|\chi\|_2^2}$, with the unitary Fourier transform. Invariance is clear. The covariance is the integral over $m \in (0,1)$, with the weight $m^2 dm$, of the covariances of free fields of mass m, each of which is reflection positive [33, Sect. 6.2]; a positive combination of reflection positive covariances is reflection positive, and so is the Gaussian measure with such a covariance [33, Sect. 6.2]. For $\chi(t,\mathbf{x}) = h(t)g(\mathbf{x})$ with $h \ge 0$ supported in $t > 0$, not zero, and $\hat{g}(0) \ne 0$, the vector $v = F(\chi)$ is orthogonal to the vacuum, and after the Fourier transform in $\mathbf{x}$,

$$\langle v, e^{-sH} v\rangle = \int_0^1 m^2\, dm \int \frac{|\hat{g}(\mathbf{k})|^2}{2\omega} e^{-\omega s} \times \Big(\int_0^\infty h(t) e^{-\omega t} dt\Big)^2 d\mathbf{k},$$

with $\omega = \sqrt{|\mathbf{k}|^2 + m^2}$ and constants absorbed in $\hat{g}$. If H had a gap $m_0 > 0$, the left side would be at most $e^{-m_0 s}\|v\|^2$. For $0 < \eta < m_0$ the part of the integral with $m, |\mathbf{k}| \le \eta/2$ is at least $c_\eta e^{-\eta s}$ with $c_\eta > 0$, a contradiction as $s \to \infty$. □

The Gaussian field of Proposition 41 is not a gauge theory; it shows only that an infrared estimate, such as a bound $\langle v_a, T_a^{\lfloor t/a\rfloor} v_a\rangle \le C_v e^{-m_* t}$ uniform in the spacing a along the continuum tuning, has to be proved in addition to the axioms. The lattice bounds of this paper are of this form only with m_* proportional to $1/a$, and by [14] such bounds give trivial limits. The routes we know to a continuum gap therefore all stop at the same two estimates: the construction of the infinite-volume continuum measure past Bałaban's small-field analysis [4, 6], which the conditional construction of [60] assumes clause by clause and which yields there a non-negative Hamiltonian without any statement on its spectrum, and a decay estimate in physical units at large scales. The finite-size criterion described in [60, Sect. 9.4] reduces a gap to a threshold condition on one box, uniformly in the cutoff; as stated there, the condition fails in every small box and would have to be checked at large scales, which the construction does not reach.

X. THE CONTINUUM STEP AT ONE PHYSICAL SCALE

The bounds of the preceding sections hold at fixed coupling and give a gap that is at least a fixed number in lattice units, and [14] shows that no such bound can produce a continuum gap. This section replaces the single-link condition of Proposition 11 by a condition on one cube of n^4 sites, averaged over its translates in the manner of Dobrushin and Shlosman [28], and proves that the condition gives a gap of order $1/n$ in lattice units. If it holds with n proportional to $1/a$, that is, in a cube of fixed physical size, the gap survives in physical units and passes to every continuum limit (Theorem 45). The two statements that remain unproved are named at the end: the cube estimate and the existence of the limit.

A. Averaged cubes

For a finite set $V \subset A$ let $\gamma_V(\cdot \mid Q)$ be the probability measure on G^V with density proportional to $\exp\bigl(N\beta \sum_{p\cap V\neq\emptyset} \operatorname{Re}\operatorname{Tr} Q_p^F\bigr)$ with respect to Haar measure, the links outside V being those of Q. Its conditional law of Q_e, $e \in V$, given the other links of V is γ_e^F. Write $d_V(Q_V, Q'_V) = \sum_{i\in V} \delta(Q_i, Q'_i)$ and, for $j \in A \setminus V$,

$$C_{V,j} = \sup \frac{W_1^{d_V}\bigl(\gamma_V(\cdot \mid Q), \gamma_V(\cdot \mid Q')\bigr)}{\delta(Q_j, Q'_j)},$$

the supremum over pairs that differ only at j; $W_1^{d_V}$ is the Kantorovich distance for d_V. The coefficient vanishes unless j shares a plaquette with a link of V.

Let $n \geq 2$ be even and let V be the set of active links (x,μ) with $x \in [0,n)^4$. The direction field has period two, so the translations by $2\mathbb{Z}^4$ preserve F, A and γ^F, and the translates $V_x = V + x$, $x \in 2\mathbb{Z}^4$, satisfy $C_{V_x, j+x} = C_{V,j}$. Every active link lies in exactly $N_n = (n/2)^4$ of them. A *class* is an orbit of $2\mathbb{Z}^4$ on A, that is, a residue of the base point modulo two together with a direction; there are at most 64 classes. For a class c put

$$K_c = \sum_{j\in c\setminus V} C_{V,j}.$$

Definition 42. For $0 < \alpha < 1$, the condition $D(n,\alpha)$ at the coupling β is $K_c \leq \alpha N_n$ for every class c.

Since $\sum_{x\in 2\mathbb{Z}^4,\, j\notin V_x} C_{V_x,j} = \sum_x C_{V,j-x}[j-x \notin V] = K_{c(j)}$ for the class $c(j)$ of j, the condition says that the total influence of a link on the translates that do not contain it is at most the fraction α of the number of translates that do. Figure 6 shows the cubes.

For $n = 2$ it is a bound on blocks of 48 links. Distances between links are those of their midpoints; a link of V and a link sharing a plaquette with V are at distance at most $\rho_n = 2n$, and c_n denotes the number of active links within distance ρ_n of a given one.

Theorem 43. *Let $\beta \geq 0$ and suppose that $D(n,\alpha)$ holds. Then statements* (i)–(iv) *of Proposition 11 hold,* (v) *holds for every $\lambda < \log(1/\alpha)/\rho_n$, and the transfer matrix of ω has spectral gap at least $\log(1/\alpha)/(2n)$.*

Proof. The proof of Proposition 11 uses the coefficients $C_{ee'}$ only through Lemma 48 and Corollary 49. Lemma 51 of Appendix B gives the same conclusions under $D(n,\alpha)$, for every $\lambda > 0$ with $\alpha e^{\lambda\rho_n} < 1$, with the constant $\theta D/(1-\theta)$ replaced by $D(1 + c_n e^{\lambda\rho_n})/(1-\theta)$ and $\theta = \alpha e^{\lambda\rho_n}$. The measures used there are invariant under every γ_{V_x} with V_x disjoint from the exceptional set: a Gibbs measure of γ^F and its conditional measures given the links of a finite set by Lemma 51(a), and the extended finite-volume measure of part (ii) because every plaquette that meets such a V_x lies in $P(\Lambda)$. Part (vi) then follows from Proposition 52 as before, with every $\lambda < \log(1/\alpha)/(2n)$. □

B. The continuum limit

Let $a_k = 2^{-k}a_0$ and let $\beta_k \geq 0$ be couplings, typically with $\beta_k \to \infty$. Suppose that the lattice states ω_k at β_k exist as in Proposition 11, with transfer matrices $\mathbb{T}_k$ on the reconstructed spaces $\mathscr{H}_k$ and vacua Ω_k (Appendix F). Let $\mathscr{T}$ be a complex vector space of test functions on $\mathbb{R}^4$ supported in $\{t > 0\}$ and, for each k, let $f \mapsto O_k(f)$ be a linear map from $\mathscr{T}$ to gauge-invariant local observables supported in positive lattice times; the example to keep in mind is $O_k(f) = Z_k \sum_x a_k^4 f(a_k x)\Phi(x)$ for a local gauge-invariant field Φ, a renormalisation constant Z_k and $\Phi(x)$ the translate of Φ to x. Write ϑ for the time reflection and τ_t for the time translation by t lattice steps.

Definition 44. A *continuum limit* of (ω_k, O_k) is a Hilbert space $\mathscr{H}$ with a unit vector Ω, a self-adjoint operator $H \geq 0$ with $H\Omega = 0$, and a linear map $f \mapsto \psi(f)$ from $\mathscr{T}$ to $\mathscr{H}$ with dense range, such that for all $f, g \in \mathscr{T}$ and all dyadic $\tau \geq 0$,

$$\langle \psi(f), e^{-\tau H}\psi(g)\rangle = \lim_{k\to\infty} \omega_k\bigl(\vartheta(\overline{O_k(f)})\,\tau_{\tau/a_k} O_k(g)\bigr),$$
$$\langle \Omega, \psi(f)\rangle = \lim_{k\to\infty} \omega_k\bigl(O_k(f)\bigr),$$

the limits taken over the k for which τ/a_k is an integer. It is *nontrivial* if $\mathscr{H} \neq \mathbb{C}\Omega$.

The Osterwalder–Schrader reconstruction [50, 51] produces such a limit from convergent renormalised Schwinger functions that satisfy the axioms; the definition records only what the next theorem uses.

Theorem 45. *Suppose that there are $\alpha < 1$, $\ell > 0$ and even n_k with $n_k a_k \leq \ell$ such that $D(n_k,\alpha)$ holds at β_k for every k. Then every continuum limit of (ω_k, O_k) satisfies*

$$\sigma(H) \subset \{0\} \cup [m_*, \infty), \qquad m_* = \frac{\log(1/\alpha)}{2\ell},$$

and 0 is a simple eigenvalue. If the limit is nontrivial, $\sigma(H) \cap [m_*,\infty) \neq \emptyset$, so H has a mass gap of at least m_*.

Proof. By Theorem 43 the state ω_k exists and $\mathbb{T}_k$ has gap at least $\log(1/\alpha)/(2n_k)$ on the complement of Ω_k. Let $f \in \mathscr{T}$, put $O = O_k(f)$ and $\varphi_k = \Psi_O - \omega_k(O)\Omega_k$. In the reconstruction of Appendix F, $\langle \Psi_F, \mathbb{T}_k^t \Psi_H\rangle = \omega_k(\vartheta(\bar F)\tau_t H)$, so $\langle \Omega_k, \Psi_O\rangle = \omega_k(O)$, and $\mathbb{T}_k\Omega_k = \Omega_k$; hence, for integers $t \geq 0$,

$$\langle \varphi_k, \mathbb{T}_k^t \varphi_k\rangle = \omega_k\bigl(\vartheta(\bar O)\,\tau_t O\bigr) - |\omega_k(O)|^2$$
$$\leq e^{-t\log(1/\alpha)/(2n_k)}\,\|\varphi_k\|^2.$$

Take $\tau \geq 0$ dyadic and k large, so that $t = \tau/a_k$ is an integer; then $t/(2n_k) = \tau/(2n_k a_k) \geq \tau/(2\ell)$, and the right side is at most $e^{-m_*\tau}\|\varphi_k\|^2$. Both sides converge by Definition 44. With $\varphi = \psi(f) - \langle\Omega, \psi(f)\rangle\Omega$ and $e^{-\tau H}\Omega = \Omega$ the limit reads

$$\langle \varphi, e^{-\tau H}\varphi\rangle \leq e^{-m_*\tau}\|\varphi\|^2.$$

The maps $f \mapsto O_k(f)$ and $f \mapsto \psi(f)$ are linear, so the vectors φ form a linear subspace, and it is dense in $\Omega^\perp$ because the range of ψ is dense. The inequality extends to $\Omega^\perp$ by continuity, and from dyadic τ to all $\tau \geq 0$ by the strong continuity of $e^{-\tau H}$. For $\varphi \in \Omega^\perp$ with spectral measure μ_φ of H this says $\int e^{-\tau E} d\mu_\varphi(E) \leq e^{-m_*\tau}\mu_\varphi([0,\infty))$ for all $\tau \geq 0$; if $\mu_\varphi([0, m_* - \eta]) > 0$ for some $\eta > 0$, the left side is at least $\mu_\varphi([0, m_* -$

(a) averaged cubes, $n = 4$ (b) one physical scale

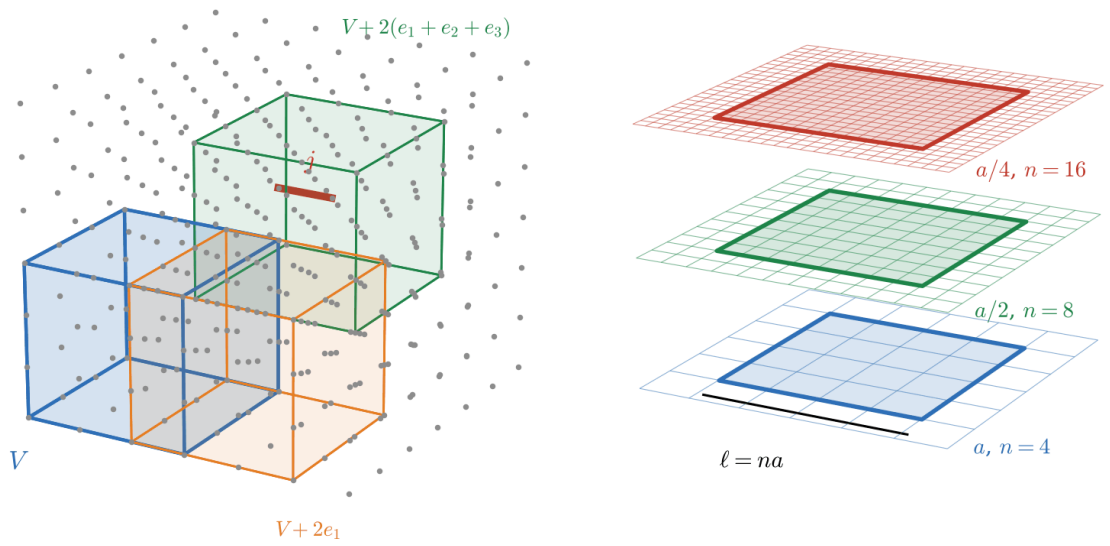


FIG. 6. (a) The cube V of side $n = 4$ and two of its translates by $2\mathbb{Z}^4$, drawn in three of the four dimensions. A link j outside V enters K_c through every translate whose boundary it touches, and $D(n,\alpha)$ compares that total with the $N_n = (n/2)^4$ translates that contain j. (b) The cube of Theorem 45: its physical side $\ell = na$ is held fixed while the spacing is halved, so n doubles at each step and the lattice gap $\log(1/\alpha)/(2n)$ is of order a/ℓ.

$\eta]) e^{-(m_*-\eta)\tau}$, which exceeds the right side for large τ. So μ_φ lives on $[m_*,\infty)$ for every $\varphi \in \Omega^\perp$, which gives both claims about 0, and the spectrum on $\Omega^\perp \neq \{0\}$ lies in $[m_*,\infty)$ and is not empty. □

The theorem bypasses the obstruction of [14]: under the hypothesis the lattice gap is at least $\log(1/\alpha)/(2n_k)$, of order a_k/ℓ in lattice units, which tends to zero, and only its value in physical units is bounded below. In the language of [14] the hypothesis implies the gap part (R2) of Hypothesis R, with $m = m_*$, and the conclusion holds for every continuum limit without a separate assumption on the gap.

C. The two residues

Theorem 45 reduces a continuum mass gap to two statements, neither of which is proved here or, to our knowledge, anywhere else.

(D) *The cube estimate.* There are $\alpha < 1$ and $\ell > 0$ such that $D(n,\alpha)$ holds with $na \le \ell$ along the continuum tuning $\beta = \beta(a) \to \infty$, uniformly in a.

(E) *The limit.* The renormalised Schwinger functions converge along $a_k = 2^{-k} a_0$ to Schwinger functions that satisfy the Osterwalder–Schrader axioms and are not those of the vacuum alone.

(D) and (E) together give the mass gap of the continuum theory; the remaining Clay requirements, Euclidean invariance and the local structure of the limit [39], belong to (E). (D) is a statement about a single cube of fixed physical size and a single coupling at each spacing; (E) is the construction problem, on which Bałaban's renormalisation group analysis [4–6] gives the ultraviolet part in finite volume.

Remark 46. A linear-response count suggests that (D) is consistent with a massive continuum theory. If the gauge-fixed field responds to a change of one link on the boundary of a cube through a Poisson kernel with decay length ξ in lattice units, the response summed over the cube is of order ξ, so K_c is of order $n^3\xi$ and $D(n,\alpha)$ asks for $n \gtrsim \xi/\alpha$, that is, $\ell \gtrsim \xi a/\alpha$, a fixed multiple of the physical correlation length. The count is not a proof, and it assumes that the gauge-fixed links decorrelate on the scale ξ. The gauge here is built on straight lines, and in gauges of axial type the gauge-fixed variables can fluctuate on all scales even when gauge-invariant observables cluster; if that happens, (D) fails in this gauge and has to be posed for block-averaged variables, where it becomes part of the renormalisation group problem.

Remark 47. The other routes we tried stop at (D) or (E), or earlier.

(a) *Disjoint blocks.* The block criterion of Lemma 50 with a partition into cubes of side n requires, for a link on a face, that its influence on the neighbouring cube be less than one. In the count of Remark 46 that influence is of order ξ, which grows like $1/a$. Averaging over the N_n translates divides it by the number of translates on whose boundary the link lies, a factor of order $1/n$, and this is why (D) is stated for averaged cubes.

(b) *Rescaling a fixed lattice gap.* Any bound uniform in lattice units, such as those of Theorems 1–7, gives trivial continuum two-point functions [14]; rescaling it produces no gap.

(c) *Assuming the gap.* Hypothesis R of [14] assumes the gap in physical units together with the limit. Theorem 45 replaces the gap part by the cube estimate (D), which concerns one box and one coupling at each spacing.

(d) *Computation.* At a fixed coupling and a fixed small n, $D(n,\alpha)$ is a finite set of Kantorovich bounds and could in principle be certified as in Propositions 34 and 35; for $n=2$ and $\mathrm{SU}(2)$ the block already has 48 links and 144 real variables. (D) asks for it at infinitely many couplings with $n\to\infty$, which no finite computation covers.

(e) *Axioms alone.* Proposition 41 shows that positivity, invariance and regularity do not force a gap, and Proposition 40 shows that the growth route to rotation invariance is closed for the action density. The infrared input therefore has to be an estimate such as (D).

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This paper rests on the work of many earlier scientists. Lattice gauge theory is due to Wilson [63] and its Hamiltonian form to Kogut and Susskind [43]; reflection positivity and the reconstruction of the transfer matrix to Osterwalder and Schrader [50, 51], and their lattice form, with the first proof of a gap at strong coupling, to Osterwalder and Seiler [52, 54]. The uniqueness method is Dobrushin's [26–28], the curvature method is that of Bakry and Émery [2, 3] in the form of Shen, Zhu and Zhu [55], the gauge fixing on a forest is Creutz's [24], and the curvature of Lie groups is Milnor's [48]. The analysis of the single-link law uses results of Barta [8], Biernacki and Krzyż [18], McShane [47], Stein and Weiss [58] and Watson [62], and the continuum problem is the one formulated by Jaffe and Witten [39] and studied by Bałaban [4–6]. The Yang–Mills manuscripts of Sussman and collaborators [59, 60] were read in the preparation of Sec. IX.

DATA AND CODE AVAILABILITY

The analytic proofs are complete as written, except the evaluations in Propositions 34 and 35 and Lemma 38 and the numerical bounds of Lemma 16, which are computations with rigorous error bounds, and the finite check of Lemma 12. The repository `creelie/4Y-M` contains the LaTeX source of this paper; the programs of that computation, in Python with python-flint (FLINT and Arb) and a C kernel for products of Taylor polynomials, with their outputs and the list of certified boxes; the programs and outputs of the earlier certificate at $\beta_W=\frac{45}{64}$; independent checks in exact rational arithmetic of Lemmas 20 and 28 and of (10); numerical checks of Lemma 19 on tori; the searches of Remark 26 and Sec. IX; and a Lean 4 certificate of the line field (Lemmas 8 and 9 for the table of Appendix A, by `decide`); and the programs of Appendix C for the plaquette blocks, with the floating-point search of Remark 18; and an exact computation of the curvatures of Table I from the root systems. This paper accompanies release v3.5.0 of the archive, https://github.com/creelie/4y-m/releases/tag/v3.5.0. The PDF of this paper is at https://github.com/creelie/4Y-M/blob/main/preprintYangMillsForestGauge/YangMillsForestGauge.pdf. The computations and the Lean, Python and shell checks were run with AI assistance (Claude Opus 5.5, Claude Fable 5.1 with Ultracode, and GPT Sol 6.1 Ultra) under the direction of Deep Bhattacharjee. The first composite behind Theorem 1 was assembled by the Anaxia engine of the BLLM system [11], release v3.0.5, as described in the footnote to that theorem; release v3.1.0 [12] records the engine's Yang–Mills runs in its failure log, whose last Yang–Mills entry is the change from single links to blocks of several links; the plaquette blocks of Sec. V, the bound of Lemma 15, their $\mathrm{SU}(3)$ version in Sec. VIII and the certified constants were found and checked by the authors. All three authors approved the final text.

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Appendix A: The line field

Table III lists, for each residue class of x modulo two, the direction $\sigma(x)$ and the permutation $\pi_x(\mu)=\sigma(x+e_\mu)$. In every row π_x fixes $\sigma(x)$ and is a three-cycle on the other directions, which is the condition of Lemma 8.

TABLE III. The line field: x modulo two (digits $x_0x_1x_2x_3$), the direction $\sigma(x)$ and the values $\sigma(x+e_\mu)$ for $\mu=0,1,2,3$.

x	$\sigma(x)$	$\mu=0$	$\mu=1$	$\mu=2$	$\mu=3$
0000	0	0	2	3	1
1000	0	0	3	1	2
0100	2	3	0	2	1
1100	3	2	0	1	3
0010	3	1	2	0	3
1010	1	3	1	0	2
0110	2	1	3	2	0
1110	1	2	1	3	0
0001	1	2	1	3	0
1001	2	1	3	2	0
0101	1	3	1	0	2
1101	3	1	2	0	3
0011	3	2	0	1	3
1011	2	3	0	2	1
0111	0	0	3	1	2
1111	0	0	2	3	1

Appendix B: The comparison lemma

Let (G,δ) be a compact group with a bi-invariant metric of diameter D, $\Omega = G^A$, and $C_{ee'}$ the coefficients of Sec. III, with row and column sums at most $\alpha < 1$ and $C_{ee'} = 0$ unless e, e' share a plaquette, so that $|e - e'| \le 1$ for the distance of link midpoints. Fix $\lambda > 0$ with $\theta = \alpha e^{\lambda} < 1$.

Lemma 48. *Let $S \subset A$, possibly infinite, and let μ_1, μ_2 be probability measures on Ω invariant under γ_e^F for every $e \notin S$. Let $T \subset A \setminus S$ be finite, $w_b = e^{-\lambda \operatorname{dist}(b,T)}$, and suppose $\sum_{z\in S} w_z < \infty$. There is a coupling of μ_1, μ_2 whose link distances $\rho(b) = \mathbb{E}\,\delta(\sigma_b, \sigma'_b)$ satisfy*

$$\sum_{b \notin S} w_b \rho(b) \le \frac{\theta D}{1-\theta} \sum_{z \in S} w_z .$$

If $S = \emptyset$ then $\mu_1 = \mu_2$.

Proof. The couplings form a weakly compact set and $\Psi(\pi) = \sum_b w_b \mathbb{E}_\pi \delta(\sigma_b, \sigma'_b)$ is weakly continuous because $\sum_b w_b < \infty$; let π^* be a minimiser. For $e \notin S$ replace the pair (σ_e, σ'_e) by a sample of an optimal coupling of $\gamma_e^F(\cdot \mid \sigma)$ and $\gamma_e^F(\cdot \mid \sigma')$, chosen measurably [61, Cor. 5.22]. Both μ_i are invariant, so the result is again a coupling, and changing the finitely many links that enter γ_e^F one at a time gives $W_1 \le \sum_{e'} C_{ee'} \delta(\sigma_{e'}, \sigma'_{e'})$. Minimality gives $\rho(e) \le \sum_{e'} C_{ee'} \rho(e')$. Multiply by w_e and sum over $e \notin S$; where $C_{ee'} \neq 0$, $w_e \le e^{\lambda} w_{e'}$, so

$$\sum_{e \notin S} w_e \rho(e) \le \theta \Big[\sum_{e' \notin S} w_{e'} \rho(e') + \sum_{z \in S} w_z \rho(z) \Big],$$

all sums are finite, and $\rho \le D$ on S. If $S = \emptyset$, $\rho \equiv 0$. □

Corollary 49. *Let μ be the Gibbs measure of γ^F, F a local function with support S_F, L_F-Lipschitz in each link, and H bounded with finite support S_H disjoint from S_F. Then*

$$|\operatorname{Cov}_\mu(F,H)| \le \|H\|_\infty L_F \frac{\theta D}{1-\theta} |S_H|\, e^{-\lambda \operatorname{dist}(S_F, S_H)} .$$

Proof. Condition on the links of S_H [42, Ch. 3]; the conditional measures $\mu^\sigma = \mu(\cdot \mid \sigma_{S_H})$ are invariant under γ_e^F for $e \notin S_H$. Lemma 48 with $S = S_H$ and $T = S_F$, for which $w_b = 1$ on S_F and $w_z \le e^{-\lambda \operatorname{dist}(S_F, S_H)}$ on S_H, gives $|\mu^\sigma(F) - \mu^{\sigma'}(F)| \le L_F \sum_{b \in S_F} \rho(b) \le L_F \frac{\theta D}{1-\theta} |S_H| e^{-\lambda \operatorname{dist}(S_F,S_H)}$ for any two boundary values. Since H depends only on σ_{S_H}, $\operatorname{Cov}_\mu(F,H) = \iint H(\sigma)\big(\mu^\sigma(F) - \mu^{\sigma'}(F)\big)\, d\mu(\sigma)\, d\mu(\sigma')$, and the bound follows. □

Lemma 50. *Let the blocks partition A, let C_{Bj} be the block coefficients of Sec. V, with $\sum_B C_{Bj} \le \alpha < 1$ for every j and $C_{Bj} = 0$ unless every link of B lies within distance 2 of j, and fix $\lambda > 0$ with $\theta = \alpha e^{2\lambda} < 1$. Let $S \subset A$ be a union of blocks and let μ_1, μ_2 be invariant under $\gamma_B^{\mathscr{B}}$ for every block $B \not\subset S$. With T, w_b as in Lemma 48 and $\sum_{z\in S} w_z < \infty$, there is a coupling with $\sum_{b \notin S} w_b \rho(b) \le \frac{\theta D}{1-\theta} \sum_{z \in S} w_z$, and $\mu_1 = \mu_2$ if $S = \emptyset$.*

Proof. Take a minimiser π^* of Ψ as before. For a block $B \not\subset S$ replace (σ_B, σ'_B) by an optimal coupling, for the metric d, of the two block laws; it is again a coupling, and changing the boundary links one at a time bounds the expected new d-distance by $\sum_j C_{Bj} \rho(j)$. Put $\bar w_B = \max_{i \in B} w_i$. Minimality gives

$$\sum_{i \in B} w_i \rho(i) \le \bar w_B \sum_j C_{Bj} \rho(j),$$

and $\bar w_B \le e^{2\lambda} w_j$ whenever $C_{Bj} \neq 0$. Summing over the blocks outside S and using the column sums,

$$\sum_{b \notin S} w_b \rho(b) \le \theta \Big[\sum_{j \notin S} w_j \rho(j) + \sum_{z \in S} w_z \rho(z) \Big],$$

which gives the claim as in Lemma 48. □

Lemma 51. *Let $n \ge 2$ be even, let V_x, $x \in 2\mathbb{Z}^4$, be the translates of Sec. X, and suppose that $D(n,\alpha)$ holds.*

(a) *If a probability measure μ on G^A is invariant under γ_e^F for every e in a finite set V, it is invariant under γ_V.*

(b) *Fix $\lambda > 0$ with $\theta = \alpha e^{\lambda \rho_n} < 1$. Let $S \subset A$, and let μ_1, μ_2 be invariant under γ_{V_x} for every x with $V_x \cap S = \emptyset$. With T, w_b as in Lemma 48 and $\sum_{z\in S} w_z < \infty$, there is a coupling with*

$$\sum_{b \notin S} w_b \rho(b) \le \frac{D(1 + c_n e^{\lambda \rho_n})}{1-\theta} \sum_{z \in S} w_z ,$$

and $\mu_1 = \mu_2$ if $S = \emptyset$.

Proof. (a) Let ν_η be the conditional law of Q_V given the links η outside V. For bounded f on G^V and bounded g of η, $\gamma_e^F(fg) = g\,\gamma_e^F f$ for $e \in V$, so invariance gives $\int g(\eta)\big(\nu_\eta(\gamma_e^F f) - \nu_\eta(f)\big)\, d\mu = 0$ for all g; taking f in a countable dense set of continuous functions, ν_η is invariant under every $\gamma_e^F(\cdot \mid \cdot, \eta)$, $e \in V$, for μ-almost every η. So is $\gamma_V(\cdot \mid \eta)$, whose conditional laws of single links are the γ_e^F. The composition P_η of the kernels γ_e^F, $e \in V$, in a fixed order has a density with respect to Haar measure on G^V bounded below by a positive constant, because each single-link density is at least $e^{-12N^2\beta}$ times its maximum over the group. A Markov kernel with such a lower bound has at most one invariant probability measure (Doeblin's condition), so $\nu_\eta = \gamma_V(\cdot \mid \eta)$.

(b) Let π^* minimise Ψ as in Lemma 48 and call x admissible if $V_x \cap S = \emptyset$. For admissible x replace $(\sigma_{V_x}, \sigma'_{V_x})$ by an optimal coupling, for d_{V_x}, of the two block laws; both μ_i are invariant, so the result is a coupling, and changing the boundary links one at a time bounds the new expected d_{V_x}-distance by $\sum_j C_{V_x,j} \rho(j)$. With $\bar w_x = \max_{i \in V_x} w_i$, minimality gives

$$\sum_{i \in V_x} w_i \rho(i) \le \bar w_x \sum_{j \notin V_x} C_{V_x,j} \rho(j),$$

and $\bar w_x \le e^{\lambda \rho_n} w_j$ whenever $C_{V_x,j} \neq 0$. Let S' be the set of links outside S that lie in a translate meeting S; each is within distance ρ_n of a link of S, so $\sum_{i \in S'} w_i \le c_n e^{\lambda \rho_n} \sum_{z \in S} w_z$. A link

outside $S\cup S'$ lies in N_n translates, all admissible. Summing over admissible x and using $\sum_{x:\, j\notin V_x} C_{V_x,j} = K_{c(j)} \le \alpha N_n$,

$$N_n \sum_{i\notin S\cup S'} w_i\rho(i) \le \theta N_n\Big[\sum_{i\notin S\cup S'} w_i\rho(i) + \sum_{i\in S\cup S'} w_i\rho(i)\Big].$$

All sums are finite and $\rho \le D$, so $\sum_{i\notin S\cup S'} w_i\rho(i) \le \frac{\theta D}{1-\theta}(1 + c_n e^{\lambda\rho_n})\sum_{z\in S} w_z$; adding $\sum_{i\in S'} w_i\rho(i) \le D c_n e^{\lambda\rho_n}\sum_{z\in S} w_z$ gives the bound. If $S=\emptyset$ then $S'=\emptyset$ and $\rho\equiv 0$. □

The covariance bound of Corollary 49 follows from part (b) in the same way, with $\theta D/(1-\theta)$ replaced by $D(1 + c_n e^{\lambda\rho_n})/(1-\theta)$, since the conditional measures μ^σ there are invariant under γ_{V_x} for $V_x \cap S_H = \emptyset$ by part (a).

Appendix C: The plaquette blocks

Table IV lists the block plaquettes of Sec. V modulo $2\mathbb{Z}^4$: the plaquette at x in the plane $\{\mu,\nu\}$ is a block plaquette exactly when x modulo two and $\{\mu,\nu\}$ appear in the table. Eight lie in the plane $\{0,1\}$ and eight in $\{2,3\}$; together they contain the 48 active links of a cell of side two, each once.

TABLE IV. The sixteen block plaquettes modulo two: the base point x (digits $x_0x_1x_2x_3$ modulo two) and the plane.

x	plane	x	plane
0000	$\{0,1\}$	0000	$\{2,3\}$
0001	$\{0,1\}$	0011	$\{2,3\}$
0010	$\{0,1\}$	0100	$\{2,3\}$
0011	$\{0,1\}$	0111	$\{2,3\}$
1100	$\{0,1\}$	1000	$\{2,3\}$
1101	$\{0,1\}$	1011	$\{2,3\}$
1110	$\{0,1\}$	1100	$\{2,3\}$
1111	$\{0,1\}$	1111	$\{2,3\}$

The program `cover.py` in the folder `strong-coupling/gauge-fixed/blocks/` of the archive finds this set by an exact-cover search on the torus of side two and checks Lemma 12 on the tori of side four and six. The program `constants.py` evaluates the bounds of Lemma 16 in ball arithmetic with python-flint at 120 bits: the series of Z_0 and of its first three derivatives in $s=r^2$ is summed to forty terms, and the tail is bounded for $s\le 5$ by twice the first omitted term $(5/4)^{40}40^3/(40!\,41!)$, because consecutive terms decrease by a factor below one half. The program `alpha.py` evaluates the constants of (3) and α at a given β_W, read exactly.

For SU(3) the folder `strong-coupling/gauge-fixed/blocks/su3/` contains `consts.py`, which evaluates the bounds of Lemma 38 on the 841 boxes of side $\frac{3}{16}$ at $r=\frac{3}{2}$ and the 784 boxes of side $\frac{19}{100}$ at $r=\frac{38}{25}$ with the moments of Appendix E, and `alpha3.py`, which reads these bounds and the cover of Proposition 34 or Proposition 35 and evaluates the constants of (15) and α_3 in ball arithmetic at 120 bits; its outputs at $\beta_W=\frac{3}{4}$ and 0.76 are listed with the bounds of each box. The cover of Proposition 35 is computed by the program of Appendix E with `--r 38/25 --thr 1.03`.

Appendix D: The blocks of Lemma 20

Diagonal blocks. Let $w_a = e^{i\phi}$. On the diagonal part, $f_p = -\cos\phi\,(y_1+y_2-y_3-y_4)^2$ and $c_p = 2(y_1y_4 - y_1y_2 + y_2y_3 - y_3y_4)$, so

$$\mathscr{F} = \sum_i \alpha_i y_i^2 + \cos\phi\,(y_1+y_2-y_3-y_4)^2 - \tfrac{9}{5}(y_1y_4 - y_1y_2 + y_2y_3 - y_3y_4)$$

with $y_k = 0$ and weights $\alpha_{k\pm1} = \frac{21}{10}$, $\alpha_{k+2} = \frac{5}{4}$. It is affine in $\cos\phi$, so positivity at $\cos\phi = \pm1$ suffices. For $k=1$ and $\cos\phi = 1$ the matrix in (y_2,y_3,y_4) is

$$\begin{pmatrix} \frac{31}{10} & -\frac{19}{10} & -1 \\ -\frac{19}{10} & \frac{9}{4} & \frac{19}{10} \\ -1 & \frac{19}{10} & \frac{31}{10} \end{pmatrix},$$

and Table V lists the leading principal minors for every position; all are positive.

TABLE V. Leading principal minors of the diagonal blocks of Lemma 20.

Fixed link	$\cos\phi = 1$	$\cos\phi = -1$
e_1, e_4	$\frac{31}{10}$, $\frac{673}{200}$, $\frac{8421}{2000}$	$\frac{11}{10}$, $\frac{53}{200}$, $\frac{21}{2000}$
e_2	$\frac{31}{10}$, $\frac{861}{100}$, $\frac{8421}{2000}$	$\frac{11}{10}$, $\frac{21}{100}$, $\frac{21}{2000}$
e_3	$\frac{9}{4}$, $\frac{673}{200}$, $\frac{8421}{2000}$	$\frac{1}{4}$, $\frac{53}{200}$, $\frac{21}{2000}$

Pair blocks. Let $w_a = e^{i(\chi+\vartheta)}$ and $w_b = e^{i(\chi-\vartheta)}$. The terms of f_p carry $w_a + \bar w_b = 2\cos\chi\, e^{i\vartheta}$ or its conjugate, and c_p carries $w_a\bar w_b = e^{2i\vartheta}$, so the Hermitian matrix $M(m,\vartheta)$ of $\mathscr{F}$, $m=\cos\chi$, is affine in m with $M(-1,\vartheta) = M(1,\vartheta+\pi)$, and $m=1$ suffices. For the fixed link e_2, in the variables (z_1,z_3,z_4) and with $c=\cos\vartheta$,

$$M = \begin{pmatrix} \frac{21}{5}+2c & -2e^{i\vartheta} & -\frac{9}{5}-2e^{i\vartheta} \\ -2e^{-i\vartheta} & \frac{21}{5}+2c & \frac{9}{5}+2e^{-i\vartheta} \\ -\frac{9}{5}-2e^{-i\vartheta} & \frac{9}{5}+2e^{i\vartheta} & \frac{5}{2}+2c \end{pmatrix}.$$

With $u=\cos^2(\vartheta/2)\in[0,1]$ its leading principal minors are

$$\frac{20u+11}{5},\quad \frac{(20u+1)(20u+21)}{25},\quad \frac{400u(40u^2-23u+4)+21}{250},$$

and $40u^2-23u+4$ has negative discriminant $529-640$. The other positions give the same determinant and the leading minors $\frac{20u+11}{5}$ or $\frac{8u+1}{2}$ and $\frac{800u^2-180u+53}{50}$, whose discriminant $32400-169600$ is negative. Every block is positive definite.

The inequality is not tight. In floating point the least eigenvalue of the whole form over $W\in \mathrm{U}(N)$, $N\le 4$, is 0.0149, which is also the least eigenvalue of the exact blocks; this slack is what the choice of the weights $\frac{21}{10}$, $\frac{5}{4}$, $\frac{9}{10}$ trades against the constant $\frac{29}{2}$.

Appendix E: The certified computation

Truncation. For a corner b of a box the moments of Lemma 32 are derivatives of Z at $(b,\bar b)$. The truncated series is $\mathbb{E}_H[A_t(UD)B_t(\bar U D')]$ with A_t the part of $\sum_\lambda c_\lambda s_\lambda(UD)$ over partitions with $\lambda_1-\lambda_3\le 34$ and $\lambda_3\le 11$, and B_t likewise. Every omitted partition has $|\lambda|>34$, because $3\cdot 12>34$, and the omitted part is at most $T(\|D\|)=\sum_{n>34}(\frac{3}{2}\|D\|)^n/n!$ in modulus. On the polydisc $|D_{ij}-b_{ij}|\le\rho$, $\|D\|\le\|b\|+3\rho$, and Cauchy's estimate bounds the α-th derivative of the omitted part by $\alpha!\rho^{-|\alpha|}T(\|b\|+3\rho)$. The derivatives of the full exponential are at most $2^{-|\alpha|}e^{\nu/2}$ with $\nu=\sum_k|b_k|$, since $|\operatorname{Tr}(Ub)|\le\sum_k|b_k|$ for diagonal b. With $\varepsilon_\alpha=2^{|\alpha|}\alpha!\rho^{-|\alpha|}T(\|b\|+3\rho)$ the truncated moment differs from the true one by at most $\varepsilon_\alpha e^{\nu/2}+(e^{\nu/2}+\varepsilon_\alpha)\varepsilon_\beta$, and ρ is chosen to minimise $\rho^{-6}T$. The largest error bound of an unnormalised moment at any corner is 1.7×10^{-11}.

Weight zero. Conjugation by the diagonal torus T preserves Haar measure and $\operatorname{Re}\operatorname{Tr}(Ub)$ for diagonal b, and multiplies u_{ij} by $t_it_j^{-1}$; a moment of a monomial on which T acts by a nontrivial character is therefore 0, and only 8278 of the 134596 monomials of degree at most six have the trivial character. The functions $\operatorname{Tr}(X)$, $e_2(X)$, $\det X$ of $X=DD'^{\top}$ and $\det D$, $\det D'$ are invariant under the same action on the variables, so after these are formed in the full ring of Taylor polynomials of degree six in the 18 variables, the series is computed in the subring of weight-zero monomials.

Rounding. The truncated series is a Taylor polynomial of total degree six in the 36 real directions of $(D-b,D'-\bar b)$, computed in double precision. Every operation is repeated on absolute values and carries an operation count n, so that the computed coefficient differs from the exact one by at most $\gamma_n=nu/(1-nu)$, $u=2^{-53}$, times the absolute-value coefficient [35, Sec. 3.1]. With C in $2^{-30}\mathbb{Z}$, the polynomials $2^{31}q_h$, $X_a(2^{31}q_h)$, $12\cdot2^{31}(\varphi_h+\Delta q_h)$ and $12\cdot2^{31}\langle\nabla\varphi_{F_k},\nabla q_h\rangle$ for the coordinate vectors h have coefficients in $\mathbb{Z}[i]$, and every partial sum in their computation is bounded, through the same operations on absolute values, by less than 2^{50}; they are therefore computed exactly. Normalisation by Z, the products with the corner coordinates, the quadratic forms and their error matrices are formed with the same bookkeeping, and the largest eigenvalues are bounded in ball arithmetic [40].

Separation. The functional c of a discarded box is the dual solution of a linear program for the gauge of the centre of the box with respect to a 72×72 grid of T. Only the final inequality is used: $\max_T\ell$ is bounded by its maximum over a 512×512 grid plus $\frac{1}{2}H\delta^2/2$, with $\delta=2\pi/512$ and $H=|c_1|+|c_2|+2|c_3|$ a bound for its Hessian on the torus, and the inequality is checked with relative margin 10^{-12}.

Running time. The moments at one corner take about one and a half seconds and the bound on one box about two and a half seconds on one core. The cover runs in about an hour on four cores and writes every certified box with its θ and its bound.

Appendix F: The transfer matrix

This appendix recalls the Osterwalder–Schrader reconstruction for the lattice [50–52] in the form used for the gaps, following our earlier paper [13].

Let ω be a state on the gauge-invariant local functions of $\mathbb{Z}^4$ that is invariant under translations and the hypercubic group and reflection positive for reflections in planes through sites and in planes bisecting links; by Proposition 11(iii)–(iv) and its analogues in the other theorems, the states of this paper are of this kind. The reconstruction gives a Hilbert space with a vector Ω, a map $F\mapsto\Psi_F$ with dense range, and a self-adjoint $\mathbb{T}$ with $0\le\mathbb{T}\le1$, $\mathbb{T}\Omega=\Omega$ and $\langle\Psi_F,\mathbb{T}^t\Psi_H\rangle=\omega(\vartheta(\bar F)\tau_tH)$, where ϑ is the time reflection and τ_t the time translation.[12]

Proposition 52. *For $m>0$ the following are equivalent:*

(a) *the spectrum of $\mathbb{T}$ on $\Omega^\perp$ lies in $[0,e^{-m}]$;*

(b) *for every F in a class of gauge-invariant local functions supported at times in a bounded interval and dense in $L^2(\omega)$ on each such support, there is $C_F<\infty$ with $0\le\omega(\vartheta(\bar F)\tau_tF)-|\omega(F)|^2\le C_Fe^{-mt}$ for all $t\ge0$.*

Under (a) *one may take $C_F=\|F\|_\infty^2$ for every bounded F.*

Proof. If (a) holds, then

$$\omega(\vartheta(\bar F)\tau_tF)-|\omega(F)|^2=\langle\Psi'_F,\mathbb{T}^t\Psi'_F\rangle$$

with $\Psi'_F=\Psi_F-\langle\Omega,\Psi_F\rangle\Omega\in\Omega^\perp$; this number lies in $[0,e^{-mt}\|\Psi_F\|^2]$, and $\|\Psi_F\|^2\le\|F\|_\infty^2$. Conversely, let σ be the spectral measure of $\mathbb{T}$ at Ψ'_F and let $\delta>0$. Then

$$\sigma\big((e^{-m+\delta},1]\big)e^{(-m+\delta)t}\le C_Fe^{-mt}$$

for all t, so $\sigma((e^{-m+\delta},1])=0$. The spectral projection of $(e^{-m},1]$ restricted to $\Omega^\perp$ therefore annihilates the vectors Ψ'_F of the class, which are dense by the footnote above, and it vanishes. □

Appendix G: The single-link law of SU(2)

This appendix proves the sharp quarter, $W_1(\nu_b,\nu_{b'})\le\frac{1}{4}|b-b'|$ for the one-link law of SU(2), which Theorems 1 and 3 use, and the exact value at $b=0$, which Theorem 2 uses. The argument is that of our earlier paper [13]; it is given in full so that the present paper is self-contained.

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[12] Ψ_F is the class of F in the completion of the positive-time local functions for the form $\langle F,H\rangle=\omega(\vartheta(\bar F)H)$, which is positive semidefinite by reflection positivity. Since $|\omega(\vartheta(\bar F)F)|\le\omega(|F|^2)$ by the Cauchy–Schwarz inequality and the invariance of ω under ϑ, functions that are close in $L^2(\omega)$ give close vectors.

1. The von Mises–Fisher law

Identify $\mathrm{SU}(2)$ with the unit quaternions, that is with the unit sphere $S^3 \subset \mathbb{R}^4$, so that $\mathrm{Re}\,\mathrm{Tr}\,U = 2x_0(U)$ and $\mathrm{Re}\,\mathrm{Tr}(UV) = 2\langle u, \bar{v}\rangle$ for the Euclidean inner product of $\mathbb{R}^4$ and quaternion conjugation $v \mapsto \bar{v}$. Fix a link e and the field off e. Every plaquette through e can be rotated cyclically so that its trace reads $\mathrm{Re}\,\mathrm{Tr}(Q_e A_p)$ with $A_p \in \mathrm{SU}(2)$, the staple of e in p. The action therefore depends on Q_e only through

$$N\beta \sum_{p \ni e} \mathrm{Re}\,\mathrm{Tr}(Q_e A_p) = \langle x, b\rangle, \qquad b = 4\beta \sum_{p \ni e} \bar{A}_p \in \mathbb{R}^4,$$

with $x = Q_e \in S^3$. Since there are six staples, each of norm one,

$$\kappa := |b| \le y := 24|\beta|. \tag{G1}$$

The conditional law of Q_e is the von Mises–Fisher law[13]

$$\nu_b(dx) = Z(\kappa)^{-1} e^{\langle x, b\rangle}\, \sigma(dx) \tag{G2}$$

on the unit sphere, with σ the normalised surface measure. We use coordinates in which $b = \kappa e_1$. The density of $t = x_1$ is proportional to $(1-t^2)^{1/2} e^{\kappa t}$ on $[-1,1]$, so $Z(\kappa) = 2I_1(\kappa)/\kappa$ with the modified Bessel function I_1, normalised so that $Z(0) = 1$.

Lemma 53. *For every $\kappa \ge 0$ the law ν_b on the unit sphere S^3 satisfies the Poincaré inequality $\mathrm{Var}_{\nu_b}(g) \le \frac{1}{3}\int |\nabla g|^2 d\nu_b$, and the constant $\frac{1}{3}$ is attained at $\kappa = 0$.*

Proof. In the polar angle $\theta \in (0,\pi)$ from e_1 the generator of the reversible diffusion of ν_b is $L = \partial_\theta^2 + b(\theta)\partial_\theta + \sin^{-2}\theta\, \Delta_{S^2}$ with $b(\theta) = 2\cot\theta - \kappa\sin\theta$. On spherical harmonics of degree ℓ on S^2 it acts as a Sturm–Liouville operator L_ℓ with potential $\ell(\ell+1)\sin^{-2}\theta$. A ground state comparison of Barta's type [8] with the trial function $\psi = \sin\theta\, e^{a\kappa\cos\theta}$ gives $-L_1\psi \ge 3\psi$ for $a = -\frac{1}{5}$, and the same trial function with $a = -\frac{2}{5}$ gives the bound 3 for the derivatives of zonal eigenfunctions; Appendix H carries out both computations. Hence every nonzero eigenvalue of $-L$ is at least 3. At $\kappa = 0$ the first eigenvalue of the Laplacian of the unit S^3 is 3. □

2. Moments

Let $A(\kappa) = \mathbb{E}_{\nu_b} x_1 = Z'(\kappa)/Z(\kappa) = I_2(\kappa)/I_1(\kappa)$.

Lemma 54. *For every $\kappa > 0$, in coordinates with $b = \kappa e_1$,*

(i) $\mathbb{E}x_2^2 = \mathbb{E}x_3^2 = \mathbb{E}x_4^2 = A/\kappa$, $\mathbb{E}(1-x_1^2) = 3A/\kappa$, *the covariance matrix of x is* $\mathrm{diag}(A', A/\kappa, A/\kappa, A/\kappa)$, *and* $A' = 1 - A^2 - 3A/\kappa$;

(ii) $\dfrac{\kappa}{2+\sqrt{\kappa^2+4}} \le A(\kappa) \le \dfrac{\kappa}{1+\sqrt{\kappa^2+9}}$;

(iii) $A' \le A/\kappa \le \big(1+\sqrt{\kappa^2+9}\big)^{-1} \le \frac{1}{4}$;

(iv) $A(\kappa) \ge \kappa/\big(\frac{3}{2} + \sqrt{\kappa^2 + 25/4}\big)$.

Proof. (i) Integrating by parts against $\frac{d}{dt}(1-t^2)^{3/2} = -3t(1-t^2)^{1/2}$ gives $\kappa\mathbb{E}(1-x_1^2) = 3\mathbb{E}x_1$, and the three coordinates x_2, x_3, x_4 are exchangeable under ν_b, with $\sum_i x_i^2 = 1$. The off-diagonal covariances vanish by the reflections $x_i \mapsto -x_i$, $i \ge 2$, and $A' = \mathrm{Var}\, x_1$ because $A = (\log Z)'$. (ii) By (i), A solves the Riccati equation $A' = 1 - A^2 - 3A/\kappa$ with $A(\kappa) = \kappa/4 + O(\kappa^3)$. Appendix H shows that the left side of (ii) is a subsolution and the right side a supersolution of this equation, with the same behaviour at 0, and concludes by comparison. (iii) The middle inequality is (ii). For the first, put $s = \sqrt{\kappa^2+4}$. By (ii), $4A/\kappa \ge 4/(2+s)$ and $1 - A^2 \le 1 - \kappa^2/(2+s)^2$, and

$$1 - \frac{\kappa^2}{(2+s)^2} \le \frac{4}{2+s} \iff (2+s)^2 - \kappa^2 \le 4(2+s)$$
$$\iff s^2 - \kappa^2 \le 4,$$

which holds with equality. So $A' = 1 - A^2 - 3A/\kappa \le A/\kappa$. (iv) is the comparison of (ii) with a third barrier, which improves the lower bound of (ii); it is carried out in the same appendix. □

3. The Lipschitz rate of the family

Let δ be the chordal metric $\delta(x,x') = |x - x'|$ on S^3 and W_1 the Kantorovich distance for δ. For $c \in \mathbb{R}^4$ and a unit vector $h \in \mathbb{R}^4$ put

$$L(c,h) = \sup_g \mathrm{Cov}_{\nu_c}\big(g, \langle h, x\rangle\big), \tag{G3}$$

the supremum over 1-Lipschitz functions g for δ. Since $\frac{d}{dt}\nu_{c+th}(g) = \mathrm{Cov}_{\nu_{c+th}}(g, \langle h, x\rangle)$, the Kantorovich–Rubinstein duality gives, for all b, b',

$$W_1(\nu_b, \nu_{b'}) \le |b - b'| \sup_{c \in [b,b']} L\Big(c, \frac{b'-b}{|b'-b|}\Big). \tag{G4}$$

By rotation invariance $L(c,h)$ depends only on $\kappa = |c|$ and on the angle φ between h and $c^\perp$, and we write $L(\kappa,\varphi)$.

Lemma 55. *For all c and h, $L(c,h) \le 1/(2\sqrt{3})$. Moreover:*

(i) *at $c = 0$, $L(0,h) = \frac{1}{4}$ for every h;*

(ii) *if $h \perp c$, then $L(c,h) = \mathbb{E}_{\nu_c} x_2^2 = A(\kappa)/\kappa \le \frac{1}{4}$;*

(iii) *if $\kappa \ge \sqrt{88}/3 \approx 3.127$, then $L(c,h) \le \frac{1}{4}$ for every h.*

---

[13] On the sphere $S^{p-1} \subset \mathbb{R}^p$ the von Mises–Fisher law with mean direction μ and concentration κ has density proportional to $e^{\kappa\langle x,\mu\rangle}$ with respect to the surface measure, and its normalising constant is $(2\pi)^{p/2} I_{p/2-1}(\kappa)/\kappa^{p/2-1}$; it is the maximum entropy law with prescribed mean resultant, and it plays on the sphere the role of the Gaussian in $\mathbb{R}^p$. For $p = 4$ the Bessel index is 1. The mean resultant length $\mathbb{E}\langle x,\mu\rangle = I_{p/2}(\kappa)/I_{p/2-1}(\kappa)$ is the function A of Lemma 54. See Mardia and Jupp [46], Ch. 9, for these facts and for the tangent-normal decomposition used in Appendix H.

Proof. A 1-Lipschitz function for δ is 1-Lipschitz for the geodesic distance, which is larger, so $|\nabla g| \le 1$ almost everywhere, and Lemma 53 gives $\operatorname{Var} g \le \frac{1}{3}$. By Lemma 54(i), (iii) the largest eigenvalue of the covariance of x is $A/\kappa \le \frac{1}{4}$, so $\operatorname{Var}\langle h,x\rangle \le A/\kappa$. The Cauchy–Schwarz inequality gives

$$L(c,h) \le \left(\tfrac{1}{3}A/\kappa\right)^{1/2} \le \left(3\left(1+\sqrt{\kappa^2+9}\right)\right)^{-1/2} \le \frac{1}{2\sqrt{3}}. \tag{G5}$$

(i) and (ii) are reflection arguments. Let R be the reflection in the hyperplane $h^{\perp}$, and suppose ν_c is R-invariant, which is the case when $c=0$ or $h \perp c$. Then $\mathbb{E}\langle h,x\rangle = 0$ and

$$\begin{aligned}\operatorname{Cov}(g,\langle h,x\rangle) &= \mathbb{E}\big[(g(x)-g(Rx))\langle h,x\rangle;\ \langle h,x\rangle > 0\big] \\ &\le \mathbb{E}\big[2\langle h,x\rangle^2;\ \langle h,x\rangle > 0\big] = \mathbb{E}\langle h,x\rangle^2,\end{aligned}$$

since $\delta(x,Rx) = 2|\langle h,x\rangle|$. Equality holds for $g = \langle h,x\rangle$, which is 1-Lipschitz. At $c=0$, $\mathbb{E}\langle h,x\rangle^2 = \frac{1}{4}$; for $h \perp c$, $\mathbb{E}\langle h,x\rangle^2 = A/\kappa$. (iii) By (G5), $L \le \frac{1}{4}$ as soon as $1+\sqrt{\kappa^2+9} \ge \frac{16}{3}$, that is $\kappa^2 \ge \frac{88}{9}$. □

The reflection settles the directions $h \perp c$ and the point $c=0$; for an oblique h the reflection no longer preserves ν_c, and the Cauchy–Schwarz bound (G5) is too weak for $\kappa < \sqrt{88}/3$. The next theorem closes this case.

Theorem 56 (The sharp quarter)**.** *For every $c \in \mathbb{R}^4$ and every unit vector $h \in \mathbb{R}^4$,*

$$L(c,h) \le \tfrac{1}{4},$$

with equality at $c=0$. Equivalently, $W_1(\nu_b,\nu_{b'}) \le \frac{1}{4}|b-b'|$ for all $b,b' \in \mathbb{R}^4$, and the constant $\frac{1}{4}$ is the best possible.

The equivalence is (G4) in one direction and the derivative formula above it in the other; the sharpness is Lemma 55(i). The metric matters.[14] The proof has four steps. A Lipschitz function on S^3 extends to the unit ball $B \subset \mathbb{R}^4$, and Green's formula in B bounds the covariance by a negative Sobolev norm of the density (Lemma 57). This norm splits into a zonal and a transverse part, both explicit series in modified Bessel functions (Lemma 58). The transverse part is at most $\frac{1}{4}$ because a certain power series has nonnegative coefficients (Proposition 60), and the zonal part is at most $\frac{1}{4}$ by a Riccati comparison (Proposition 61).

4. Harmonic extension

Let σ be the normalised surface measure of S^3 and H_k the space of spherical harmonics of degree k, the restrictions to S^3 of harmonic homogeneous polynomials of degree k on $\mathbb{R}^4$. The spaces H_k are orthogonal in $L^2(\sigma)$ and span it; we write u_k for the component of u in H_k, and for u with $\int u\,d\sigma = 0$ we put

$$\mathscr{Q}(u) = \sum_{k\ge 1} \frac{1}{k}\,\|u_k\|^2_{L^2(\sigma)}. \tag{G6}$$

This is $\langle u, \Lambda^{-1}u\rangle$ for the Dirichlet-to-Neumann operator Λ of B, which acts on H_k as multiplication by k.

Lemma 57. *Let u be a real-analytic function on S^3 with $\int u\,d\sigma = 0$ and let g be 1-Lipschitz for the chordal metric. Then*

$$\int_{S^3} g\,u\,d\sigma \le \tfrac{1}{2}\,\mathscr{Q}(u)^{1/2}.$$

Proof. The chordal metric is the Euclidean metric of $\mathbb{R}^4$ restricted to S^3, so McShane's extension $G(y) = \inf_{x\in S^3}\big(g(x) + |x-y|\big)$ is 1-Lipschitz on $\mathbb{R}^4$ and equals g on S^3 [47]; by Rademacher's theorem $|\nabla G| \le 1$ almost everywhere. Let $\psi = \sum_{k\ge1} k^{-1} r^k u_k(\omega)$ in polar coordinates $y = r\omega$. Each term is a harmonic polynomial, the series converges with all derivatives on the closed ball because u is real-analytic,[15] and $\partial_r \psi = u$ on S^3. Green's formula for the Lipschitz function G and the smooth harmonic function ψ reads $\int_{S^3} G\,\partial_r\psi\,dS = \int_B \nabla G\cdot\nabla\psi\,dy$. With $|S^3| = 2\pi^2$ and $|B| = \pi^2/2$ the Cauchy–Schwarz inequality gives

$$\int g\,u\,d\sigma = \frac{1}{2\pi^2}\int_B \nabla G\cdot\nabla\psi\,dy \le \frac{|B|^{1/2}}{2\pi^2}\Big(\int_B |\nabla\psi|^2 dy\Big)^{1/2}.$$

Green's formula once more gives $\int_B |\nabla\psi|^2 = \int_{S^3}\psi\,\partial_r\psi\,dS = 2\pi^2\sum_k k^{-1}\|u_k\|^2 = 2\pi^2\mathscr{Q}(u)$, and the right side becomes $\frac{1}{2}\mathscr{Q}(u)^{1/2}$. □

The lemma is a form of Thomson's principle. By Kantorovich–Rubinstein duality, $\sup_g \int g u\,d\sigma$ is the least cost of a flow in $\mathbb{R}^4$ that carries the positive part of $u\,d\sigma$ to its negative part, the cost being the total length travelled by the mass.[16] The harmonic field $\nabla\psi$ is one such flow, through the interior of the ball, and the lemma computes its cost. At $c=0$ the flow is optimal: for $u = \langle h,x\rangle$ one has $\psi = \langle h,y\rangle$, the field $\nabla\psi = h$ is constant, the Cauchy–Schwarz inequality is an equality, and $\mathscr{Q}(u) = \|u\|^2 = \frac{1}{4}$.

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[14] In the geodesic metric of S^3 the statement is false already at $c=0$: the function $\arcsin x_2$ is geodesically 1-Lipschitz and its covariance with x_2 under the uniform law is $\frac{8}{9\pi} \approx 0.283 > \frac{1}{4}$. A proof therefore has to use that the chordal distance between two points is shorter than their geodesic distance, that is, it has to see the sphere from the ball it bounds. The proof below does exactly this: it extends a chordally Lipschitz function to the ball and integrates by parts there.

[15] For real-analytic u the norms $\|u_k\|$ decay geometrically in k, and the supremum of a function in H_k is at most $(\dim H_k)^{1/2} = k+1$ times its $L^2(\sigma)$ norm, so the series for ψ and for each of its derivatives converge uniformly on $\bar B$.

[16] This is the Beckmann formulation of the transport problem with cost $|x-y|$: the optimal cost equals the least $\int |V|$ over vector measures V on $\mathbb{R}^4$ with divergence $u\,d\sigma$. Restricting the flow to the closed ball does not change the infimum, since projecting onto $\bar B$ shortens paths. The gradient $\nabla\psi$ on B is one admissible flow. It minimises the energy $\int |V|^2$ rather than the cost $\int |V|$, and the Cauchy–Schwarz inequality converts one into the other at the price of the factor $|B|^{1/2}$.

5. Spectral decomposition

Fix $c=\kappa e_1$ with $\kappa>0$ and write $h=\cos\varphi\, e_1+\sin\varphi\, e_2$, which is no loss of generality by rotation invariance. Let $\rho=d\nu_c/d\sigma=e^{\kappa x_1}/Z(\kappa)$, $Z(\kappa)=\int e^{\kappa x_1}d\sigma=2I_1(\kappa)/\kappa$, and

$$u=\big(\langle h,x\rangle-\mathbb{E}_{\nu_c}\langle h,x\rangle\big)\rho,$$

so that $\mathrm{Cov}_{\nu_c}(g,\langle h,x\rangle)=\int g u\,d\sigma$ and $\int u\,d\sigma=0$. Let U_n be the Chebyshev polynomials of the second kind and put

$$d_n(\kappa)=\frac{(n+1)\,I_{n+1}(\kappa)}{I_1(\kappa)},\qquad n\ge 0. \tag{G7}$$

Lemma 58. *With this notation*

$$\mathscr{Q}(u)=\cos^2\varphi\,\mathscr{Q}_z(\kappa)+\sin^2\varphi\,\mathscr{Q}_x(\kappa),$$

where

$$\mathscr{Q}_z=\sum_{n\ge1}\frac{d_n'(\kappa)^2}{n},\qquad \mathscr{Q}_x=\frac{1}{3\kappa^2}\sum_{n\ge1}(n+2)\,d_n(\kappa)^2.$$

Proof. The functions $U_n(x_1)$, $n\ge0$, are the zonal harmonics: $U_n(x_1)\in H_n$, and they are orthonormal in $L^2(\sigma)$ (Appendix I). The expansion $e^{\kappa t}=\frac{2}{\kappa}\sum_n(n+1)I_{n+1}(\kappa)U_n(t)$ of the same appendix gives $\rho=\sum_n d_n(\kappa)U_n(x_1)$, with $d_0=1$.

Write $u=\cos\varphi\,u_z+\sin\varphi\,u_x$ with $u_z=(x_1-A)\rho$ and $u_x=x_2\rho$; here $\mathbb{E}_{\nu_c}x_1=A$ and $\mathbb{E}_{\nu_c}x_2=0$. The reflection $x_2\mapsto-x_2$ maps each H_k to itself, fixes u_z and reverses u_x, so the components of u_z and u_x in each H_k are orthogonal and the cross term in $\mathscr{Q}(u)$ vanishes.

Since $A=Z'/Z$, we have $u_z=\partial_\kappa\rho=\sum_n d_n'(\kappa)U_n(x_1)$, and the component of u_z in H_n is $d_n'U_n(x_1)$, of norm $|d_n'|$. This gives $\mathscr{Q}_z$.

For u_x, let $y(t)=\cos t\,e_1+\sin t\,e_2$. The law $\nu_{\kappa y(t)}$ has density $\rho_t=e^{\kappa\langle y(t),x\rangle}/Z(\kappa)=\sum_n d_n(\kappa)U_n(\langle y(t),x\rangle)$, and $\partial_t\rho_t|_{t=0}=\kappa x_2\rho$. The function $U_n(\langle y(t),x\rangle)$ lies in H_n for every t, hence so does its derivative $x_2U_n'(x_1)$ at $t=0$, and the component of u_x in H_n is $\kappa^{-1}d_n x_2U_n'(x_1)$. Its norm follows from the eigenvalue $n(n+2)$ of the Laplacian of S^3 on H_n. For $f=U_n(x_1)$,

$$\begin{aligned}n(n+2)=n(n+2)\|f\|^2&=\int|\nabla_{S^3}f|^2d\sigma\\&=\int U_n'(x_1)^2(1-x_1^2)\,d\sigma,\end{aligned}$$

since the tangential gradient of x_1 has squared length $1-x_1^2$. Given x_1, the point (x_2,x_3,x_4) is uniform on the sphere of radius $(1-x_1^2)^{1/2}$, so $\mathbb{E}[x_2^2\mid x_1]=\frac13(1-x_1^2)$ and $\|x_2U_n'(x_1)\|^2=\frac13 n(n+2)$. Summing $n^{-1}\kappa^{-2}d_n^2\cdot\frac13 n(n+2)$ over $n\ge1$ gives $\mathscr{Q}_x$. □

By Lemmas 57 and 58,

$$L(\kappa,\varphi)\le\tfrac12\max\big(\mathscr{Q}_z(\kappa),\mathscr{Q}_x(\kappa)\big)^{1/2}, \tag{G8}$$

and Theorem 56 follows once both parts are at most $\frac14$. Both are $\frac14$ at $\kappa=0$, by continuity from the values $d_1\sim\kappa/2$, $d_1'\to\frac12$ and $d_n=O(\kappa^n)$.

6. Three Bessel sums

The series of Lemma 58 reduce to closed forms. Write I_j for $I_j(\kappa)$.

Lemma 59. *For every $\kappa>0$,*

$$\sum_{j\ge1}j^2I_j^2=\frac{\kappa}{4}I_1(2\kappa),\qquad \sum_{j\ge1}j^3I_j^2=\frac{\kappa^2}{4}\big(I_0^2+I_1^2\big),$$
$$I_1(2\kappa)=2\sum_{j\ge1}I_{j-1}I_j.$$

The first and third identities follow from Parseval's formula for $e^{\kappa\cos\theta}=\sum_{j\in\mathbb{Z}}I_je^{ij\theta}$, and the second from the recurrences of I_j by telescoping; Appendix I gives the proofs.

7. The transverse part

Proposition 60. *$\mathscr{Q}_x(\kappa)\le\frac14$ for every $\kappa\ge0$.*

Proof. By (G7), $\mathscr{Q}_x=(3\kappa^2I_1^2)^{-1}\sum_{j\ge2}j^2(j+1)I_j^2$, and the sum equals $\sum_{j\ge1}(j^3+j^2)I_j^2-2I_1^2$. By Lemma 59, $\mathscr{Q}_x\le\frac14$ is equivalent to

$$\mathscr{D}(\kappa):=(8+2\kappa^2)I_1^2-\kappa^2I_0^2-\kappa I_1(2\kappa)\ge0. \tag{G9}$$

Each term is an entire function of κ that is even after the factor κ^2 is removed, so $\mathscr{D}(\kappa)=\sum_{n\ge0}\mathscr{D}_n\kappa^{2n}$. Lemma 63 of Appendix I computes the coefficients: $\mathscr{D}_0=\mathscr{D}_1=0$, and for $n=m+1\ge2$

$$\mathscr{D}_n=\frac{N_m}{4^m\,m!\,(m+1)!\,(m+1)(m+2)},$$

with

$$\begin{aligned}N_m=\binom{2m}{m}\big(4(2m+1)+(m-1)(m+1)(m+2)\big)\\-4^m(m+1)(m+2).\end{aligned}$$

The numerator N_m vanishes for $m=0,1,2$ and is positive for $m\ge3$ (Lemma 63). So every coefficient of $\mathscr{D}$ is nonnegative and $\mathscr{D}\ge0$. □

The first nonzero coefficient is $\mathscr{D}_4=\frac{1}{2304}$, so $\frac14-\mathscr{Q}_x=\mathscr{D}/(12\kappa^2I_1^2)$ is of order κ^4 at 0: the transverse part leaves the value $\frac14$ very slowly, which is why no estimate that loses a constant can prove the proposition.

8. The zonal part

Let $A=I_2/I_1$ as before, so that $d_1=2A$ and, by Lemma 54(i), $A'=1-A^2-3A/\kappa$.

Proposition 61. *For $0<\kappa\le\sqrt{88}/3$,*

$$\mathscr{Q}_z(\kappa)\le\frac14+4A'^2+\frac{A^2}{4}-\frac{A}{2\kappa}-\frac{2A^2}{\kappa^2}<\frac14.$$

Proof. The higher modes. With $r_\nu = I_{\nu+1}/I_\nu$ the recurrence $I'_\nu = I_{\nu+1} + \frac{\nu}{\kappa} I_\nu$ gives

$$\frac{d'_n}{d_n} = \frac{I'_{n+1}}{I_{n+1}} - \frac{I'_1}{I_1} = \frac{n}{\kappa} + r_{n+1} - r_1.$$

By Lemma 64 of Appendix I, $r_{n+1} \le r_1$, and $r_1 = A \le \kappa/4$ by Lemma 54(iii). Hence $-n/\kappa \le n/\kappa - \kappa/4 \le d'_n/d_n \le n/\kappa$ for $n \ge 2$ and $\kappa^2 \le 16$, so $d'^2_n \le n^2 d_n^2/\kappa^2$. With $d'_1 = 2A'$,

$$\mathscr{Q}_z \le 4A'^2 + \frac{1}{\kappa^2}\sum_{n\ge 2} n\, d_n^2 = 4A'^2 + \frac{S_3 - S_2 - 4I_2^2}{\kappa^2 I_1^2},$$

where $S_2 = \sum_{j\ge1} j^2 I_j^2$ and $S_3 = \sum_{j\ge1} j^3 I_j^2$; the second form uses $n\, d_n^2 I_1^2 = (j-1) j^2 I_j^2$ with $j = n+1$.

The closed form. The recurrences $I_0 - I_2 = \frac{2}{\kappa} I_1$ and $I_1 - I_3 = \frac{4}{\kappa} I_2$ give $I_0/I_1 = 2/\kappa + A$ and $I_2 I_3/I_1^2 = A - 4A^2/\kappa$. By Lemma 59, $S_3/(\kappa^2 I_1^2) = \frac{1}{4}(2/\kappa + A)^2 + \frac{1}{4}$, and, keeping three terms of the third identity,

$$\frac{S_2}{\kappa^2 I_1^2} \ge \frac{I_0 I_1 + I_1 I_2 + I_2 I_3}{2\kappa I_1^2} = \frac{1}{2\kappa}\Big(\frac{2}{\kappa} + 3A - \frac{4A^2}{\kappa}\Big).$$

Collecting terms, $\mathscr{Q}_z \le \frac{1}{4} + 4A'^2 + T(A)$ with $T(a) = \frac{a^2}{4} - \frac{a}{2\kappa} - \frac{2a^2}{\kappa^2}$, which is the first inequality.

The barrier. Let $s = \sqrt{\kappa^2 + 25/4}$ and $A_\flat = \kappa/(\frac{3}{2} + s)$. By Lemma 54(iv), $A_\flat \le A \le \kappa/4$. The function $a \mapsto 1 - a^2 - 3a/\kappa$ decreases on $a \ge 0$, and $A' \ge 0$ because $A' = \operatorname{Var} x_1$, so $0 \le A' \le \alpha := 1 - A_\flat^2 - 3A_\flat/\kappa$. For $\kappa^2 \le 12$ the quadratic T decreases on $[0, \kappa/4]$, since

$$T(a) - T(a') = (a' - a)\,\frac{2\kappa - (a + a')(\kappa^2 - 8)}{4\kappa^2}$$

and $(a + a')(\kappa^2 - 8) \le \frac{\kappa}{2}(\kappa^2 - 8) \le 2\kappa$ there. So $\mathscr{Q}_z \le \frac{1}{4} + 4\alpha^2 + T(A_\flat)$. With $\kappa^2 = s^2 - \frac{25}{4}$ and $A_\flat/\kappa = 2/(2s+3)$ both terms are rational in s: $\alpha = 16/(2s+3)^2$, and

$$4\alpha^2 + T(A_\flat) = \frac{(2s-5)\,p(s)}{4(2s+3)^4}, \qquad p(s) = 8s^3 + 28s^2 - 98s - 695. \tag{G10}$$

For $0 < \kappa \le \sqrt{88}/3$ we have $\frac{5}{2} < s \le \sqrt{577}/6 < 4.004$. On $s \ge \frac{5}{2}$ the cubic p increases, since $p'(s) = 24s^2 + 56s - 98 > 0$, and $p(4.004) < -120$. So the right side of (G10) is negative. □

9. Proof of Theorem 56

At $c = 0$ the theorem is Lemma 55(i), and for $|c| \ge \sqrt{88}/3$ it is Lemma 55(iii). For $0 < |c| < \sqrt{88}/3$, Propositions 60 and 61 give $\mathscr{Q}_x, \mathscr{Q}_z \le \frac{1}{4}$, and (G8) gives $L \le \frac{1}{4}$. □

Appendix H: Computations for the single-link law

1. Barta's inequality

Let $\rho(\theta) = \sin^2\theta\, e^{\kappa\cos\theta}$ on $(0,\pi)$, let V be a continuous potential, and let

$$T = -\rho^{-1}\partial_\theta\, \rho\, \partial_\theta + V.$$

If $\psi > 0$ is smooth on $(0,\pi)$ with $\rho\psi^2 = O(\sin^4\theta)$ at the end points and $T\psi \ge \lambda_0\psi$, then

$$\int (|h'|^2 + Vh^2)\rho\, d\theta \ge \lambda_0 \int h^2 \rho\, d\theta$$

for every smooth h vanishing to first order at 0 and π. Indeed, writing $h = \psi u$ and integrating by parts, the boundary terms $\rho\psi\psi' u^2$ vanish and

$$\int (|h'|^2 + Vh^2)\rho = \int \rho\psi^2 |u'|^2 + \int \frac{T\psi}{\psi} h^2 \rho \;\ge\; \lambda_0 \int h^2 \rho.$$

This is the inequality of Barta [8] in the form of a ground state transformation.

2. The computations of Lemma 53

The generator of the reversible diffusion of ν_b on S^3 is $L = \Delta_{S^3} + \kappa \nabla x_1 \cdot \nabla$. In the polar angle θ from e_1 the round metric is $d\theta^2 + \sin^2\theta\, g_{S^2}$, so $\Delta_{S^3} = \partial_\theta^2 + 2\cot\theta\,\partial_\theta + \sin^{-2}\theta\,\Delta_{S^2}$, and $\nabla x_1 \cdot \nabla = -\sin\theta\,\partial_\theta$. On $h(\theta)Y_\ell$ with a spherical harmonic Y_ℓ of degree ℓ on S^2 the operator acts as $L_\ell = \partial_\theta^2 + b\,\partial_\theta - \ell(\ell+1)\sin^{-2}\theta$, $b = 2\cot\theta - \kappa\sin\theta = \rho'/\rho$, so $-L_\ell = T$ with $V = \ell(\ell+1)\sin^{-2}\theta$.

Degree $\ell \ge 1$. Take $\psi = \sin\theta\, e^{a\kappa\cos\theta}$. Then $\psi'/\psi = \cot\theta - a\kappa\sin\theta$ and $\psi''/\psi = (\psi'/\psi)^2 + (\psi'/\psi)'$, and a short computation gives

$$\frac{-L_1\psi}{\psi} = 3 + (1+5a)\kappa\cos\theta - a(1+a)\kappa^2\sin^2\theta.$$

For $a = -\frac{1}{5}$ this is $3 + \frac{4}{25}\kappa^2\sin^2\theta \ge 3$. Since $V = \ell(\ell+1)\sin^{-2}\theta \ge 2\sin^{-2}\theta$ for $\ell \ge 1$, Barta's inequality gives $-L_\ell \ge 3$ on every sector with $\ell \ge 1$.

Degree $\ell = 0$. Let $g(\theta)$ be a zonal eigenfunction, $-L_0 g = \lambda g$, g not constant, so that $g'(0) = g'(\pi) = 0$. Differentiating $g'' + bg' = -\lambda g$ gives, for $h = g'$,

$$-\big(h'' + bh'\big) - b'h = \lambda h, \qquad -b' = \frac{2}{\sin^2\theta} + \kappa\cos\theta,$$

so h is an eigenfunction of T with $V = 2\sin^{-2}\theta + \kappa\cos\theta$, with eigenvalue λ, and $h \not\equiv 0$ vanishes at both ends. With the same ψ,

$$\frac{T\psi}{\psi} = 3 + (2+5a)\kappa\cos\theta - a(1+a)\kappa^2\sin^2\theta,$$

which for $a = -\frac{2}{5}$ is $3 + \frac{6}{25}\kappa^2\sin^2\theta \ge 3$. Barta's inequality applied to h gives $\lambda \ge 3$. Together with the case $\ell \ge 1$ every nonzero eigenvalue of $-L$ is at least 3, which is Lemma 53; the spectrum is discrete because S^3 is compact and the drift is smooth.

3. Moments

The density of $t = x_1$ under ν_b is proportional to $(1-t^2)^{1/2}e^{\kappa t}$. For every smooth ϕ on $[-1,1]$,

$$\frac{d}{dt}\Big[(1-t^2)^{3/2}e^{\kappa t}\Big] = (1-t^2)^{1/2}e^{\kappa t}\big(\kappa(1-t^2)-3t\big),$$

and integration by parts gives $\mathbb{E}[\phi'(t)(1-t^2)] = \mathbb{E}\big[\phi(t)\big(3t-\kappa(1-t^2)\big)\big]$. With $\phi = 1$ this is $\kappa\mathbb{E}(1-x_1^2) = 3\mathbb{E}x_1$, the identity of Lemma 54(i). With $\phi = t^k$ it gives the recursion

$$m_{k+2} = m_k - \frac{k+3}{\kappa}\,m_{k+1} + \frac{k}{\kappa}\,m_{k-1}, \qquad m_k = \mathbb{E}x_1^k,$$

from $m_0 = 1$ and $m_1 = A$; all polynomial moments of ν_b are rational in κ and $A(\kappa)$.[17]

4. The Riccati barriers

By Lemma 54(i), A solves $A' = \mathscr{F}(\kappa,A) := 1 - A^2 - 3A/\kappa$ on $(0,\infty)$, and $A(\kappa) = \frac{\kappa}{4} - \frac{\kappa^3}{96} + O(\kappa^5)$ from the series of I_1 and I_2. Let $s = \sqrt{\kappa^2+4}$, $u = \sqrt{\kappa^2+9}$, $\phi_- = \kappa/(2+s)$ and $\phi_+ = \kappa/(1+u)$. Direct differentiation gives

$$\phi_-' - \mathscr{F}(\kappa,\phi_-) = -\frac{\kappa^2}{s(s+2)^2} \le 0,$$

$$\phi_+' - \mathscr{F}(\kappa,\phi_+) = \frac{(u-3)^2}{u(1+u)^2} \ge 0,$$

and $\phi_\pm = \frac{\kappa}{4} + O(\kappa^3)$.[18] The comparison is singular at 0 because of the term $3A/\kappa$, and we carry it out with an integrating factor. Let $D = A - \phi_-$. Then

$$\begin{aligned} D' &= \mathscr{F}(\kappa,A) - \mathscr{F}(\kappa,\phi_-) - \big(\phi_-' - \mathscr{F}(\kappa,\phi_-)\big) \\ &= -\Big(A + \phi_- + \frac{3}{\kappa}\Big)D + R, \end{aligned}$$

with $R = \kappa^2/(s(s+2)^2) \ge 0$. Put $\mu(\kappa) = \kappa^3\exp\int_0^\kappa (A+\phi_-)$. Then $(\mu D)' = \mu R \ge 0$, and $\mu D = O(\kappa^6) \to 0$ as $\kappa \to 0$, so $\mu D \ge 0$ and $A \ge \phi_-$. The same argument with $D = \phi_+ - A$ and $R = (u-3)^2/(u(1+u)^2)$ gives $A \le \phi_+$. This is Lemma 54(ii).

For (iv) let $s = \sqrt{\kappa^2+25/4}$ and $\phi_\flat = \kappa/(\frac{3}{2}+s)$. Since $s^2 - \frac{9}{4} = \kappa^2+4$, the same differentiation gives

$$\phi_\flat' - \mathscr{F}(\kappa,\phi_\flat) = -\frac{5\,(s-\frac{5}{2})}{2s\,(\frac{3}{2}+s)^2} \le 0,$$

so $\phi_\flat$ is a subsolution, and $\phi_\flat = \frac{\kappa}{4} - \frac{\kappa^3}{80} + O(\kappa^5)$.[19] Hence $A - \phi_\flat = \frac{\kappa^3}{480} + O(\kappa^5)$, and the integrating-factor argument above, with $D = A - \phi_\flat$, $R = \frac{5(s-5/2)}{2s(3/2+s)^2} \ge 0$ and $\mu = \kappa^3\exp\int_0^\kappa (A+\phi_\flat)$, gives $\mu D = O(\kappa^6)$ at 0, $(\mu D)' = \mu R \ge 0$, and so $A \ge \phi_\flat$. Since $\frac{3}{2} + \sqrt{\kappa^2+25/4} < 2 + \sqrt{\kappa^2+4}$ for $\kappa > 0$, this improves the lower bound of (ii).

Appendix I: Bessel functions

This appendix proves the auxiliary statements used in Appendix G for Theorem 56. Throughout, $I_j = I_j(\kappa)$ with $\kappa > 0$, and $I_{-j} = I_j$ for integers j. We use the series

$$I_\nu(\kappa) = \sum_{i\ge 0} \frac{(\kappa/2)^{2i+\nu}}{i!\,(i+\nu)!},$$

the integral $I_j(\kappa) = \pi^{-1}\int_0^\pi e^{\kappa\cos\theta}\cos j\theta\,d\theta$, the Fourier series $e^{\kappa\cos\theta} = \sum_{j\in\mathbb{Z}} I_j e^{ij\theta}$ and the recurrences

$$I_{j-1} - I_{j+1} = \frac{2j}{\kappa}I_j, \quad I_j' = I_{j+1} + \frac{j}{\kappa}I_j, \tag{I1}$$

all of which are in Watson's treatise [62], §§3.7, 3.71, 2.22. All series below converge absolutely, because $0 < I_j \le (\kappa/2)^j e^\kappa/j!$.

1. Zonal harmonics

Under σ the coordinate x_1 has the density $\frac{2}{\pi}(1-t^2)^{1/2}$ on $[-1,1]$, the area of the two-sphere $\{x_1 = t\}$ being proportional to $1-t^2$ and its normal speed to $(1-t^2)^{-1/2}$. The Chebyshev polynomials $U_n(\cos\theta) = \sin((n+1)\theta)/\sin\theta$ satisfy

$$\int U_n U_m\,d\sigma = \frac{2}{\pi}\int_0^\pi \sin((n+1)\theta)\sin((m+1)\theta)\,d\theta = \delta_{nm},$$

and $U_n = C_n^{(1)}$ is the Gegenbauer polynomial of index $\frac{4-2}{2} = 1$, so $U_n(\langle y,x\rangle)$ is the zonal harmonic of degree n on S^3 with pole y: $|x|^n U_n(\langle y,x\rangle/|x|)$ is a harmonic polynomial on $\mathbb{R}^4$ [58], Ch. IV, §2.

Lemma 62. *$e^{\kappa t} = \frac{2}{\kappa}\sum_{n\ge 0}(n+1)I_{n+1}(\kappa)U_n(t)$ in $L^2\big(\frac{2}{\pi}(1-t^2)^{1/2}dt\big)$, and uniformly on $[-1,1]$.*

Proof. With $t = \cos\theta$ and $2\sin\theta\sin((n+1)\theta) = \cos n\theta - \cos(n+2)\theta$, the n-th coefficient is

$$\begin{aligned} \frac{2}{\pi}\int_0^\pi e^{\kappa\cos\theta}\sin((n+1)\theta)\sin\theta\,d\theta &= I_n - I_{n+2} \\ &= \frac{2(n+1)}{\kappa}I_{n+1}, \end{aligned}$$

[17] For instance $m_2 = 1 - 3A/\kappa$, $m_3 = A - 3/\kappa + 12A/\kappa^2$ and $m_4 = m_2 - 5m_3/\kappa + 2A/\kappa$. The joint moments with x_2, x_3, x_4 follow from exchangeability: $\mathbb{E}[\phi(x_1)x_2^2] = \frac{1}{3}\mathbb{E}[\phi(x_1)(1-x_1^2)]$.

[18] For the first identity write $\phi_- = (s-2)/\kappa$, which follows from $s^2 - 4 = \kappa^2$; then $\phi_-' = (\kappa^2/s - s + 2)/\kappa^2$ and $1 - \phi_-^2 - 3\phi_-/\kappa = (\kappa^2 - (s-2)^2 - 3(s-2))/\kappa^2$, and the difference simplifies with $s^2 = \kappa^2+4$. The second is the same computation with $\phi_+ = \kappa(u-1)/(u^2-1) = \kappa(u-1)/(\kappa^2+8)$ in place of ϕ_- and $u^2 = \kappa^2+9$ in place of $s^2 = \kappa^2+4$. After clearing the denominator $\kappa^2(1+u)^2$ the numerator becomes $\kappa^2(u-3)^2/u$, which is the stated residual.

[19] With $\phi_\flat = \kappa(s-\frac{3}{2})/(\kappa^2+4)$ one has $\phi_\flat' = (\frac{3}{2}s + \frac{25}{4})/(s(\frac{3}{2}+s)^2)$, using $ds/d\kappa = \kappa/s$; and $\mathscr{F}(\kappa,\phi_\flat) = 1 - \phi_\flat^2 - 3/(\frac{3}{2}+s)$. Clearing the denominator $s(\frac{3}{2}+s)^2$ and replacing κ^2 by $s^2 - \frac{25}{4}$ leaves $-\frac{5}{2}s + \frac{25}{4}$ in the numerator, which is the stated residual. The expansion follows from $s = \frac{5}{2} + \frac{\kappa^2}{5} + O(\kappa^4)$.

by (I1). Since $|U_n| \le n+1$ and I_{n+1} decays factorially, the series converges uniformly. □

2. Proof of Lemma 59

Let $F(\theta) = e^{\kappa\cos\theta} = \sum_j I_j e^{ij\theta}$. Then

$$F'(\theta) = -\kappa\sin\theta\, F(\theta) = \sum_j ijI_j e^{ij\theta},$$

and Parseval's formula gives

$$\sum_{j\in\mathbb{Z}} j^2 I_j^2 = \frac{\kappa^2}{2\pi}\int_0^{2\pi} \sin^2\theta\, e^{2\kappa\cos\theta} d\theta$$
$$= \frac{\kappa^2}{2}\big(I_0(2\kappa) - I_2(2\kappa)\big) = \frac{\kappa}{2} I_1(2\kappa),$$

the last step by (I1) at the argument 2κ. Halving gives the first identity. The coefficient of $e^{i\theta}$ in $F^2 = e^{2\kappa\cos\theta}$ is $I_1(2\kappa) = \sum_{j\in\mathbb{Z}} I_j I_{1-j}$, and the terms with $j \le 0$ repeat those with $j \ge 1$; this is the third identity.

For the second, $jI_j = \frac{\kappa}{2}(I_{j-1} - I_{j+1})$ by (I1), so

$$\sum_{j\ge1} j^3 I_j^2 = \frac{\kappa^2}{4}\sum_{j\ge1} j\big(I_{j-1}^2 + I_{j+1}^2 - 2I_{j-1}I_{j+1}\big).$$

Shifting indices, $\sum_{j\ge1} j(I_{j-1}^2 + I_{j+1}^2) = I_0^2 + \sum_{m\ge1} 2mI_m^2$, and $\sum_{m\ge1} 2mI_m^2 = \kappa\sum_{m\ge1}(I_{m-1}I_m - I_mI_{m+1}) = \kappa I_0 I_1$ by telescoping. For the mixed term write $j = \frac12((j-1)+(j+1))$ and apply (I1) to $(j-1)I_{j-1}$ and to $(j+1)I_{j+1}$, using $I_{-1} = I_1$ at $j = 1$:

$$jI_{j-1}I_{j+1} = \frac{\kappa}{4}\big[(I_{j-1}I_j - I_jI_{j+1})$$
$$+ (I_{j-2}I_{j+1} - I_{j-1}I_{j+2})\big].$$

Both brackets telescope, to I_0I_1 and to $I_{-1}I_2 = I_1I_2$, so $\sum_{j\ge1} jI_{j-1}I_{j+1} = \frac{\kappa}{4}(I_0I_1 + I_1I_2)$. Altogether

$$\sum_{j\ge1} j^3 I_j^2 = \frac{\kappa^2}{4}\Big(I_0^2 + \frac{\kappa}{2}I_1(I_0 - I_2)\Big) = \frac{\kappa^2}{4}\big(I_0^2 + I_1^2\big).$$ □

3. The coefficients of $\mathscr{D}$

Lemma 63. *Write $\mathscr{D}(\kappa) = \sum_{n\ge0}\mathscr{D}_n\kappa^{2n}$ for the function of (G9). Then $\mathscr{D}_0 = \mathscr{D}_1 = 0$, and for $m \ge 1$*

$$\mathscr{D}_{m+1} = \frac{N_m}{4^m\, m!\,(m+1)!\,(m+1)(m+2)},$$

with $N_m = \binom{2m}{m}\big(4(2m+1) + (m-1)(m+1)(m+2)\big) - 4^m(m+1)(m+2)$. Moreover $N_m = 0$ for $m \le 2$ and $N_m > 0$ for $m \ge 3$.

Proof. The coefficients. Multiplying the series of I_0 and I_1 and collecting powers,

$$I_0^2 = \sum_{m\ge0}\alpha_m\kappa^{2m}, \qquad I_1^2 = \sum_{m\ge0}\beta_m\kappa^{2m+2},$$
$$\kappa I_1(2\kappa) = \sum_{m\ge0}\frac{\kappa^{2m+2}}{m!\,(m+1)!},$$

where

$$\alpha_m = \frac{1}{4^m}\sum_{i=0}^{m}\frac{1}{i!^2(m-i)!^2},$$
$$\beta_m = \frac{1}{4^{m+1}}\sum_{i=0}^{m}\frac{1}{i!\,(i+1)!\,(m-i)!\,(m-i+1)!}.$$

By Vandermonde's identity, $\sum_i \binom{m}{i}^2 = \binom{2m}{m}$ and $\sum_i \binom{m}{i}\binom{m+2}{i+1} = \binom{2m+2}{m+1}$, so

$$\alpha_m = \frac{\binom{2m}{m}}{4^m\, m!^2}, \qquad \beta_m = \frac{\binom{2m+2}{m+1}}{4^{m+1} m!\,(m+2)!}.$$

The coefficient of κ^{2m+2} in $\mathscr{D}$ is $\mathscr{D}_{m+1} = 8\beta_m + 2\beta_{m-1} - \alpha_m - 1/(m!(m+1)!)$, with $\beta_{-1} = 0$; for $m = 0$ it is $\frac84 - 1 - 1 = 0$, and $\mathscr{D}_0 = 0$ because every term of $\mathscr{D}$ is $O(\kappa^2)$. For $m \ge 1$ put $\Delta = 4^m\, m!\,(m+1)!\,(m+1)(m+2)$ and $b = \binom{2m}{m}$. With $\binom{2m+2}{m+1} = \frac{2(2m+1)}{m+1}b$,

$$8\beta_m = \frac{4(2m+1)\,b}{\Delta}, \qquad \alpha_m = \frac{(m+1)^2(m+2)\,b}{\Delta},$$
$$2\beta_{m-1} = \frac{2m(m+1)(m+2)\,b}{\Delta},$$
$$\frac{1}{m!(m+1)!} = \frac{4^m(m+1)(m+2)}{\Delta},$$

and $2m - (m+1) = m-1$ gives $\mathscr{D}_{m+1} = N_m/\Delta$.

The sign. Directly, $N_0 = N_1 = N_2 = 0$ ($2-2$, $24-24$, $192-192$), $N_3 = 80$, $N_4 = 1140$ and $N_5 = 10416$. For $m \ge 6$ we first show $16^m \le 4m\binom{2m}{m}^2$ for $m \ge 1$ by induction: it is an equality at $m = 1$, and $(m+1)\binom{2m+2}{m+1} = 2(2m+1)\binom{2m}{m}$ gives

$$4(m+1)\binom{2m+2}{m+1}^2 = \frac{16(2m+1)^2}{m+1}\binom{2m}{m}^2$$
$$\ge \frac{16(2m+1)^2}{4m(m+1)}16^m \ge 16^{m+1}.$$

Since $(m-1)^2 \ge 4m$ for $m \ge 6$, it follows that $\binom{2m}{m}^2(m-1)^2 \ge 16^m$, that is $\binom{2m}{m}(m-1) \ge 4^m$, and

$$N_m = \left(\binom{2m}{m}(m-1) - 4^m\right)(m+1)(m+2)$$
$$+ 4(2m+1)\binom{2m}{m} > 0.$$

□

4. Monotonicity of Bessel ratios

Lemma 64. *For $n \ge 1$ and $\kappa > 0$, $I_{n+2}(\kappa)/I_{n+1}(\kappa) \le I_2(\kappa)/I_1(\kappa)$.*

Proof. We use a lemma of Biernacki and Krzyż [18]: if $f=\sum a_j x^j$ and $g=\sum b_j x^j$ have positive coefficients and a_j/b_j is nonincreasing in j, then f/g is nonincreasing on $x>0$. Indeed

$$f'g-fg' = \sum_{i,j}(i-j)a_ib_jx^{i+j-1} = \frac{1}{2}\sum_{i,j}(i-j)b_ib_j\Big(\frac{a_i}{b_i}-\frac{a_j}{b_j}\Big)x^{i+j-1}\le 0.$$

Take $x=\kappa^2$, $f=I_{n+1}(\kappa)/\kappa^{n+1}$ and $g=I_1(\kappa)/\kappa$. Then $a_j/b_j=2^{-n}(j+1)!/(j+n+1)!$ decreases in j, so $I_{n+1}/(\kappa^n I_1)$ is nonincreasing in κ. Its logarithmic derivative is, by the second recurrence of (I1),

$$\frac{n+1}{\kappa}+\frac{I_{n+2}}{I_{n+1}}-\frac{n}{\kappa}-\frac{1}{\kappa}-\frac{I_2}{I_1} = \frac{I_{n+2}}{I_{n+1}}-\frac{I_2}{I_1}\le 0.$$

□

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